AKADEMIKA

# BEL-COT

## DISCRETE COMPONENTS TRAINER

## EXPERIMENTAL MANUAL

**…. A MESSAGE FROM**

Discrete Component Trainer is the first of its kind using actual techniques of discrete components applications. It is our vision to furnish products, which are reliable and convenient to use. AKADEMIKA offers a high degree of expertise in developing efficient products.

## Company's Motto

- Light years ahead – refers to leadership.

We strive to be the best in what we do and maintain high standards in the areas of

• Design • Quality • Value • Delivery • Support

We are indeed light years ahead of the competition in this field. We guarantee our valued customers great satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete and comprehensive warranty.

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# SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.

- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.

- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.

- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.

- The kits are liable to malfunction/ underperforms if it is not operated under following conditions:

➢ Ambient temperature: Between 0 to 45 ° C
➢ Relative humidity: Between 20 to 80 %

- Avoid any immediate/significant change of temperature and humidity

# WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

**Conditions and Limitations**

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

# INDEX

# INTRODUCTION

Discrete component trainer is designed mainly to study characteristics of basic semiconductor devices such as diodes, BJT, FET, MOSFET, UJT, PUT, DIAC, TRIAC, SCR, IGBT, LDR, Photo transistor, photo diode etc. To achieve the flexibility in the design various resources such as resistor, capacitor, diode and Potentiometer banks of different values are provided on board.

Using some of the onboard resources user can design small applications. Relay and opto-coupler is provided to study their functionality and characteristics. On boards variable regulated positive and negative power supply is available to cater the need of different supply voltages required in the design. The trainer system is provided with descriptive experimental manual.

# TECHNICAL SPECIFICATIONS

1) Dual Isolated power supply +35V,-12V, GND and +35VA,-12VA, AGND 500mA.
2) Dual DC 0 to 30V, 0 to 30VA Variable On-Board power supply.
3) Basic component study like resistors, capacitors, inductors and potentiometer.
4) 6 different types of diode characteristics study like General purpose diode, Ge Diode, FSD, Zener Diode, LED and Varactor Diode.
5) 5 different types of transistors characteristics study like BJT, JFET, MOSFET, UJT and PUT.
6) 4 different types of power component study like DIAC, TRIAC, SCR and IGBT.
7) Opto-copular and relay study.
8) 16 Resistors bank with different wattage.
9) 6 Capacitors bank.
10) 2 inductors.
11) 1 Potentiometer.
12) 3 diodes.
13) 1 Bread board.

# EXPERIMENT NO. 1

# EXPERIMENT NO.1

**AIM:-**
TO STUDY THE WORKING OF RESISTORS

**OBJECTIVE:-**
1. To study the color codes, types of resistors.
2. Working of resistors in series and parallel combination.

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meters as current and volt meters.

**THEORY:-**
A **resistor** is a two-terminal electronic component that produces a voltage across its terminals that is proportional to the electric current passing through it in accordance with Ohm's law: V = IR

Resistors are elements of electrical networks and electronic circuits and are ubiquitous in most electronic equipment. Practical resistors can be made of various compounds and films, as well as resistance wire (wire made of a high-resistivity alloy, such as nickel/chrome).

The primary characteristics of a resistor are the resistance, the tolerance, maximum working voltage and the power rating. Other characteristics include temperature coefficient, noise, and inductance. Less well-known is critical resistance, the value below which power dissipation limits the maximum permitted current flow, and above which the limit is applied voltage. Critical resistance depends upon the materials constituting the resistor as well as its physical dimensions; it's determined by design.

**DC resistance:**
The resistance R of a conductor of uniform cross section can be computed as

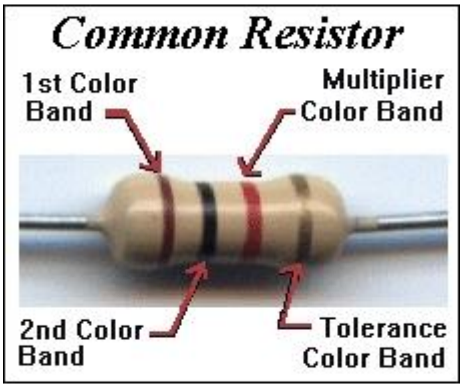
$$R = \frac{\ell \cdot \rho}{A}$$

Where, $\ell$ is the length of the conductor, measured in meters (m)
A is the cross-sectional area, measured in square meters ($m^2$)
ρ (Greek: rho) is the electrical resistivity (also called specific electrical resistance) of the material, measured in Ohm · meter (Ω m).
Resistivity is a measure of the material's ability to oppose electric current.

**AC resistance:**
If a wire conducts high-frequency alternating current then the effective cross sectional area of the wire is reduced because of the skin effect. If several conductors are together, then due to proximity effect, the effective resistance of each is higher than if that conductor were alone.
All modern resistors can be classified into four broad groups;

1. Carbon Composition Resistor - Made of carbon dust or graphite paste, low wattage values.
2. Film or Cermet Resistor - Made from conductive metal oxide paste, very low wattage values.
3. Wire-Wound Resistors. - Metallic bodies for heatsink mounting, very high wattage ratings.
4. Semiconductor Resistors - High frequency/precision surface mount thin film technology.

There are a large variety of fixed and variable resistor types with different construction styles available for each group, with each one having its own particular Characteristics, Advantages and Disadvantages. To include all types would make this section very large so I shall limit it to the most commonly used, and readily available general purpose resistor types.




Resistors are color coded. To read the color code of a common 4 band 1K ohm resistor with a 5% tolerance, start at the opposite side of the GOLD tolerance band and read from left to right. Write down the corresponding number from the color chart below for the 1st color band (BROWN). To the right of that number, write the corresponding number for the 2nd band (BLACK). Now multiply that number (you should have 10) by the corresponding multiplier number of the 3rd band (RED) (100). Your answer will be 1000 or 1K. It's that easy.

If a resistor has 5 color bands, write the corresponding number of the 3rd band to the right of the 2nd before you multiply by the corresponding number of the multiplier band. If you only have 4 color bands that include a tolerance band, ignore this column and go straight to the multiplier.

The tolerance band is usually gold or silver, but some may have none. Because resistors are not the exact value as indicated by the color bands, manufactures have included a tolerance color band to indicate the accuracy of the resistor. Gold band indicates the resistor is within 5% of what is indicated. Silver = 10% and None = 20%. Others are shown in the chart below. The 1K ohm resistor in the example (left), may have an actual measurement any where from 950 ohms to 1050 ohms. If a resistor does not have a tolerance band, start from the band closest to a lead. This will be the 1st band. If you are unable to read the color bands, then you'll have to use your multimeter. Be sure to zero it out first!

**Resistor Color Codes**

| Band Color | 1st Band # | 2nd Band # | *3rd Band # | Multiplier x | Tolerances ± % |
|---|---|---|---|---|---|
| Black | 0 | 0 | 0 | 1 | |
| Brown | 1 | 1 | 1 | 10 | ± 1% |
| Red | 2 | 2 | 2 | 100 | ± 2 % |
| Orange | 3 | 3 | 3 | 1000 | |
| Yellow | 4 | 4 | 4 | 10,000 | |
| Green | 5 | 5 | 5 | 100,000 | ± 0.5 % |
| Blue | 6 | 6 | 6 | 1,000,000 | ± 0.25 % |
| Violet | 7 | 7 | 7 | 10,000,000 | ± 0.10 % |
| Grey | 8 | 8 | 8 | 100,000,000 | ± 0.05 % |
| White | 9 | 9 | 9 | 1,000,000,000 | |
| Gold | | | | 0.1 | ± 5 % |
| Silver | | | | 0.01 | ± 10 % |
| None | | | | | ± 20 % |

## Resistor Combinations:

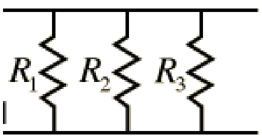
The combination rules for any number of resistors in series or parallel can be derived with the use of Ohm's Law, the voltage law, and the current law.


Series

$$\boxed{R_{equivalent} = R_1 + R_2 + R_3 + \dots}$$

$$R_{equivalent} = \frac{V}{I} = \frac{V_1 + V_2 + V_3 + \dots}{I} = \frac{V_1}{I_1} + \frac{V_2}{I_2} + \frac{V_3}{I_3} + \dots = R_1 + R_2 + R_3 + \dots$$

Series key idea: The current is the same in each resistor by the current law.




Parallel

$$\boxed{\frac{1}{R_{equivalent}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots}$$

Parallel: $$\frac{V}{R_{equivalent}} = I = I_1 + I_2 + I_2 + \dots = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + \dots$$

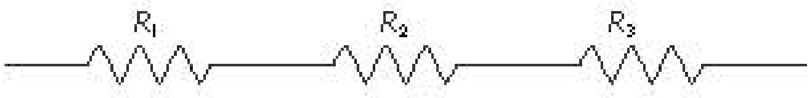
$$\frac{1}{R_{equivalent}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

Parallel key idea: The voltage is the same across each resistor by the voltage law.

**<u>PROCEDURE</u>:-**

1. Identify the resistors value, which are used for experiment using color code.
2. Make the connections on COT-01 board as shown in fig. below for series and parallel combination.

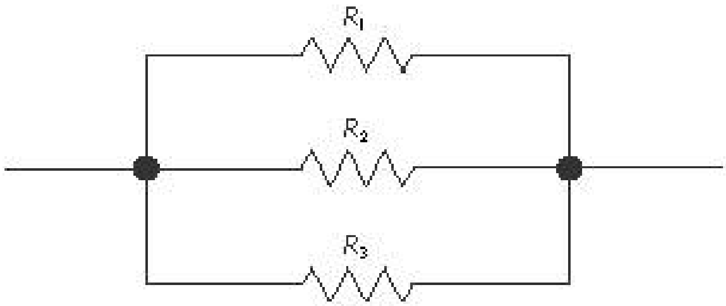
**1. Series Combination :-**




Resistors in series

Total resistance $R$ given by $R = R_1 + R_2 + R_3$

**2. Parallel Combination :-**




Resistors in parallel

Total resistance $R$ given by $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

3. Calculate the series and parallel combination of resistors value.
4. Use the multi-meter to read the series and parallel combination of resistor value.
5. Verify the calculated and observed value.

**DESIGN:-**

Consider
R1 = 1K resistor.
R2 = 2K resistor.
R3 = 10K resistor.

**OBSERVATION TABLE:-**

| Sr. no. | Combination | Calculated value | Observed Value |
|---|---|---|---|
| 1 | Series ($R_{equivalent}$) | 13 KΩ | 12.91 KΩ |
| 2 | Parallel ($R_{equivalent}$) | 625 Ω | 624 Ω |

**CONCLUSION:-**

The calculated and observed values of the resistor combinations are almost equal.

# EXPERIMENT NO. 2

# EXPERIMENT NO.2

**AIM:-**
TO STUDY THE WORKING OF CAPACITORS

**OBJECTIVE:-**
1. To study the types of capacitors.
2. Working of capacitors in series and parallel combination.

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. LCR meter.

**THEORY:-**
A **capacitor** or **condenser** is a passive electronic component consisting of a pair of conductors separated by a dielectric (insulator). When a potential difference (voltage) exists across the conductors, an electric field is present in the dielectric. This field stores energy and produces a mechanical force between the conductors. The effect is greatest when there is a narrow separation between large areas of conductor; hence capacitor conductors are often called plates.

An ideal capacitor is characterized by a single constant value, capacitance, which is measured in farads. This is the ratio of the electric charge on each conductor to the potential difference between them. In practice, the dielectric between the plates passes a small amount of leakage current. The conductors and leads introduce an equivalent series resistance and the dielectric has an electric field strength limit resulting in a breakdown voltage.

A capacitor consists of two conductors separated by a non-conductive region. The non-conductive substance is called the dielectric medium, although this may also mean a vacuum or a semiconductor depletion region chemically identical to the conductors. A capacitor is assumed to be self-contained and isolated, with no net electric charge and no influence from an external electric field. The conductors thus contain equal and opposite charges on their facing surfaces, and the dielectric contains an electric field. The capacitor is a reasonably general model for electric fields within electric circuits.
An ideal capacitor is wholly characterized by a constant capacitance *C*, defined as the ratio of charge ±*Q* on each conductor to the voltage *V* between them:

$$C = \frac{Q}{V}$$

Sometimes charge buildup affects the mechanics of the capacitor, causing the capacitance to vary. In this case, capacitance is defined in terms of incremental changes:

$$C = \frac{\mathrm{d}q}{\mathrm{d}v}$$

In SI units, a capacitance of one farad means that one coulomb of charge on each conductor causes a voltage of one volt across the device.
A capacitor is a passive electronic component that stores energy in the form of an electrostatic field. In its simplest form, a capacitor consists of two conducting plates separated by an insulating material called the dielectric. The capacitance is directly proportional to the surface areas of the plates, and is inversely proportional to the

separation between the plates. Capacitance also depends on the dielectric constant of the substance separating the plates.

The standard unit of capacitance is the farad, abbreviated F. This is a large unit; more common units are the microfarad, abbreviated µF (1 µF $=10^{-6}$F) and the pico-farad, abbreviated pF (1 pF = $10^{-12}$ F).

A table giving translations of previous commonly used multiples is as follows:

| **Preferred** | **In pf** | **In nf** | **In µf** |
|---|---|---|---|
| 1pf | 1 | 0.001 | 0.000001 |
| 10pf | 10 | 0.01 | 0.00001 |
| 100pf | 100 | 0.1 | 0.0001 |
| 1nf | 1000 | 1 | 0.001 |
| 10nf | 10000 | 10 | 0.01 |
| 100nf | 100000 | 100 | 0.1 |
| 1µf | 1000000 | 1000 | 1 |

Types of Capacitors:-
1) Metalized film Capacitors: - It uses a very thin aluminum foil, interlink with a very thin plastic dielectric. For high voltage capacitors, polyester, polystyrene plastic materials are used. Dielectric strength – is the ability of an inductor to resist the flow of current when voltage is applied across it.

2) Ceramic Capacitors: - Ceramic capacitors are used when small values of capacitance and large value of linkage current are acceptable. Ceramic capacitors are inexpensive and consist of a thin ceramic dielectric that is metalized on each side and coated with a thick protective layer usually applied by Deepings the component in an electrified protective material. Both plastics film and ceramic capacitors are available in ranges of 10pfto1uf.

3) Electrolytic Capacitors:- Electrolytic capacitors are high voltage capacitors, they give high value of capacitance in a small component at the expenses of wide tolerance in the marked value typically between -25 to 50% and the necessity of connecting the capacitor so that one terminal is always positive. The most commonly used types of electrolytic capacitors is aluminum electrolytic capacitor. Electrolytic capacitors must not be subjected to voltages in the wrong direction.

4) Variable Capacitors: - These capacitors are variable in values. Either air or thin mica sheet are used as dielectric and therefore variable capacitors is usually low, there two types of variable capacitors namely

a) Rotary
b) Compression type

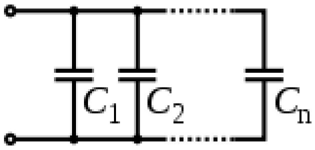
For capacitors in series


Several capacitors in series.
Connected in series, the schematic diagram reveals that the separation distance, not the plate area, adds up. The capacitors each store instantaneous charge build-up equal to that of every other capacitor in the series. The total voltage difference from end to end is apportioned to each capacitor according to the inverse of its capacitance. The entire series acts as a capacitor smaller than any of its components.

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$$

Capacitors are combined in series to achieve a higher working voltage, for example for smoothing a high voltage power supply.

For capacitors in parallel:



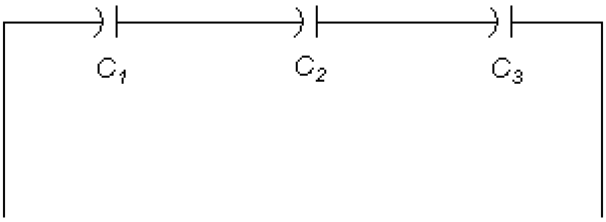

Several capacitors in parallel:
Capacitors in a parallel configuration each have the same applied voltage. Their capacitances add up. Charge is apportioned among them by size. Using the schematic diagram to visualize parallel plates, it is apparent that each capacitor contributes to the total surface area.

$$C_{eq} = C_1 + C_2 + \cdots + C_n$$

**PROCEDURE:-**

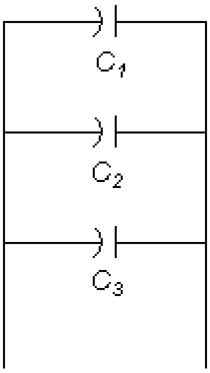
1. Identify the capacitor value, which are used for experiment using no. on it.
2. Make the connections on COT-01 board as shown below for series and parallel combination.

1. Series Combination :-




For series-connected capacitors the equivalent capacitance can be expressed as, 1/C = 1/C1 + 1/C2 + . . + 1/Cn

2. Parallel Combination :




For parallel-connected capacitors the equivalent capacitance can be expressed as, C = C1 + C2 + . . + Cn
Where, C = capacitance (Farad, F, μF)

3. Calculate the series and parallel combination of capacitor value.
4. Use the multi-meter to read the series and parallel combination of capacitor value.
5. Verify the calculated and observed value.

**DESIGN:-**

Consider
C1 = .1uf capacitor.
C2 = 1uf capacitor.
C3 = 100uf capacitor.

**OBSERVATION TABLE:-**

| Sr. no. | Combination | Calculated value | Observed Value |
|---|---|---|---|
| 1 | Series ($C_{equivalent}$) | 90.826 nf | 92 nf |
| 2 | Parallel ($C_{equivalent}$) | 101.1 μf | 105 μf |

**CONCLUSION:-**

The calculated and observed values of the capacitors combinations are almost equal.

# EXPERIMENT NO. 3

# EXPEIMENT NO.3

**AIM:-**
TO STUDY THE WORKING OF INDUCTORS

**OBJECTIVE:-**
1. To study the types of Inductors and color code of inductor.
2. Working of inductor in series and parallel combination.

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. LCR meter.

**THEORY:-**
An inductor is a passive electronic component that stores energy in the form of a magnetic field. In its simplest form, an inductor consists of a wire loop or coil. The inductance is directly proportional to the number of turns in the coil. Inductance also depends on the radius of the coil and on the type of material around which the coil is wound.

For a given coil radius and number of turns, air cores result in the least inductance. Materials such as wood, glass, and plastic - known as dielectric materials - are essentially the same as air for the purposes of inductor winding. Ferromagnetic substances such as iron, laminated iron, and powdered iron increase the inductance obtainable with a coil having a given number of turns. In some cases, this increase is on the order of thousands of times. The shape of the core is also significant. Toroidal (donut-shaped) cores provide more inductance, for a given core material and number of turns, than solenoidal (rod-shaped) cores.

The standard unit of inductance is the henry, abbreviated H. This is a large unit. More common units are the micro henry, abbreviated µH (1 µH =$10^{-6}$H) and the millihenry, abbreviated mH (1 mH =$10^{-3}$ H). Occasionally, the nanohenry (nH) is used (1 nH = $10^{-9}$ H).

It is difficult to fabricate inductors onto integrated circuit (IC) chips. Fortunately, resistors can be substituted for inductors in most microcircuit applications. In some cases, inductance can be simulated by simple electronic circuits using transistors, resistors, and capacitors fabricated onto IC chips.
Inductors are used with capacitors in various wireless communications applications. An inductor connected in series or parallel with a capacitor can provide discrimination against unwanted signals. Large inductors are used in the power supplies of electronic equipment of all types, including computers and their peripherals. In these systems, the inductors help to smooth out the rectified utility AC, providing pure, battery-like DC.

**Inductor's Voltage:**

$$v = L\frac{di}{dt}$$

**Inductor's Current:**

$$i = \frac{1}{L}\int_{t_1}^{t_2} v(t)\,dt + i\,(t_1)$$

Note that usually, the voltage will fall off to zero as the magnetic field is set up, reaching zero at an infinite time after the voltage is applied, so the integral of voltage tends to a finite limit, meaning that the current is also a finite quantity.

**Inductor's Impedance:**

For an ideal lossless inductor the impedance of an inductor at any given angular frequency ω is given by,

Z = jωL where j is $\sqrt{-1}$ and L is the inductance.
Z = sL
Z = ωL /_90

In practice, all conductor has a resistance hence for non-ideal lossless inductor

$Z_L = R_L + Z$

$Z = R_L + j\omega L$ where j is $\sqrt{-1}$ and L is the inductance.
$Z = R_L + sL$
Z = R_L/-0 + ωL /_90

**Quality factor:**

Quality factor denoted as Q is defined as the ability to store energy to the sum total of all energy losses within the component,

$$Q = \frac{X}{R}$$

Where,

- Q = quality factor (no units)
- R = total resistance associated with energy losses (in ohms)
- X = reactance (in ohms). ($X_L = 2\pi fL$ for inductors; for capacitors; f is the frequency of interest).

$$X_C = \frac{1}{2\pi fC}$$

**Types of Inductors:**

Inductors are classified by the type of core and the winding. Kinds of inductors are: coupled inductors, multi-layer chip inductors, wire wound ceramic core inductors, molded chip inductors, power inductors, shielded power inductors, molded inductors, conformal coated inductors, and wide band chokes.

**1) Coupled inductors**: In these types of conductors, the magnetic flux of one inductor is linked to another conductor. These are used in special applications where mutual inductance is required like in transformers.

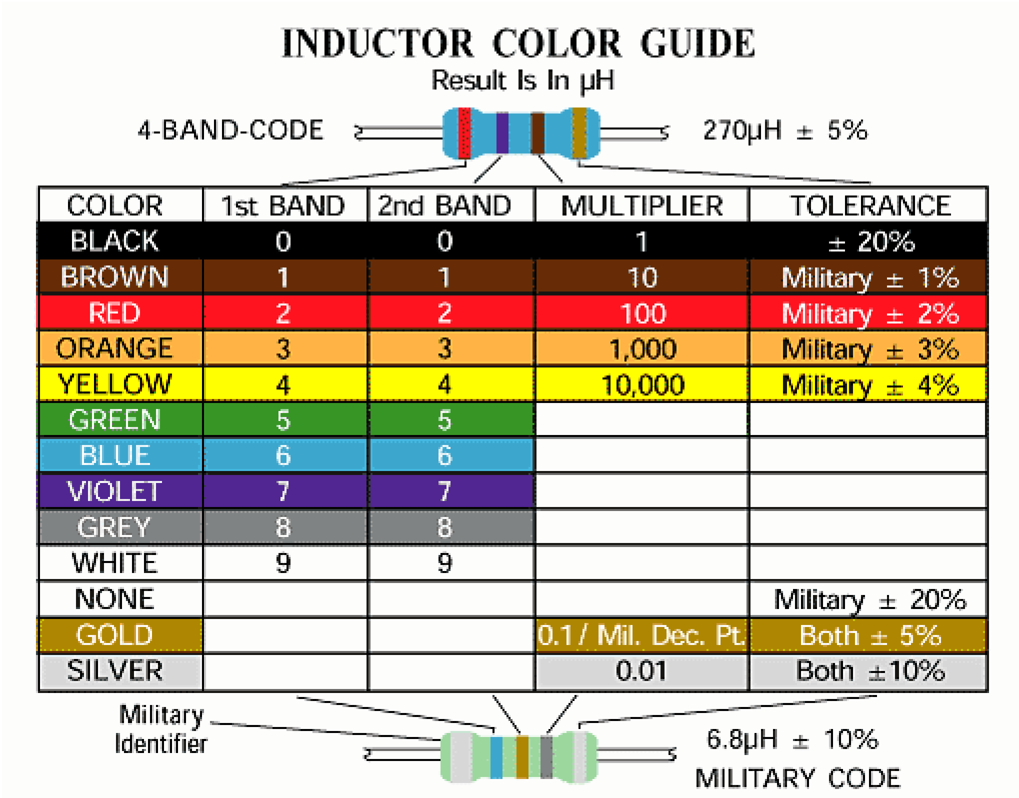
**2) Multi layer inductors**: In this type, the coil is wound in multiple layers with insulation between each layer. This provides very high inductance.

**3) Ceramic core inductors**: The core is made of a ceramic, which is a dielectric material and this type F conductor has high linearity, low Hysteresis and low distortion.

**4) Molded inductors**: These are low value inductors used in printed circuit boards. Usually bar or cylindrical in shape, they have windings on a core and the whole assembly is molded in plastic or ceramic insulation. Terminations are given at the ends and they come in two types: through mount and surface mount.

Other types are variations of the above types. Shielded types are shielded to avoid EMI and RFI.

**Color Coding:**
Small capacity molded inductors are given a color code for easy identification when they are mounted in circuits. Four-color bands are given in circles around the inductor. The bands are called 1st band, 2nd band, the third band is the multiplier and the fourth band indicates the tolerance. Colors used are: black, brown, red, orange, yellow, green, blue, violet, gray, white, no color, gold and silver. Colors in the first and second band have a specific value of inductance, the third band is a multiplier and the fourth band stands for the +- inductance tolerance like 1%, 2%, 3%, 20%, etc.




| COLOR | 1st BAND | 2nd BAND | MULTIPLIER | TOLERANCE |
|---|---|---|---|---|
| BLACK | 0 | 0 | 1 | ± 20% |
| BROWN | 1 | 1 | 10 | Military ± 1% |
| RED | 2 | 2 | 100 | Military ± 2% |
| ORANGE | 3 | 3 | 1,000 | Military ± 3% |
| YELLOW | 4 | 4 | 10,000 | Military ± 4% |
| GREEN | 5 | 5 | | |
| BLUE | 6 | 6 | | |
| VIOLET | 7 | 7 | | |
| GREY | 8 | 8 | | |
| WHITE | 9 | 9 | | |
| NONE | | | | Military ± 20% |
| GOLD | | | 0.1 / Mil. Dec. Pt. | Both ± 5% |
| SILVER | | | 0.01 | Both ±10% |

<u>Inductors In A Series Circuit</u>
The inductance in a series circuit is the sum of the inductances
Lt = L1 + L2 + ... + Ln

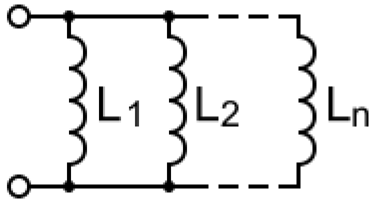
<u>Inductors In A Parallel Circuit</u>
The total inductance in a parallel circuit is less than the lowest inductance.
1/Lt = 1/L1 + 1/L2 + ... + 1/Ln

**<u>PROCEDURE</u>:-**
1. Identify the inductor value, which are used for experiment using color code on it.
2. Make the connections on COT-01 board as shown below for series and parallel combination.

### 1. Series Combination :-


$$L_{eq} = L_1 + L_2 + \cdots + L_n$$

**2. Parallel Combination :-**




$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \cdots + \frac{1}{L_n}$$

3. Calculate the series and parallel combination of inductor value.
4. Use the multi-meter to read the series and parallel combination of inductor value.
5. Verify the calculated and observed value.

**DESIGN:-**

Consider

L1 = L2 = 1 mH Inductor.

**OBSERVATION TABLE:-**

| Sr. no. | combination | Calculated value | Observed Value |
|---|---|---|---|
| 1 | Series ($L_{equivalent}$) | 2 mH | 2.08 mH |
| 2 | Parallel ($L_{equivalent}$) | 0.5 mH | 0.521 mH |

**CONCLUSION:-**

The calculated and observed values of the inductors combinations are almost equal.

# EXPERIMENT NO. 4

# EXPERIMENT NO.4

**AIM:-**
TO STUDY THE WORKING OF POTENTIOMETER

**OBJECTIVE:-**
To study the types and working of potentiometer as a Voltage divider

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter.

**THEORY:-**
A **potentiometer** (colloquially known as a "**pot**") is a three-terminal resistor with a sliding contact that forms an adjustable voltage divider. If only two terminals are used (one side and the wiper), it acts as a **variable resistor** or **rheostat**. Potentiometers are commonly used to control electrical devices such as volume controls on audio equipment. Potentiometers operated by a mechanism can be used as position transducers, for example, in a joystick.

Potentiometers are rarely used to directly control significant power (more than a watt). Instead they are used to adjust the level of analog signals (e.g. volume controls on audio equipment), and as control inputs for electronic circuits. For example, a light dimmer uses a potentiometer to control the switching of a TRIAC and so indirectly control the brightness of lamps.

Potentiometers are sometimes provided with one or more switches mounted on the same shaft. For instance, when attached to a volume control, the knob can also function as an on/off switch at the lowest volume.

**Types of potentiometers:**
**1) Low-power types:**
A potentiometer is constructed using a flat graphite annulus as the resistive element, with a sliding contact (wiper) sliding around this annulus.

**2) Linear potentiometers;**
A linear pot has a resistive element of constant cross-section, resulting in a device where the resistance between the wiper and one end terminal is proportional to the distance between them.

**3) Logarithmic potentiometers:**
A log pot has a resistive element that either 'tapers' in from one end to the other, or is made from a material whose resistivity varies from one end to the other. Most (cheaper) "log" pots are actually not logarithmic, but use two regions of different, but constant, resistivity to approximate a logarithmic law.

**4) High-power types:**
A **rheostat** is essentially a potentiometer, but is usually much larger, designed to handle much higher voltage and current. Sometimes a rheostat is made from resistance wire

wound on a heat resisting cylinder with the slider made from a number of metal fingers that grip lightly onto a small portion of the turns of resistance wire.

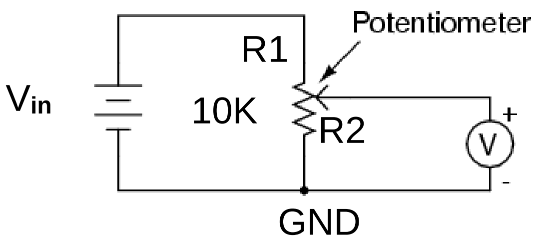
**5) Digital control:**

Digitally controlled potentiometers can be used in analogue signal processing circuits to replace potentiometers.

The same idea can be used to create digital volume controls, attenuators, or other controls under digital control. The DCP should not be confused with the digital to analogue converter (DAC) which actually creates an analogue signal from a digital one.

**PROCEDURE:-**

1. Identify the different terminals of potentiometer like two fixed and one variable.
2. Make the connections on COT-01 board as shown below for voltage divider design.




**$V_{OUT}$ = R2 / (R1+R2) X $V_{in}$**

3. Apply Vin 5V DC as input to potentiometer.
4. Calculate output voltage using voltage divider rule.
5. Observe the $V_{out}$ readings on multi-meter.
6. Verify the calculated and observed value.

**DESIGN:-**

Consider, Pot = 10KΩ

**OBSERVATION TABLE:-**

POT = 10.22KΩ

| Sr. no. | Pot value | Resistor R2 value w.r.t. GND | Calculated value (Vout in V) | Observed Value (Vout in V) |
|---|---|---|---|---|
| 1 | Minimum position | 0.5 Ω | 0.00024V | 0 V |
| 2 | Center position | 5.193 KΩ | 2.54 V | 2.496 V |
| 3 | Maximum position | 10.21KΩ | 4.995 V | 5.042 V |

**CONCLUSION:-**

The calculated and observed values of the inductors combinations are almost equal.

# EXPERIMENT NO. 5

# EXPERIMENT NO.5

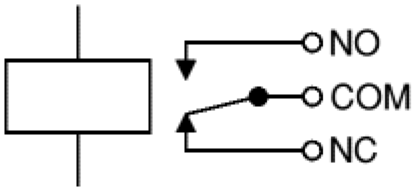
**AIM:-**
TO STUDY THE WORKING OF RELAY

**OBJECTIVE:-**
To study the working and parameters of relay

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter.

**THEORY:-**
A relay is a simple **electromechanical switch** made up of an electromagnet and a set of contacts. A relay is an electrically operated switch. Current flowing through the coil of the relay creates a magnetic field which attracts a lever and changes the switch contacts. The coil current can be on or off so relays have two switch positions and they are double throw (changeover) switches.

Relays allow one circuit to switch a second circuit which can be completely separate from the first. For example a low voltage battery circuit can use a relay to switch a 230V AC mains circuit. There is no electrical connection inside the relay between the two circuits; the link is magnetic and mechanical.
The coil of a relay passes a relatively large current, typically 30mA for a 12V relay, but it can be as much as 100mA for relays designed to operate from lower voltages. Relays are usually SPDT or DPDT but they can have many more sets of switch contacts, for example relays with 4 sets of changeover contacts are readily available.



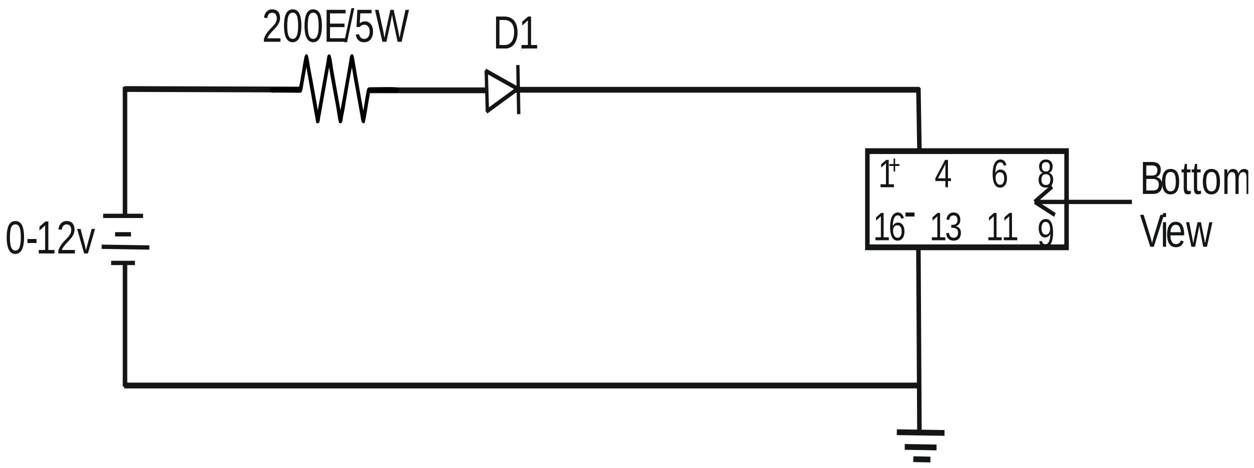
Circuit symbol for a relay

The relay's switch connections are usually labeled COM, NC and NO:
- COM = Common, always connect to this; it is the moving part of the switch.
- NC = Normally Closed, COM is connected to this when the relay coil is off.
- NO = Normally Open, COM is connected to this when the relay coil is on.

Electromagnetic Relay Basic Construction:
An iron core has a wire wound around it. A movable arm is provided and is mounted in such a way that it is in contact with one of the terminals 'NC' to the movable arm a terminal is provided called pole 'P' There are two terminals provided to the coil when the coil is not activated. The arm will be held in the position by the spring. Any circuit connected between pole and NC will be ON and circuit connected between pole and 'NO' will be OFF. When the relay is activated by applying sufficient voltage to the coil and the required current flows through the coil .The iron core gets magnetized and attracts the arm. Due to this the contact 'NC' breaks and contact 'NC' is made. Now the circuit connected between pole and NC switches OFF and the circuit connected between poles and 'NO' becomes ON.

**PROCEDURE:-**

1. Using Multi-meter, determine the terminal connections of the coil, no. of contacts, NO / NC contacts.
2. Measure the coil resistance using ohm-meter. Make a note in observation table.
3. Make the connections on COT-01 board as shown in practical circuit for relay operation study.




4. Apply Vin 12V DC as supply to relay.
5. Now slowly increase the dc voltage observing the ohm-meter. The ohm-meter will read Zero resistance. Observe and record the minimum current requires tripping the relay. This is called Pickup-current of the relay.
6. Slowly reduce the Dc voltage, observing the reading on the ohm-meter. When relay resets, the ohm-meter will read infinite resistance. Observe and record the value of current at which relay is resets. This is called Reset current of the relay.
7. Repeat the 4 and 5 steps to get the average (approximately correct) value.

**DESIGN:-**

Consider, Rin = 200 E/5W and Diode = 1N4148

**OBSERVATION TABLE:-**

| Sr. no. | Descriptions | Observed value |
|---|---|---|
| 1 | Coil Resistance | 698 Ω |
| 2 | Pick-up Current | 9.5 mA @ 9.18 V for 200E |
| 3 | Reset Current | 5.2 mA @ 4.9 V for 200E |
| 4 | Contact resistance | 0.2 Ω |

**CONCLUSION:-**

Parameters of the relay are studied and verify with the data-sheet of the used relay.

# EXPERIMENT NO. 6

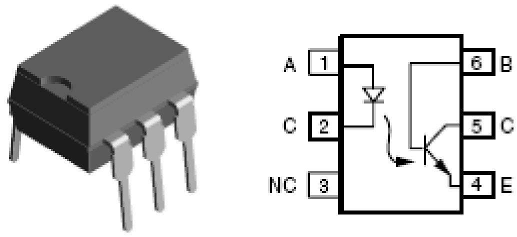
# EXPERIMENT NO.6

**AIM:-**
TO STUDY THE FREQUENCY RESPONSE OF OPTO-COUPLER

**OBJECTIVE:-**
To study the frequency response of opto-coupler

**EQUIPMENTS:**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Function Generator.
5. Oscilloscope.

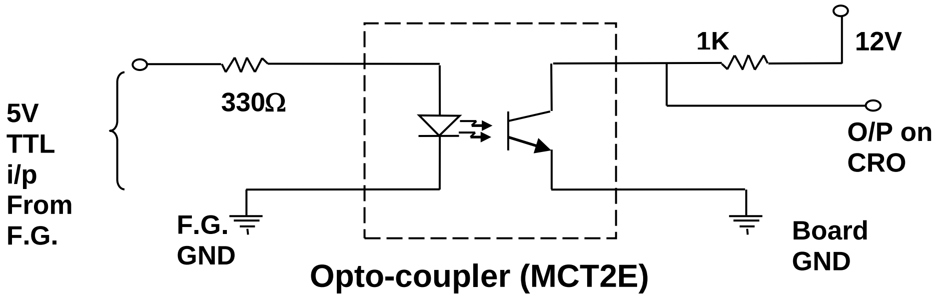
**THEORY:-**
Optical couplers, also referred to as opto-couplers, are well known devices used to direct light from one light source to a light receiving member. Optical couplers are the heart of an optical communication network. Optical fiber technology is used in a variety of applications such as telecommunication, computer, and medical applications.




Optocouplers are used to electrically isolate an input signal from a corresponding output signal. Optocouplers, also referred to as optically coupled isolator devices or optical coupler circuits; provide isolation between different circuit portions which operate at vastly different voltages. The opto-coupler offers an advantage of providing electrical isolation between the two circuits, thus reducing interface problems. Optocouplers have been used for electrical isolation in systems such as computers, power supplies, telecommunications, and controllers.
Optocouplers usually include a light-emitting diode (LED) and a light-responsive transistor (light sensor) such as a phototransistor or a photodiode. Electrical isolation occurs because information is transmitted using light emitted by the LED and received by the light-responsive transistor. When the current driving the LED is changed, the amount of light that is emits also changes proportionally and consequently also the electrical resistance of the photo-resistor.

**PROCEDURE:-**
1) Make the connections as shown in the circuit diagram on COT-01 board.

**Opto-coupler characteristics**




**Opto-coupler (MCT2E)**

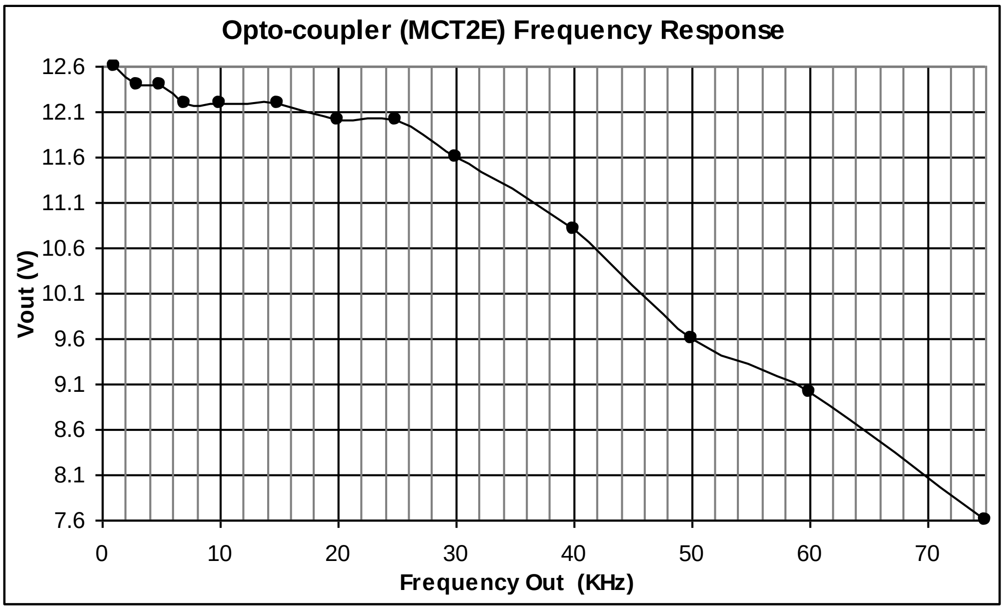
2) Give TTL input from function generator as per given in the observation table.
3) Vary the TTL signal input and note down the readings as per shown in the observation table.
4) Plot the graph for opto-coupler's frequency response.

**DESIGN:-**
Consider, Rin = 330 E, Rout = 1K

**OBSERVATION TABLE:-**

| SR. NO. | FREQUENCY INPUT (KHZ) | FREQUENCY OUT (KHZ) | VOUT (V) | REMARKS |
|---|---|---|---|---|
| 1 | 1 | 1 | 12.6 | - |
| 2 | 3 | 3 | 12.4 | - |
| 3 | 5 | 5 | 12.4 | - |
| 4 | 7 | 7 | 12.2 | - |
| 5 | 10 | 10 | 12.2 | - |
| 6 | 15 | 15 | 12.2 | DUTY CYCLE CHANGE |
| 7 | 20 | 20 | 12.0 | DUTY CYCLE CHANGE |
| 8 | 25 | 25 | 12.0 | DUTY CYCLE CHANGE |
| 9 | 30 | 30 | 11.6 | DUTY CYCLE CHANGE |
| 10 | 40 | 40 | 10.8 | DUTY CYCLE CHANGE (MORE $T_{ON}$ TIME) |
| 11 | 50 | 50 | 9.6 | DUTY CYCLE CHANGE (MORE $T_{ON}$ TIME) |
| 12 | 60 | 60 | 9 | DUTY CYCLE CHANGE (MORE $T_{ON}$ TIME) |
| 13 | 75 | 75 | 7.6 | DUTY CYCLE CHANGE (MORE $T_{ON}$ TIME) |

**Frequency Response Graphs:-**




## CONCLUSION:-

Opto-coupler optically isolates the output signal (circuit) from input signal (circuit).

# EXPERIMENT NO. 7

# EXPERIMENT NO.7

**AIM:-**
TO STUDY THE CHARACTERISTICS OF Si-DIODE

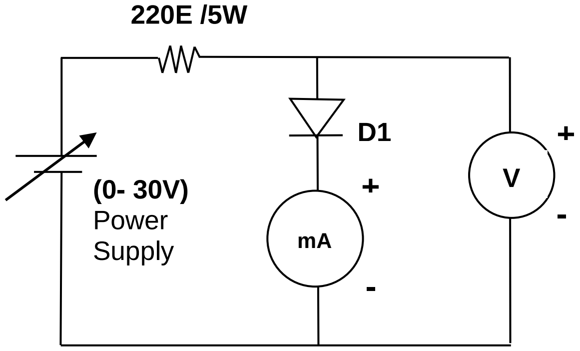
**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF Si-DIODE

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multimeter.

**THEORY:-**
In electronics, a **diode** is a two-terminal electronic component that conducts electric current in only one direction. The term usually refers to a **semiconductor diode**, the most common type today, which is a crystal of semiconductor connected to two electrical terminals, a P-N junction.
The most common function of a diode is to allow an electric current in one direction (called the diode's forward direction) while blocking current in the opposite direction (the reverse direction). This unidirectional behavior is called rectification, and is used to convert alternating current to direct current, and remove modulation from radio signals in radio receivers.
A Semiconductor Diode is a P-N junction Diode. It conducts only in one direction. It is a unidirectional device.
**Forward bias:** When Anode (P-side) of the Diode is connected to battery positive terminal and cathode (N-side) is connected to negative terminal the Diode is said to be forward biased. The forward resistance of the Diode is small.
**Reverse bias**: When anode of the Diode is connected to battery negative terminal and cathode is connected to positive terminal of the Diode is said to be reverse biased. The reverse resistance of a Diode is very high.

**PROCEDURE:-**
1. Make the connections as shown in following circuit diagram on COT-01 board.

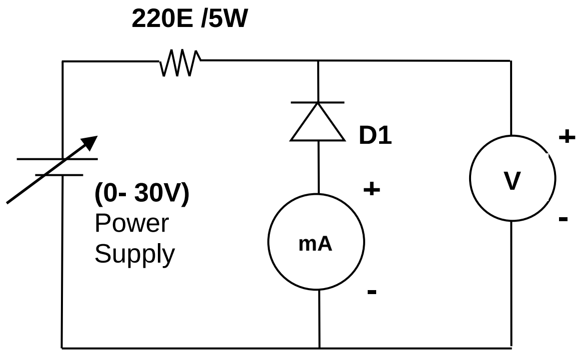
**Forward Bias Characteristics**




2. Apply dc voltage to forward bias the diode and vary the voltages as per the observation table.

3. Note down the corresponding forward voltage $V_F$, across the diode and the forward current, $I_F$.
4. Similarly, make the connections as shown in following circuit diagram on COT-01 board.

**Reverse Bias Characteristics**



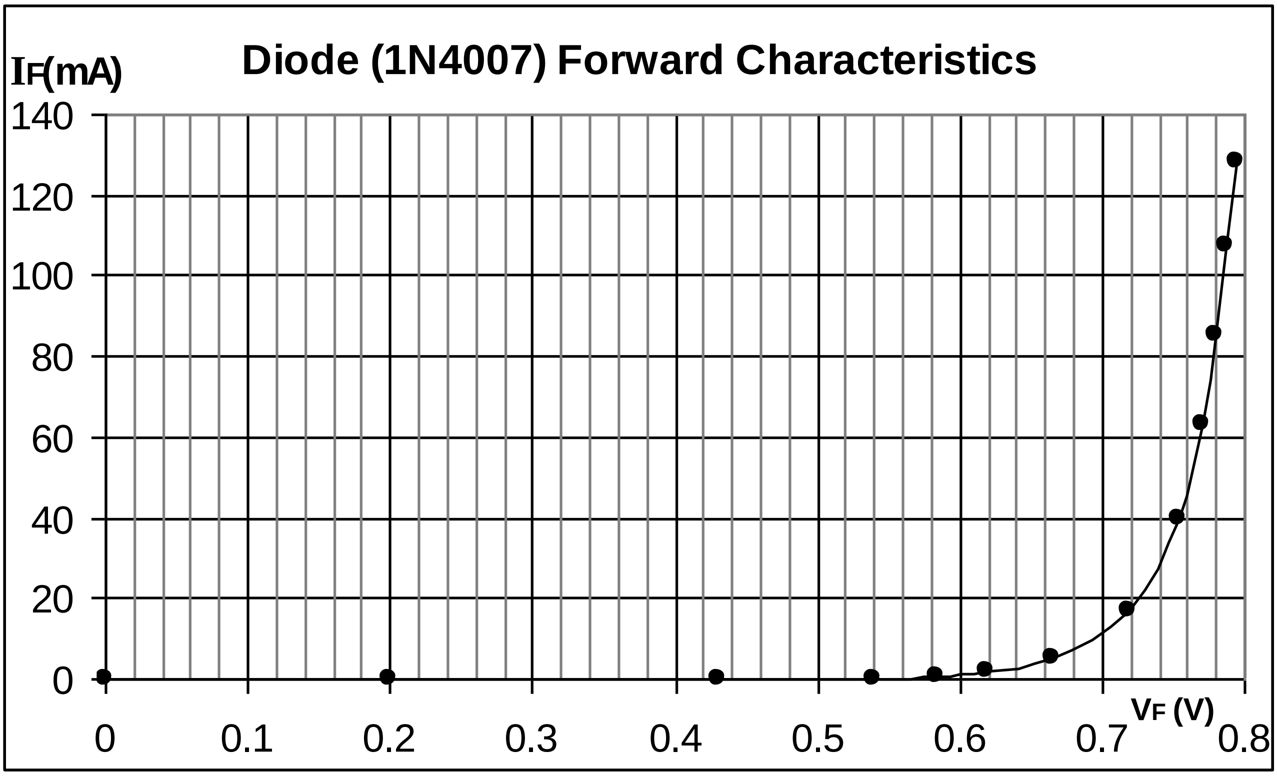

5. Apply dc voltage to reverse bias the diode and vary the voltages as per the observation table.
6. Note down the corresponding reverse bias voltage $V_R$, across the diode and the reverse bias current, $I_R$.
7. Plot the forward and reverse biased voltage Vs current graph.

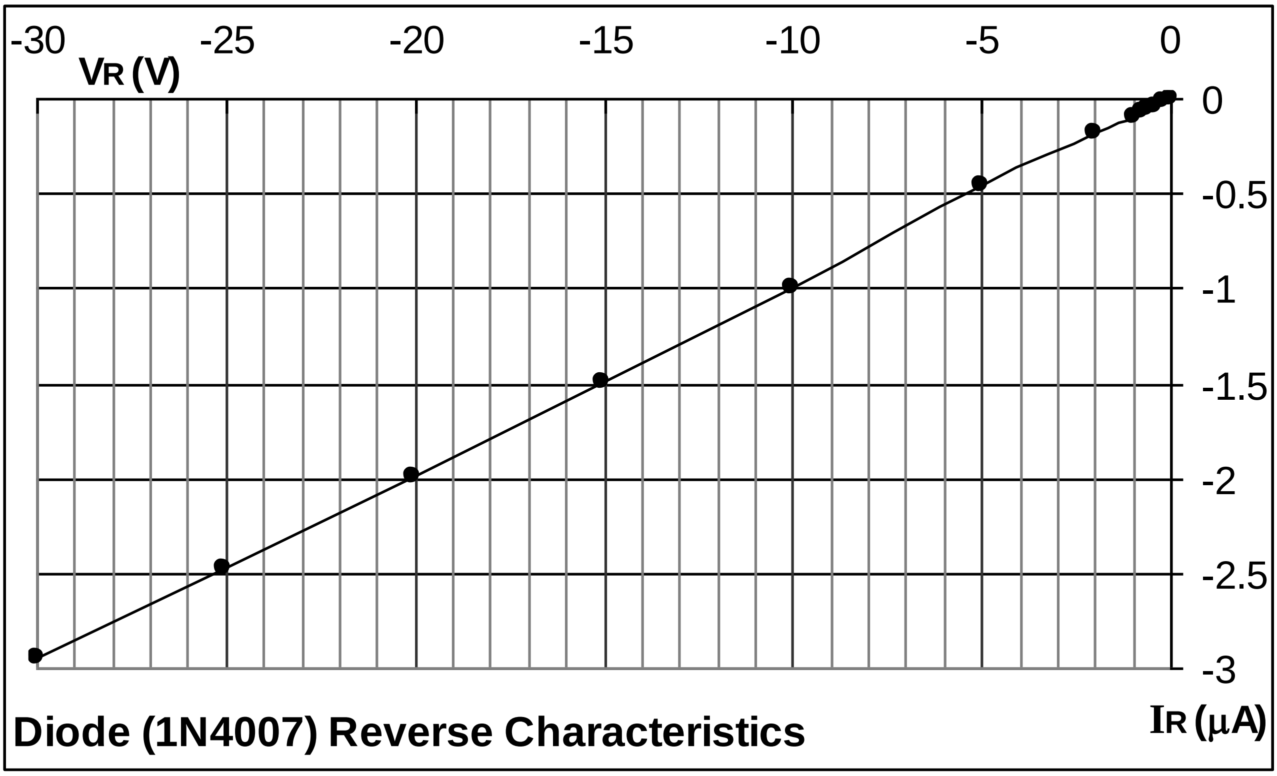
**DESIGN:-**
Diode = 1N4007
Resistance = 220E / 5W

**OBSERVATION TABLE:-**

| Sr. no. | Dc Voltage Source | Diode Forward Characteristics | | Diode Reverse Characteristics | |
|---|---|---|---|---|---|
| | Vin | $V_F$(V) | $I_F$(mA) | $V_R$(V) | $I_R$(μA) |
| 1. | 0 | 0 | 0 | 0 | 0 |
| 2 | 0.2 | 0.2 | 0 | 0.2 | -0.02 |
| 3 | 0.4 | 0.43 | 0.02 | 0.4 | -0.04 |
| 4 | 0.6 | 0.539 | 0.304 | 0.6 | -0.06 |
| 5 | 0.8 | 0.584 | 0.812 | 0.8 | -0.07 |
| 6 | 1 | 0.619 | 1.748 | 1 | -0.09 |
| 7 | 2 | 0.666 | 5 | 2 | -0.18 |
| 8 | 5 | 0.718 | 17.2 | 5 | -0.45 |
| 9 | 10 | 0.753 | 39.7 | 10.01 | -0.99 |
| 10 | 15 | 0.770 | 62.9 | 15.04 | -1.49 |
| 11 | 20 | 0.779 | 85.5 | 20.05 | -1.99 |
| 12 | 25 | 0.787 | 107.3 | 25.03 | -2.47 |
| 13 | 30 | 0.794 | 128.1 | 30.01 | -2.95 |

**Voltage Vs Current graphs:-**



**Calculations from Graph:**

1) Static forward Resistance Rdc = $V_f$ / $I_f$ Ω
2) Dynamic forward Resistance rac = $\Delta V_f$ / $\Delta I_f$ Ω
3) Static Reverse Resistance Rdc =$V_r$ / $I_r$ Ω
4) Dynamic Reverse Resistance rac = $\Delta V_r$ / $\Delta I_r$ Ω

**CONCLUSION:-**

In forward bias after $V_F$=0.6 as voltage increases large current starts flowing through the diode and small amount of current flows when diode is in reverse bias. Thus the VI characteristic of PN junction diode is verified.

1. Cut in voltage = ……… V
2. Static forward resistance = ………. Ω
3. Dynamic forward resistance = ………. Ω

# EXPERIMENT
# NO. 8

# EXPERIMENT NO.8

**AIM:-**
**TO STUDY THE CHARACTERISTICS OF FSD- DIODE**

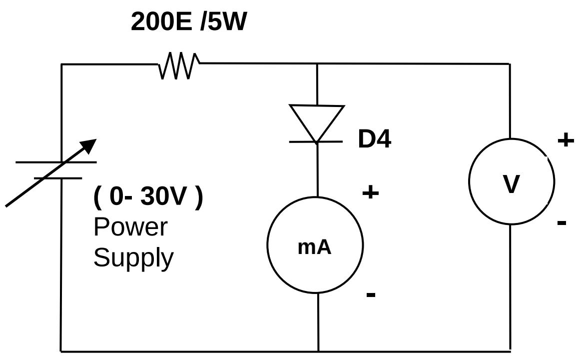
**OBJECTIVE:**
TO STUDY THE CHARACTERISTICS OF FSD-DIODE

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Function Generator.
5. Oscilloscope.
6. Multimeter.

**THEORY:-**
Fast Switching Diode (FSD), its voltage-current characteristic similar to normal silicon diode but its switching response is very fast as compare to normal silicon diode.
**Forward bias:** When Anode (P-side) of the Diode is connected to battery positive terminal and cathode (N-side) is connected to negative terminal the Diode is said to be forward biased. The forward resistance of the Diode is small.
**Reverse bias**: When anode of the Diode is connected to battery negative terminal and cathode is connected to positive terminal of the Diode is said to be reverse biased. The reverse resistance of a Diode is very high.

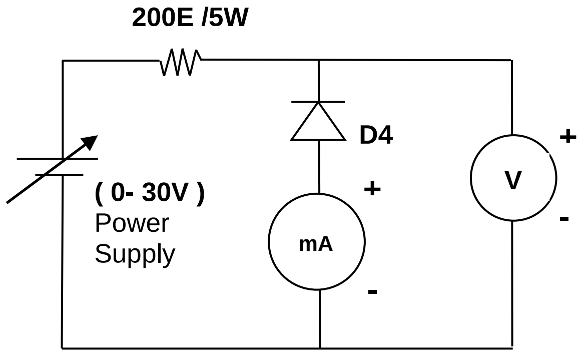
**PROCEDURE:-**
1. Make the connections as shown in following circuit diagram on BEL-COT board.

**Forward Bias Characteristics**



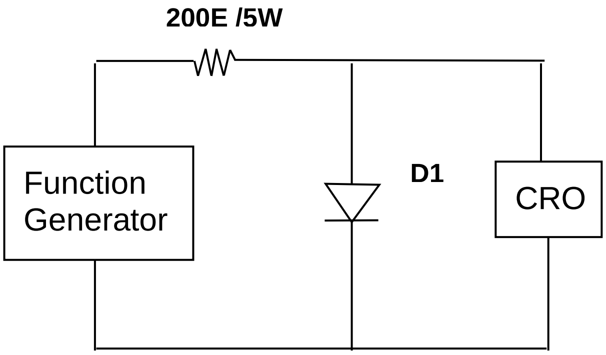

2. Apply dc voltage to forward bias the diode and vary the voltages as per the observation table.
3. Note down the corresponding forward voltage $V_F$, across the diode and the forward current, $I_F$.
4. Similarly, make the connections as shown in following circuit diagram on COT-01 board.

## Reverse Bias Characteristics




5. Apply dc voltage to reverse bias the diode and vary the voltages as per the observation table.
6. Note down the corresponding reverse bias voltage $V_R$, across the diode and the reverse bias current, $I_R$.
7. Plot the forward and reverse biased voltage Vs current graph.
8. Make the connections as shown in following circuit diagram on BEL-COT board.

## Frequency response diagram



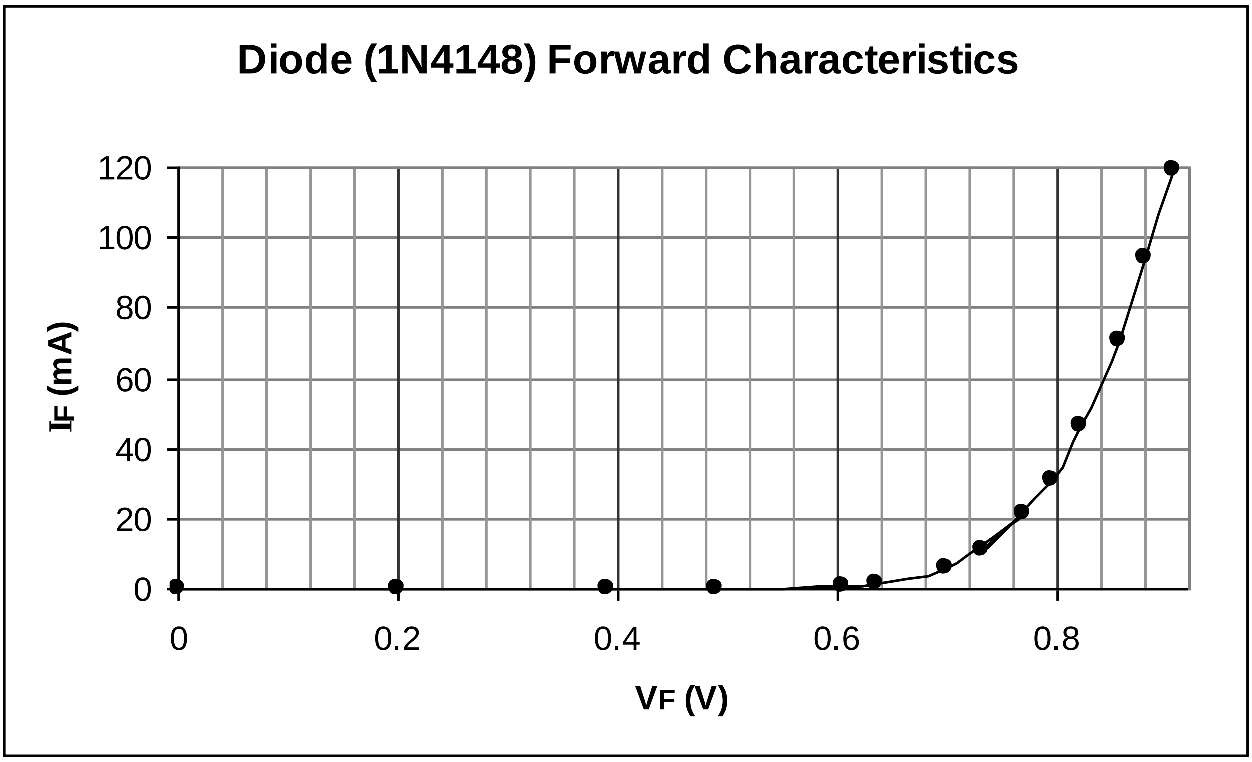

9. Now apply the TTL signal across the diode and note down its response as per the observation table.
10. Replace the 1N4148 with 1N4007. Repeat the procedure no.9.
11. Plot the frequency response for the diode.

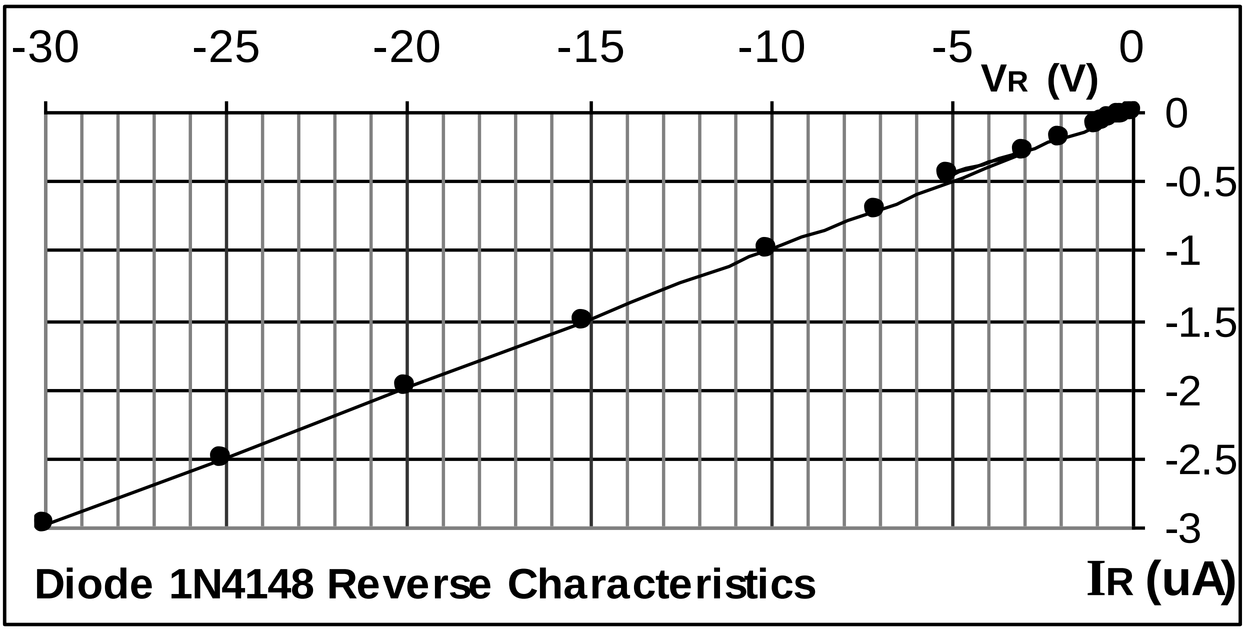
**DESIGN:-**
DIODE=1N4148
RESISTANCE = 200E / 5W

**OBSERVATION TABLE:-**

| Sr. no. | Dc Voltage Source | Diode Forward Characteristics | | Diode Reverse Characteristics | |
|---|---|---|---|---|---|
| | Vin | $V_F$(V) | $I_F$ | $V_R$(V) | $I_R$(μA) |
| 1. | 0 | 0 | 0 | 0 | 0 |
| 2 | 0.2 | 0.2 | 0.26 μA | 0.239 | 0.02 |
| 3 | 0.4 | 0.39 | 11.13 μA | 0.401 | 0.03 |
| 4 | 0.6 | 0.49 | 90. 2 μA | 0.599 | 0.05 |
| 5 | 0.8 | 0.605 | 0.948 mA | 0.802 | 0.07 |
| 6 | 1.0 | 0.636 | 1.771 mA | 1.001 | 0.09 |
| 7 | 2.0 | 0.699 | 6.2 mA | 1.999 | 0.18 |
| 8 | 3.0 | 0.731 | 11.31 mA | 3.022 | 0.28 |
| 9 | 5.0 | 0.770 | 21.24 mA | 5.102 | 0.46 |
| 10 | 7.0 | 0.795 | 30.98 mA | 7.05 | 0.70. |
| 11 | 10. | 0.822 | 46.14 mA | 10.03 | 1.00 |
| 12 | 15. | 0.857 | 71.03 mA | 15.16 | 1.50 |
| 13 | 20 | 0.881 | 94.07 mA | 20.05 | 1.99 |
| 14 | 25 | 0.905 | 119.15 mA | 25.12 | 2.50 |
| 15 | 30 | 0.924 | 142.12 mA | 29.96 | 2.97 |

**Voltage Vs Current graphs:-**

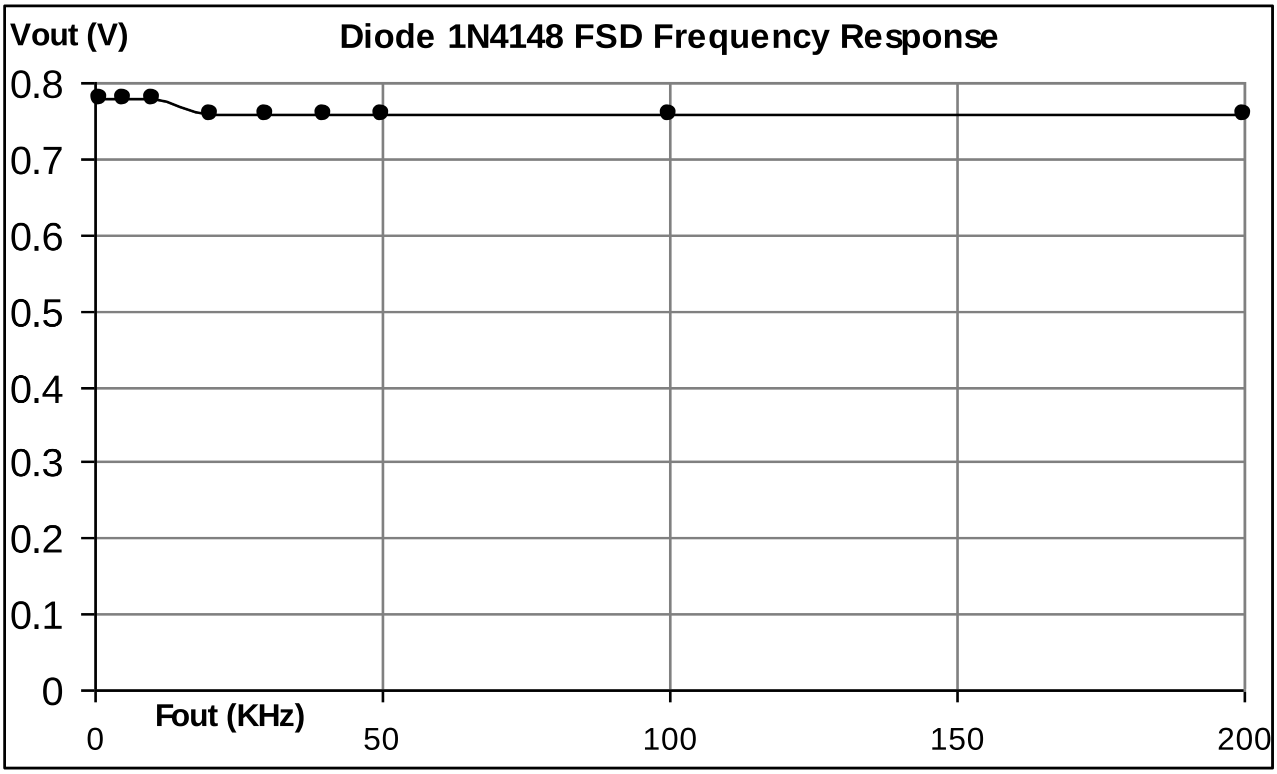





**Frequency response of 1n4148 and 1n4007 diode:-**

| Sr. no. | Frequency input (TTL signal) | 1N4148 diode | | 1N4007 diode | |
|---|---|---|---|---|---|
| | Fin (KHz) | Fout (KHz) | Vout (V) | Fout (KHz) | Vout (V) |
| 1. | 1 | 1 | 0.78 | 1 | 0.72 |
| 2 | 5 | 5 | 0.78 | 5 | 0.72 (Duty cycle change) |
| 3 | 10 | 10 | 0.78 | 10 | More Duty cycle change |
| 4 | 20 | 20 | 0.76 | 20 | Distorted |
| 5 | 30 | 30 | 0.76 | 30 | Signal lost |
| 6 | 40 | 40 | 0.76 | | |
| 7 | 50 | 50 | 0.76 | | |
| 8 | 100 | 100 | 0.76 | | |
| 9 | 200 | 200 | 0.76 | | |
| 10 | 300 | 300 | 0.76 (Distortion @ failing edge) | | |
| 11 | 400 | 400 | 0.76 (Distortion @ failing edge) | | |
| 12 | 500 | 500 | 0.76 (Distortion @ failing edge) | | |
| 13 | 1000 | 1000 | 0.76 (Distortion @ failing edge = not 50% duty cycle) | | |

**Frequency response Graph of 1N4148:-**




**Calculations from Graph:**
1) Static forward Resistance Rdc = Vf/If Ω
2) Dynamic forward Resistance rac = ΔVf/ΔIf Ω
3) Static Reverse Resistance Rdc =Vr/Ir Ω
4) Dynamic Reverse Resistance rac = ΔVr/ΔIr Ω

**<u>CONCLUSION</u>:-**
In forward bias after $V_F$=0.7 as voltage increases large current starts flowing through the diode and small amount of current flows when diode is in reverse bias. Thus the VI characteristic of PN junction diode is verified.
1. Cut in voltage = ……… V
2. Static forward resistance = ………. Ω
3. Dynamic forward resistance = ………. Ω
Frequency response of 1n4148 diode is very fast as compare to normal 1n4007 diode.

# EXPERIMENT NO. 9

# EXPERIMENT NO.9

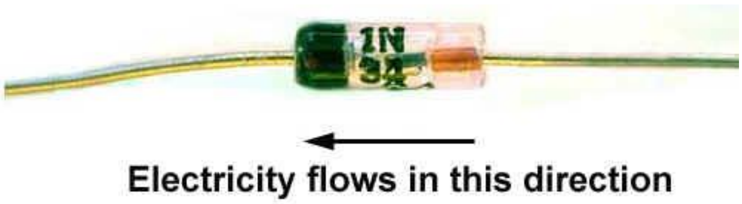
**AIM:-**
TO STUDY THE CHARACTERISTICS OF Ge-DIODE

**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF Ge-Diode

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multimeter.

**THEORY:-**
Diodes are a very important component of most alternative energy generating systems. Most diodes are made of silicon because of its ease of processing and stability, however they have one disadvantage: a silicon diode has a forward voltage drop of around 0.7 volts. This means that if you had a 7.0 Volt rated solar panel charging a battery via a silicon diode, only 7 - 0.7 = 6.3 Volts would be seen by the battery - the remainder is lost as heat in the diode.
Germanium diodes find some use since Ge has much smaller band gap energy than Si, producing lower forward voltages. However, this smaller band gap also makes Ge less useful at higher temperatures due to a higher leakage current.

**Germanium Diodes**

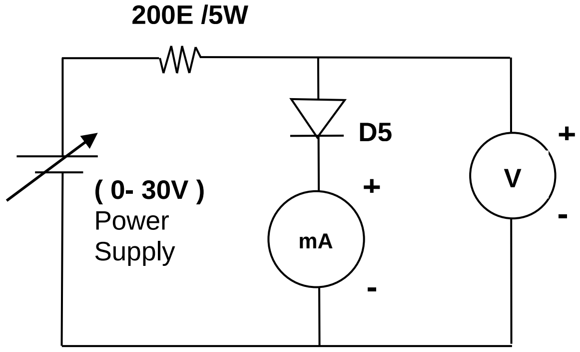



A Germanium Diode (such as the 1N34 pictured above) will typically have a forward voltage drop of just 0.3 volts which means they are much more efficient. Older germanium diodes had a larger leakage of current at a reverse voltage, but now American Micro semiconductor and others supply a range of improved low current leakage germanium diodes
**Forward bias:** When Anode (P-side) of the Diode is connected to battery positive terminal and cathode (N-side) is connected to negative terminal the Diode is said to be forward biased. The forward resistance of the Diode is small.
**Reverse bias**: When anode of the Diode is connected to battery negative terminal and cathode is connected to positive terminal of the Diode is said to be reverse biased. The reverse resistance of a Diode is very high.

**PROCEDURE:-**

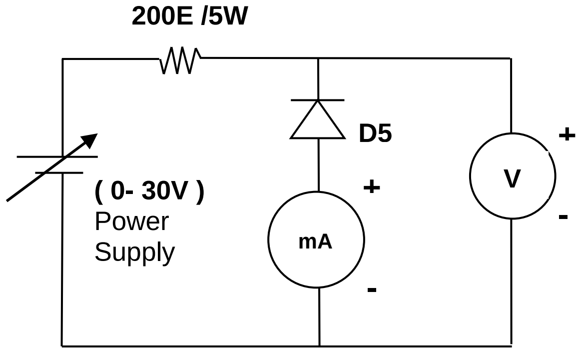
1. Make the connections as shown in following circuit diagram on BEL-COT board.

**Forward Bias Characteristics**




2. Apply dc voltage to forward bias the diode and vary the voltages as per the observation table.
3. Note down the corresponding forward voltage $V_F$, across the diode and the forward current, $I_F$.
4. Similarly, make the connections as shown in following circuit diagram on BEL-COT board.

**Reverse Bias Characteristics**




5. Apply dc voltage to reverse bias the diode and vary the voltages as per the observation table.
6. Note down the corresponding reverse bias voltage $V_R$, across the diode and the reverse bias current, $I_R$.
7. Plot the forward and reverse biased voltage Vs current graph.

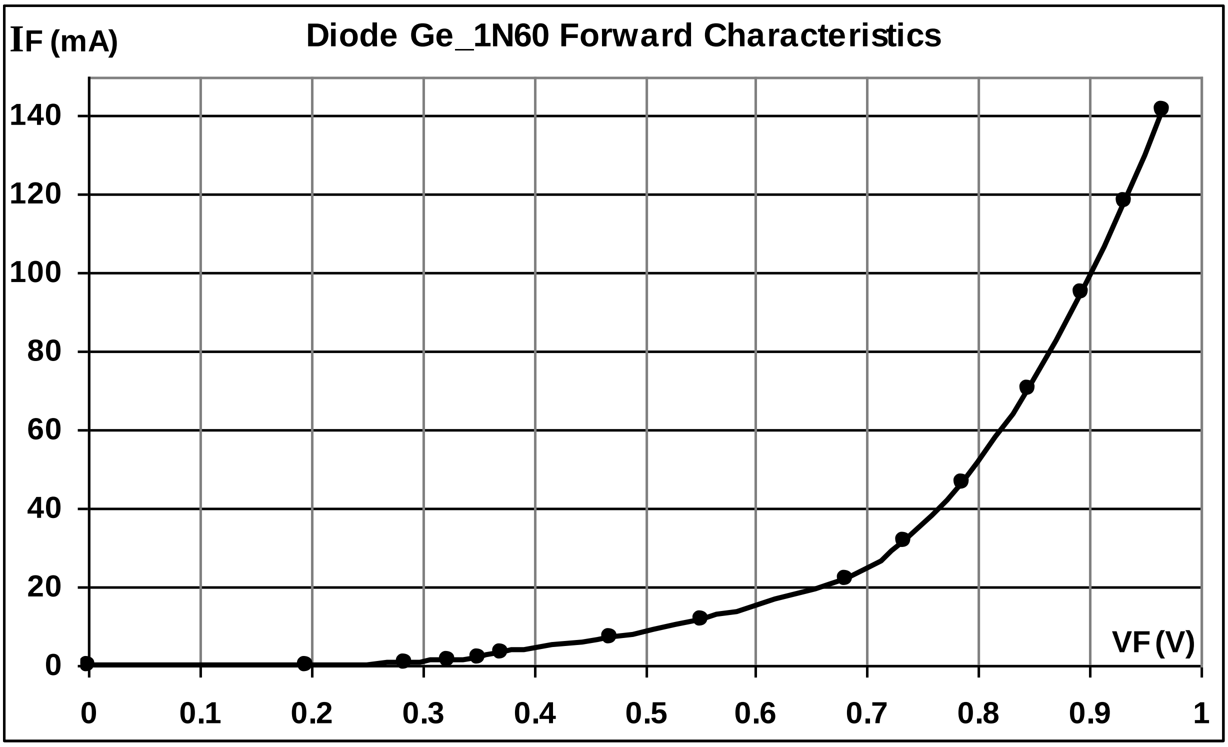
**DESIGN:-**

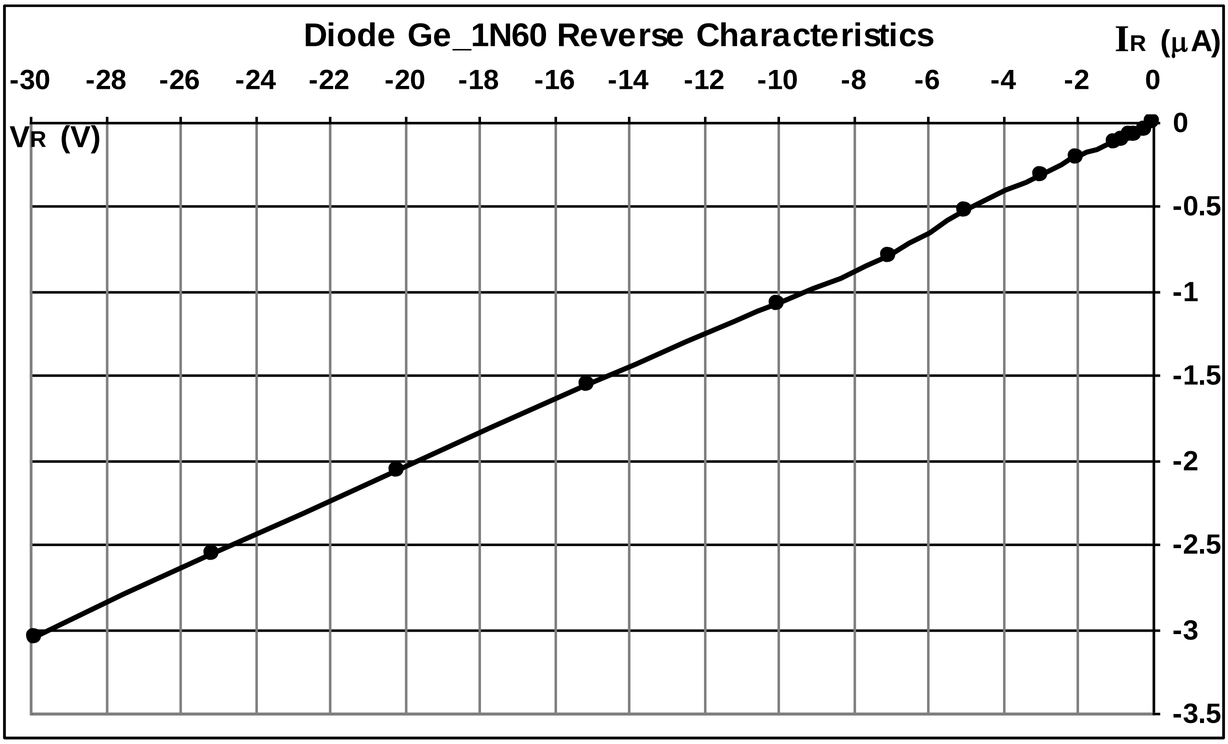
Diode =Ge_1N60
Resistance = 200E/5W

## OBSERVATION TABLE:-

| Sr. no. | Dc Voltage Source | Diode Forward Characteristics | | Diode Reverse Characteristics | |
|---|---|---|---|---|---|
| | Vin | $V_F$(V) | $I_F$(mA) | $V_R$(V) | $I_R$(μA) |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 2 | 0.2. | 0.195 | 0.027. | 0.208 | 0.04 |
| 3 | 0.4 | 0.285 | 0.551 | 0.499 | 0.08 |
| 4 | 0.6 | 0.324 | 1.361 | 0.609 | 0.08 |
| 5 | 0.8 | 0.350 | 2.167 | 0.810 | 0.10 |
| 6 | 1.0 | 0.372 | 3.020 | 1.024 | 0.12 |
| 7 | 2.0 | 0.469 | 7.365 | 2.004 | 0.21 |
| 8 | 3.0 | 0.551 | 11.652 | 2.993 | 0.32 |
| 9 | 5.0 | 0.681 | 21.82 | 5.004 | 0.52 |
| 10 | 7.0 | 0.734 | 31.58 | 7.06 | 0.79 |
| 11 | 10.0 | 0.785 | 46.29. | 10.02 | 1.07 |
| 12 | 15.0 | 0.845 | 70.47 | 15.10 | 1.56 |
| 13 | 20.0 | 0.894 | 94.76 | 20.20 | 2.07 |
| 14 | 25.0 | 0.931 | 118.64 | 25.10. | 2.56 |
| 15 | 30.0 | 0.965 | 141.39 | 29.89 | 3.05 |

### Voltage Vs Current graphs:-






**Calculations from Graph:**

1) Static forward Resistance Rdc = Vf/If Ω
2) Dynamic forward Resistance rac = ΔVf/ΔIf Ω
3) Static Reverse Resistance Rdc =Vr/Ir Ω
4) Dynamic Reverse Resistance rac = ΔVr/ΔIr Ω

**CONCLUSION:-**

In forward bias after $V_F$=0.6 as voltage increases large current starts flowing through the diode and small amount of current flows when diode is in reverse bias. Thus the VI characteristic of PN junction diode is verified.

1. Cut in voltage = ……… V
2. Static forward resistance = ………. Ω
3. Dynamic forward resistance = ………. Ω

# EXPERIMENT NO. 10

# EXPERIMENT NO.10

**AIM:-**
TO STUDY THE CHARACTERISTICS OF ZENER DIODE

**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF Zener-Diode

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter

**THEORY:-**
A Zener diode is a type of diode that permits current not only in the forward direction like a normal diode, but also in the reverse direction if the voltage is larger than the breakdown voltage known as "Zener knee voltage" or "Zener voltage". The device was named after Clarence Zener, who discovered this electrical property.

A conventional solid-state diode will not allow significant current if it is reverse-biased below its reverse breakdown voltage. When the reverse bias breakdown voltage is exceeded, a conventional diode is subject to high current due to avalanche breakdown. Unless this current is limited by external circuitry, the diode will be permanently damaged. In case of large forward bias (current in the direction of the arrow), the diode exhibits a voltage drop due to its junction built-in voltage and internal resistance. The amount of the voltage drop depends on the semiconductor material and the doping concentrations.

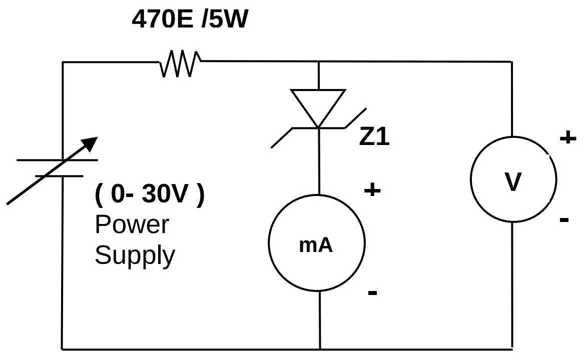
A Zener diode exhibits almost the same properties, except the device is specially designed so as to have a greatly reduced breakdown voltage, the so-called Zener voltage. A reverse-biased Zener diode will exhibit a controlled breakdown and allow the current to keep the voltage across the Zener diode at the Zener voltage.A heavily doped P-N junction Diode which has sharp break down voltage is called Zener Diode. It is normally operated in reverse bias.

**Forward Bias**: When Anode of the Diode is connected to battery positive terminal and cathode is connected to negative terminal the Diode is said to be forward biased.
**Reverse Bias:** When Anode of the Diode is connected to battery negative terminal and cathode of the Diode is connected to the Battery positive terminal the Diode is said to be reverse biased.
**Break down Voltage**: At some reverse voltage the voltage across the Zener Diode remains constant and current through it increases sharply. This voltage is known as Zener break down voltage ($V_Z$).

**PROCEDURE:-**

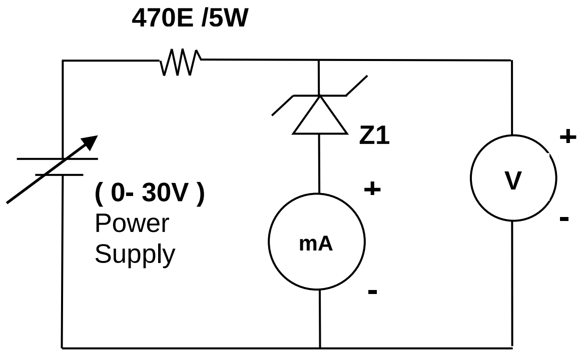
1. Make the connections as shown in following circuit diagram on BEL-COT board.

**Forward Bias Characteristics**




2. Apply dc voltage to forward bias the diode and vary the voltages as per the observation table.
3. Note down the corresponding forward voltage $V_F$, across the diode and the forward current, $I_F$.
4. Similarly, make the connections as shown in following circuit diagram on BEL-COT board.

**Reverse Bias Characteristics**




5. Apply dc voltage to reverse bias the diode and vary the voltages as per the observation table.
6. Note down the corresponding reverse bias voltage $V_R$, across the diode and the reverse bias current, $I_R$.
7. Plot the forward and reverse biased voltage Vs current graph.

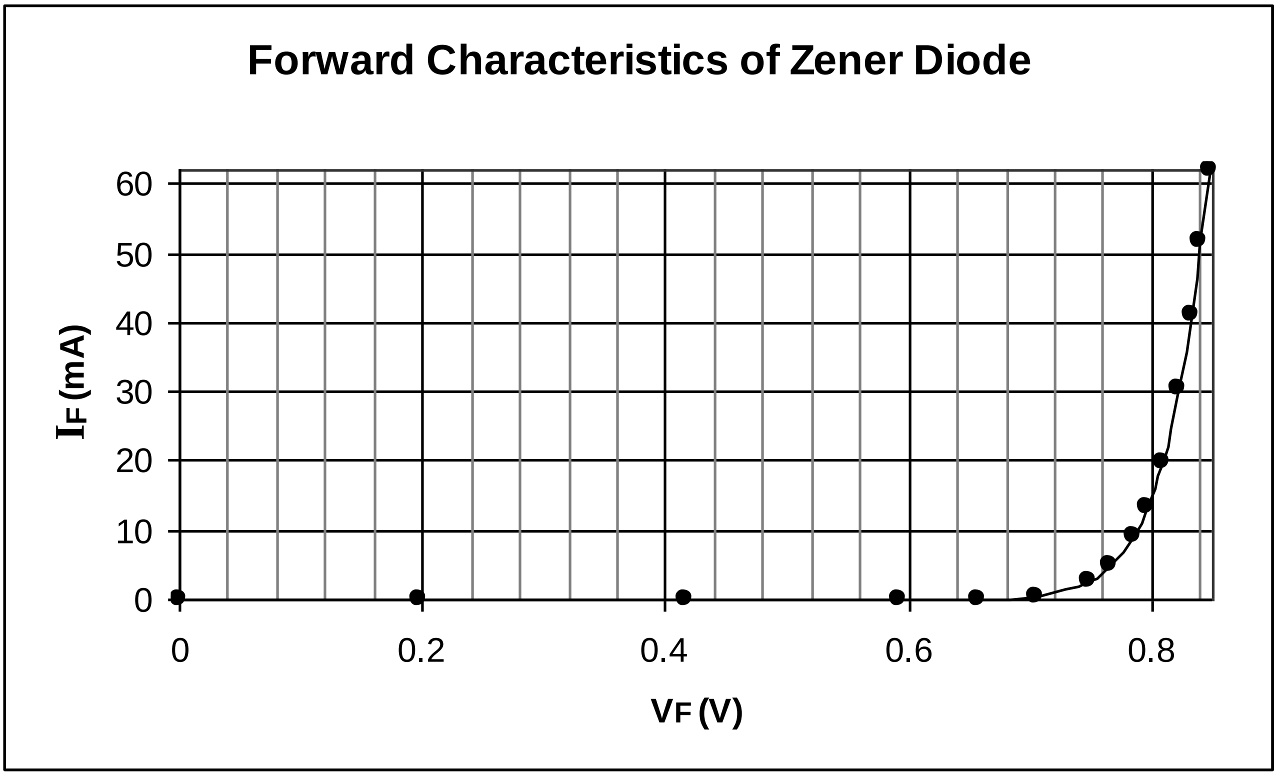
**DESIGN:-**

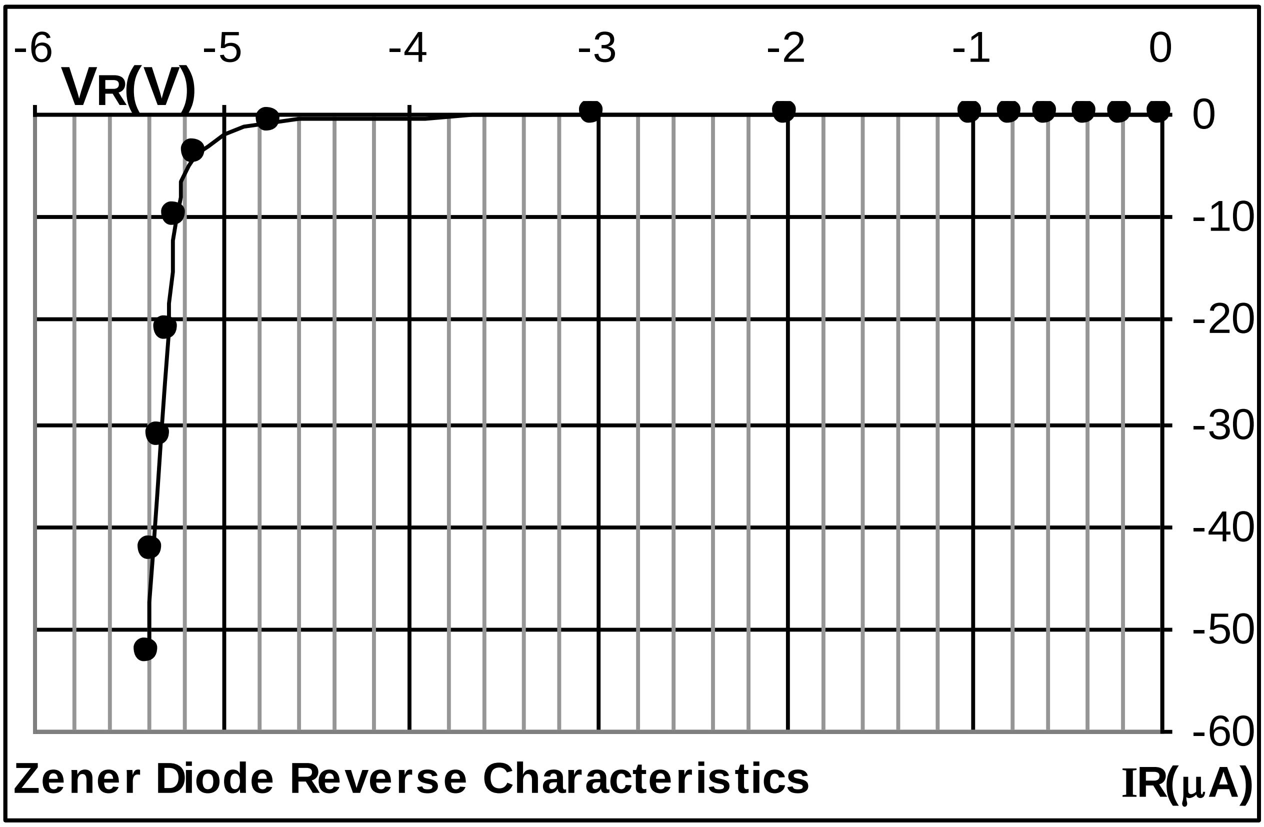
Diode= zener 5.1V
Resistance=470E/5W

## OBSERVATION TABLE:-

| Sr. no. | Dc Voltage Source | Diode Forward Characteristics | | Diode Reverse Characteristics | |
|---|---|---|---|---|---|
| | Vin | $V_F$(V) | $I_F$ | $V_R$(V) | $I_R$ |
| 1. | 0 | 0 | 0 | 0 | 0 |
| 2 | 0.2 | 0.198 | 0.02 μA | 0.21 | 0.02 μA |
| 3 | 0.4 | 0.417 | 0.05 μA | 0.397 | 0.04 μA |
| 4 | 0.6 | 0.591 | 10.70 μA | 0.609 | 0.05 μA |
| 5 | 0.8 | 0.656 | 100 μA | 0.795 | 0.07 μA |
| 6 | 1.0 | 0.704 | 0.56 mA | 1.007 | 0.09 μA |
| 7 | 2.0 | 0.748 | 2.646 mA | 1.993 | 0.22 μA |
| 8 | 3.0 | 0.766 | 4.81 mA | 3.020 | 3.60 μA |
| 9 | 5.0 | 0.785 | 9.02 mA | 4.750 | 0.638 mA |
| 10 | 7.0 | 0.796 | 13.29 mA | 5.136 | 3.982 mA |
| 11 | 10.0 | 0.809 | 19.87 mA | 5.237 | 10.036 mA |
| 12 | 15,0 | 0.822 | 30.48 mA | 5.297 | 20.83 mA |
| 13 | 20.0 | 0.833 | 41.18 mA | 5.336 | 31.42 mA |
| 14 | 25.0 | 0.840 | 51.92 mA | 5.373 | 42.44 mA |
| 15 | 30.0 | 0.847 | 62.05 mA | 5.399 | 52.31 mA |

**Voltage Vs Current graphs:-**






**Calculations from Graph:**
Cut in voltage = ---------- (v)
Break down voltage = ------------(v)

**<u>CONCLUSION</u>:-**
The forward and reverse bias characteristics of a Zener Diode are obtained and the graph is drawn. The Zener break down voltage is found to be VZ= 5.1V. The zener diode characteristics have been plotted.
1. Cut in voltage = ……… V
2 Break down voltage = ------------(v)

# EXPERIMENT NO. 11

# EXPERIMENT NO.11

**AIM:-**
TO STUDY THE CHARACTERISTICS OF LED

**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF LED

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter.

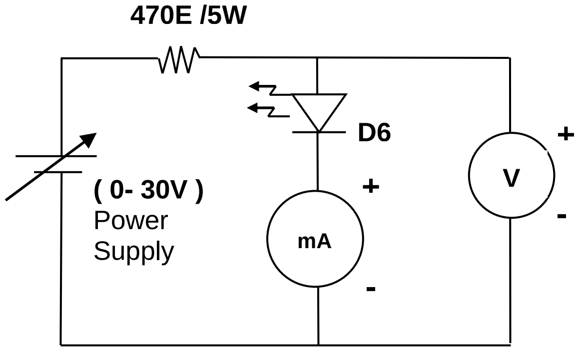
**THEORY:-**
A light-emitting diode (LED) is a semiconductor light source. LEDs are used as indicator lamps in many devices, and are increasingly used for lighting. LEDs emitted low-intensity red light, but modern versions are available across the visible, ultraviolet and infrared wavelengths, with very high brightness.

The LED is based on the semiconductor diode. When a diode is forward biased (switched on), electrons are able to recombine with holes within the device, releasing energy in the form of photons. This effect is called electroluminescence and the color of the light (corresponding to the energy of the photon) is determined by the energy gap of the semiconductor. An LED is usually small in area (less than 1 $mm^2$), and integrated optical components are used to shape its radiation pattern and assist in reflection. LEDs present many advantages over incandescent light sources including lower energy consumption, longer lifetime, improved robustness, smaller size, faster switching, and greater durability and reliability.

**Forward bias:** When Anode (P-side) of the LED is connected to battery positive terminal and cathode (N-side) is connected to negative terminal the LED is said to be forward biased.
**Reverse bias**: When anode of the LED is connected to battery negative terminal and cathode is connected to positive terminal of the LED is said to be reverse biased.

**PROCEDURE:-**
1. Make the connections as shown in following circuit diagram on BEL-COT board.

**Forward Bias Characteristics**

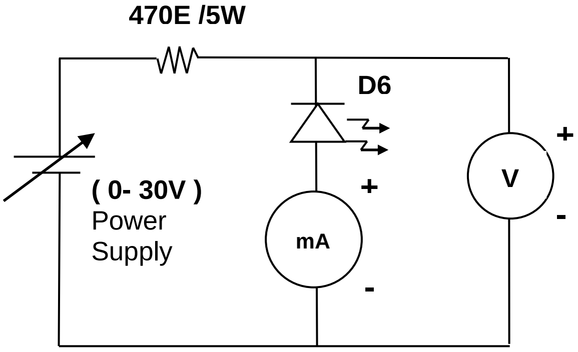



2. Apply dc voltage to forward bias the LED and vary the voltages as per the

observation table.

3. Note down the corresponding forward voltage $V_F$, across the LED and the forward current, $I_F$.
4. Similarly, make the connections as shown in following circuit diagram on COT-01 board.

**Reverse Bias Characteristics**




5. Apply dc voltage to reverse bias the LED and vary the voltages as per the observation table.
6. Note down the corresponding reverse bias voltage $V_R$, across the LED and the reverse bias current, $I_R$.
7. Plot the forward and reverse biased voltage Vs current graph.

**DESIGN:-**

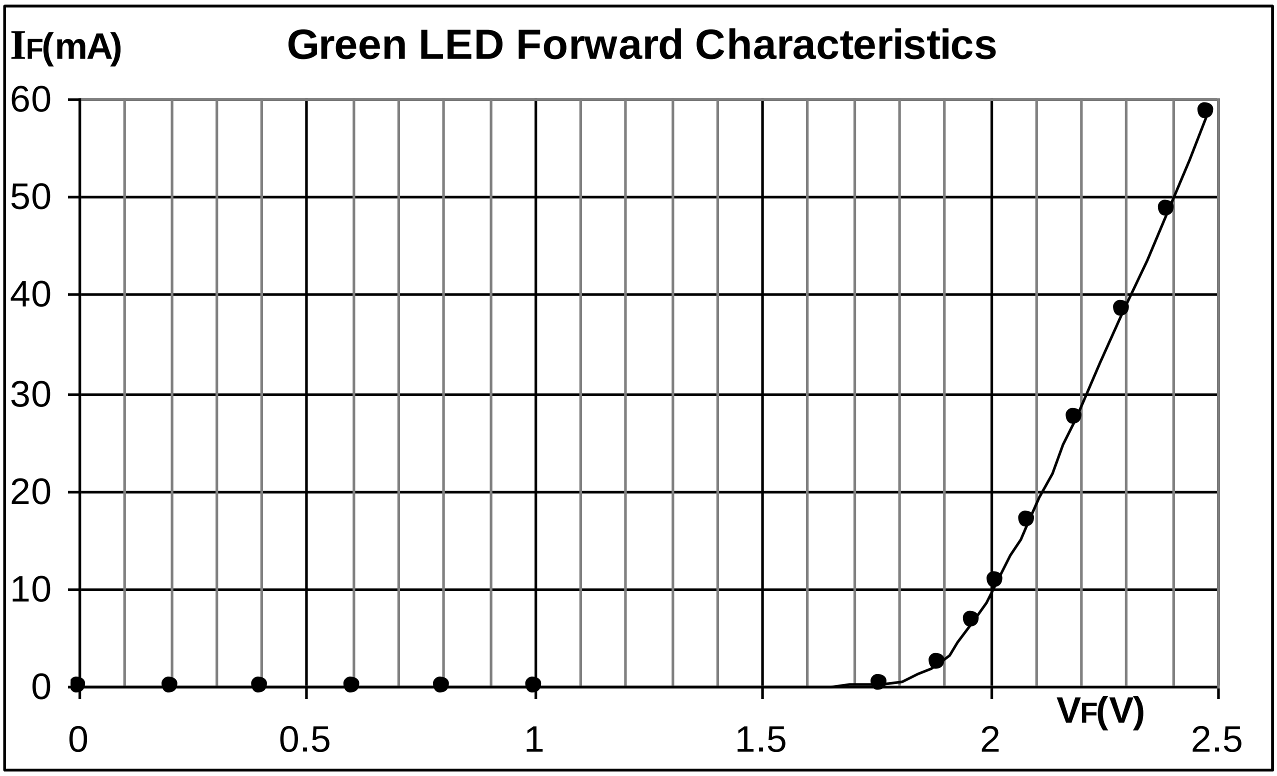
Diode = Green LED 5mm

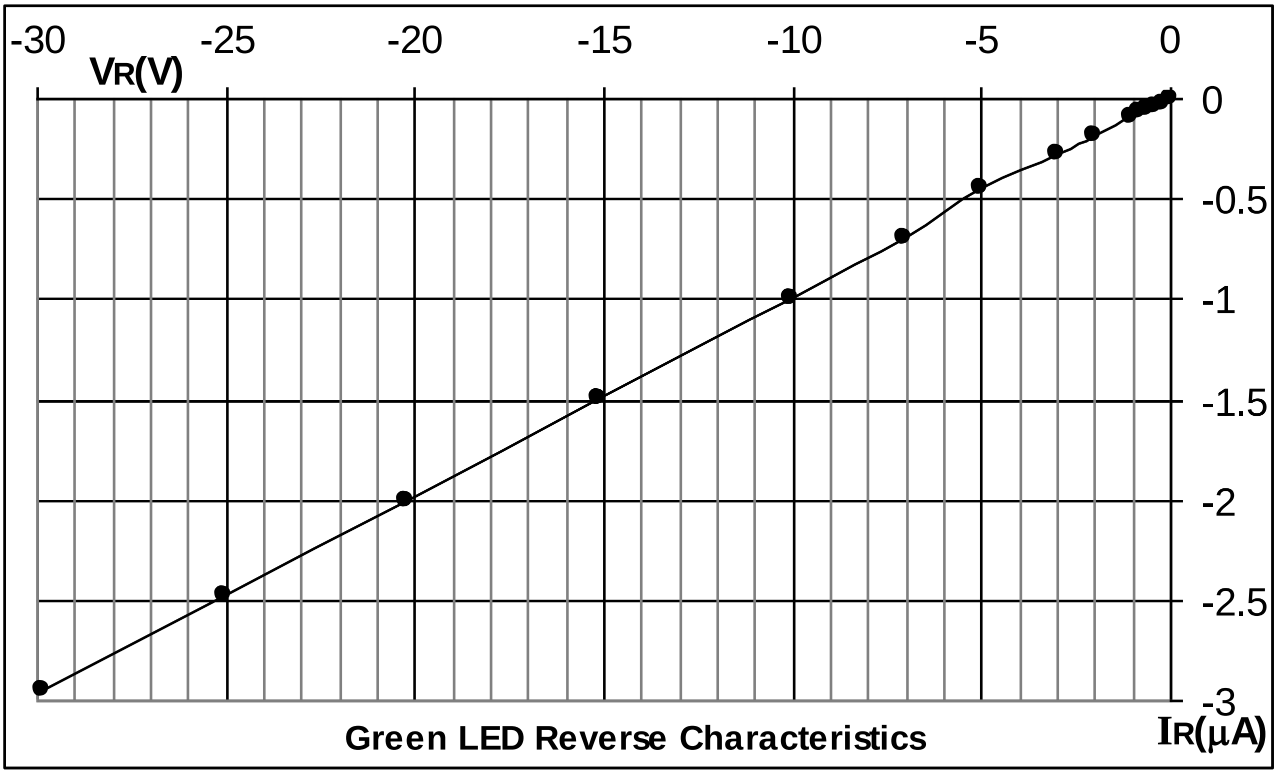
Resistance = 470E/5W

**OBSERVATION TABLE:-**

| Sr. no. | Dc Voltage Source | Diode Forward Characteristics | | Diode Reverse Characteristics | |
|---|---|---|---|---|---|
| | Vin | $V_F$(V) | $I_F$ | $V_R$(V) | $I_R$(μA) |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 2 | 0.2 | 0.2 | 0.02 μA | 0.198 | 0.02 |
| 3 | 0.4 | 0.4 | 0.03 μA | 0.414 | 0.04 |
| 4 | 0.6 | 0.6 | 0.05 μA | 0.596 | 0,05 |
| 5 | 0.8 | 0.8 | 0.06 μA | 0.806 | 0.07 |
| 6 | 1 | 1.002 | 0.09 μA | 1.024 | 0.09 |
| 7 | 2 | 1.758 | 161.52 μA | 1.995 | 0.18 |
| 8 | 3 | 1.884 | 2.4 mA | 3.017 | 0.27 |
| 9 | 5 | 1.961 | 6.68 mA | 5.011 | 0.44 |
| 10 | 7 | 2.015 | 10.82 mA | 7.04 | 0.69 |
| 11 | 10 | 2.083 | 16.98 mA | 10.06 | 0.99 |
| 12 | 15 | 2.190 | 27.49 mA | 15.11 | 1.49 |
| 13 | 20 | 2.294 | 38.39 mA | 20.20 | 2 |
| 14 | 25 | 2.388 | 48.64 mA | 25.06 | 2.47 |
| 15 | 30 | 2.478 | 58.54 mA | 29.88 | 2.95 |

**Voltage Vs Current graphs:-**




**<u>CONCLUSION</u>:-**

In forward bias after $V_F$=1.8V as voltage increases large current starts flowing through the diode and small amount of current flows when diode is in reverse bias.

# EXPERIMENT NO. 12

# EXPERIMENT NO.12

**AIM:-**
TO STUDY THE CHARACTERISTICS OF VARACTOR DIODE

**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF VARACTOR DIODE

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter

**THEORY:-**
The varactor is a device whose reactance can be varied in a controlled manner with a bias voltage. It is a *p-n* junction with a special impurity profile, and its capacitance variation is very sensitive to reverse-biased voltage. varactor is a diode that changes its level of capacitance depending on the level of reverse bias applied to the diode. Also known as a varicap diode.

Varactors are operated reverse-biased so no current flows, but since the thickness of the depletion zone varies with the applied bias voltage, the capacitance of the diode can be made to vary. Generally, the depletion region thickness is proportional to the square root of the applied voltage; and capacitance is inversely proportional to the depletion region thickness. Thus, the capacitance is inversely proportional to the square root of applied voltage.

Varactors are widely used in parametric amplification, harmonic generation, mixing, detection, and voltage-variable tuning applications.

**Capacitor Vs Reverse voltage Characteristics:-**
When the steady DC voltage is applied, the capacitance of varactor diode varies depending upon the voltage applied it holds that capacitance as long as the voltage is maintained. The reverse dc voltage sees varactor an infinite resistance since the leakage current is negligible. This is a distinct advantage, as the loading of the source may for practical purposes be regarded as zero. Any dc voltage applied to varactor for control or bias purposes must be free of ripple, in order to maintain a non-varying junction capacitance at each voltage setting.

In as much as varactor dc controls current is practically zero, at reverse voltages, the voltage may be applied through a fixed high resistance without voltage loss across resistor. This provides an isolation resistor, which eliminates the effect of hand capacitance or of capacitance in the voltage source, which would otherwise shunt the varactor.

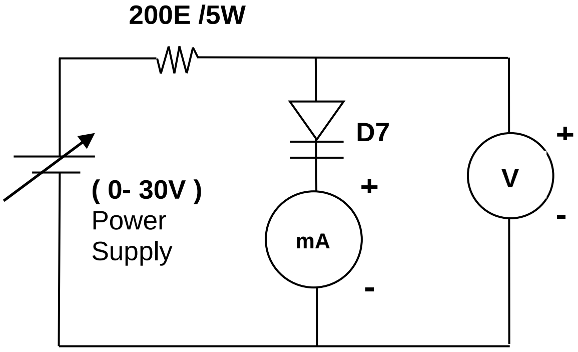
In circuit diagram power supply is adjustable, steady dc voltage R1 a high and resistance X1 the varactor. A variable capacitance corresponding to the capacitance voltage curve of varactor used is obtained at output terminals. If the circuit or device in which this variable capacitance is to be used contains a dc voltage the varactor must be protected from its effects by means of a blocking capacitor C1.

With very low leakage varactors, there is theoretically no limit to the value of isolating resistor R1. A practical limitation is imposed, however by the time Constant (t) resulting from series connections of this resistance and the varactor Capacitance (t = RC. where t is in microsecond. R in ohms and C is microfarads).
The time constant is important when the varactor must respond to the control signal within some desired time after application of the signal. Thus the time constant of 100 MΩ isolating resistor and 100 pF varactor is 10 mS. If the varactor must respond in a shorter period than that, a lower resistance must be used.

**PROCEDURE:-**

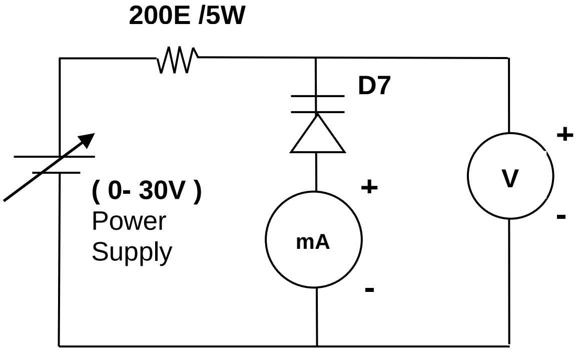
1. Make the connections as shown in following circuit diagram on BEL-COT board.

**Forward Bias Characteristics**



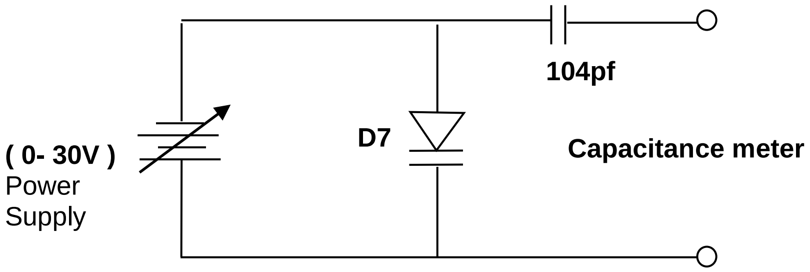

2. Apply dc voltage to forward bias the Varactor diode and vary the voltages as per the observation table.
3. Note down the corresponding forward voltage $V_F$, across the Varactor diode and the forward current, $I_F$.
4. Similarly**,** make the connections as shown in following circuit diagram on BEL-COT board.

**Reverse Bias Characteristics**




5. Apply dc voltage to reverse bias the Varactor diode and vary the voltages as per the observation table.
6. Note down the corresponding reverse bias voltage $V_R$, across the Varactor diode and the reverse bias current, $I_R$.
7. Plot the forward and reverse biased voltage Vs current graph.
8. Make the connections as shown in following circuit diagram on BEL-COT board.

## Capacitor Vs Reverse voltage Characteristics



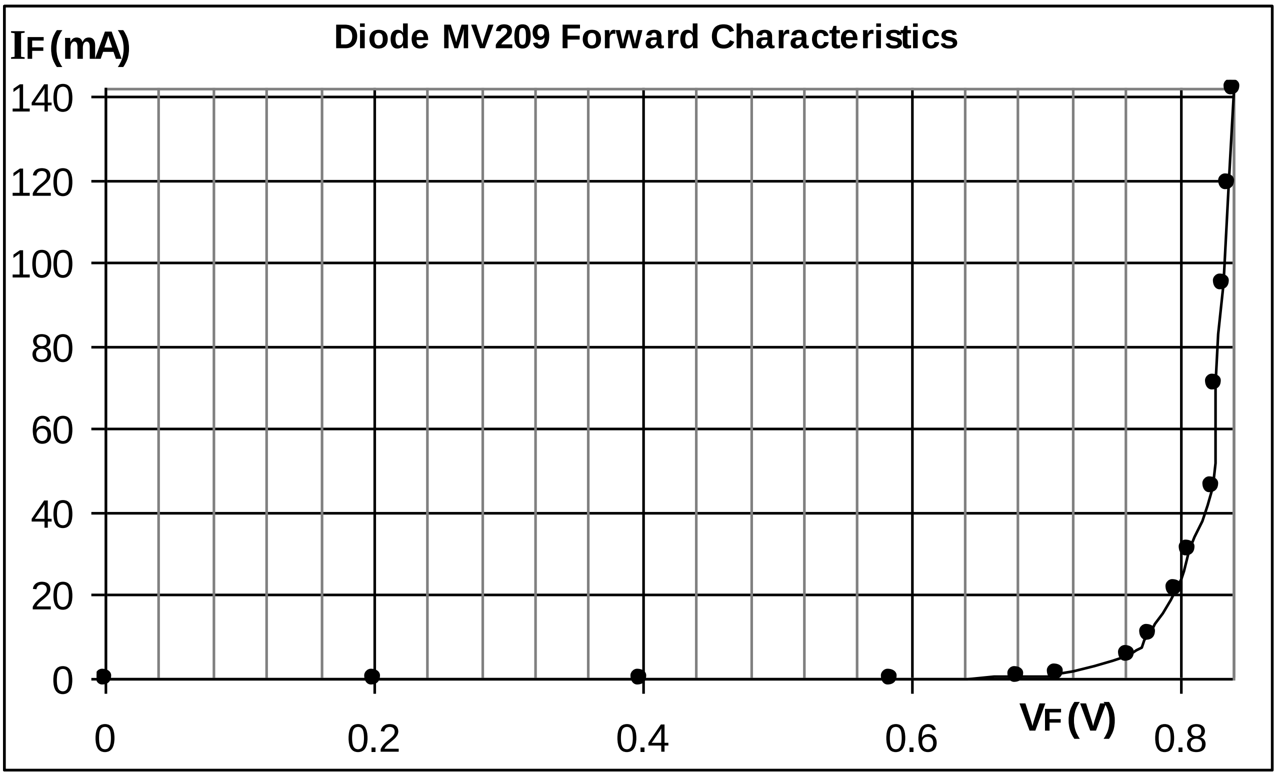

9. Again apply dc reverse voltage to the varactor diode and note down the corresponding capacitance across it, as per the observation table.
10. Plot the capacitive response for the varactor diode.

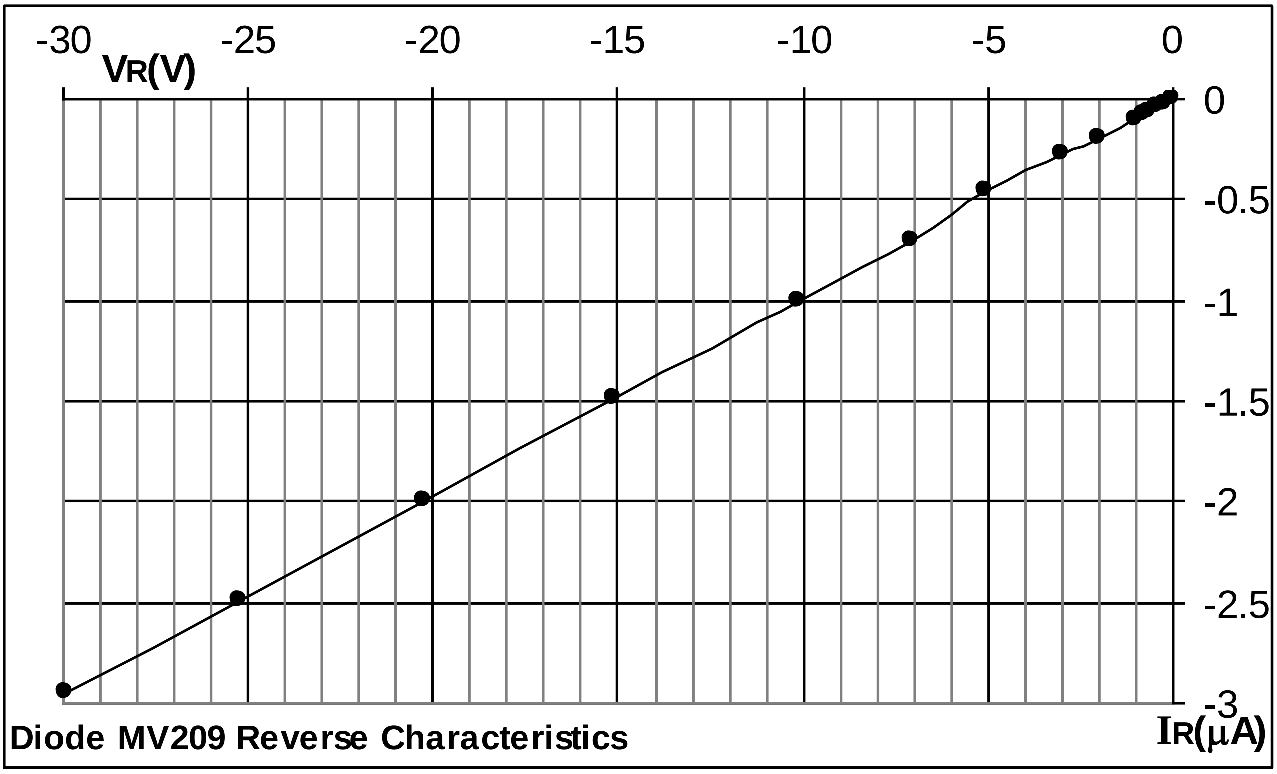
**DESIGN:-**
Diode =MV209
Resistance = 200E/5W

**OBSERVATION TABLE:-**

| Sr. no. | Dc Voltage Source | Diode Forward Characteristics | | Diode Reverse Characteristics | |
|---|---|---|---|---|---|
| | Vin | $V_F$(V) | $I_F$ | $V_R$(V) | $I_R$(μA ) |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 2 | 0.2 | 0.2 | 0.02 μA | 0.2 | 0.02 |
| 3 | 0.4 | 0.398 | 0.06 μA | 0.402 | 0.04 |
| 4 | 0.6 | 0.584 | 17.66 μA | 0.606 | 0.06 |
| 5 | 0.8 | 0.680 | 0.4 μA | 0.800 | 0.08 |
| 6 | 1.0 | 0.709 | 1.0 mA | 1.006 | 0.10 |
| 7 | 2.0 | 0.761 | 5.96 mA | 2.009 | 0.19 |
| 8 | 3.0 | 0.778 | 10.65 mA | 3.022 | 0.27 |
| 9 | 5.0 | 0.796 | 21.35 mA | 5.028 | 0.45 |
| 10 | 7.0 | 0.806 | 31.15 mA | 7.04 | 0.70 |
| 11 | 10.0 | 0.825 | 46.07 mA | 10.12 | 1.00 |
| 12 | 15.0 | 0.827 | 70.81 mA | 15.09 | 1.49 |
| 13 | 20.0 | 0.833 | 95.40 mA | 20.26 | 2.00 |
| 14 | 25.0 | 0.836 | 119 mA | 25.20 | 2.49 |
| 15 | 30.0 | 0.840 | 142 mA | 29.93 | 2.95 |

**Voltage Vs Current graphs:-**



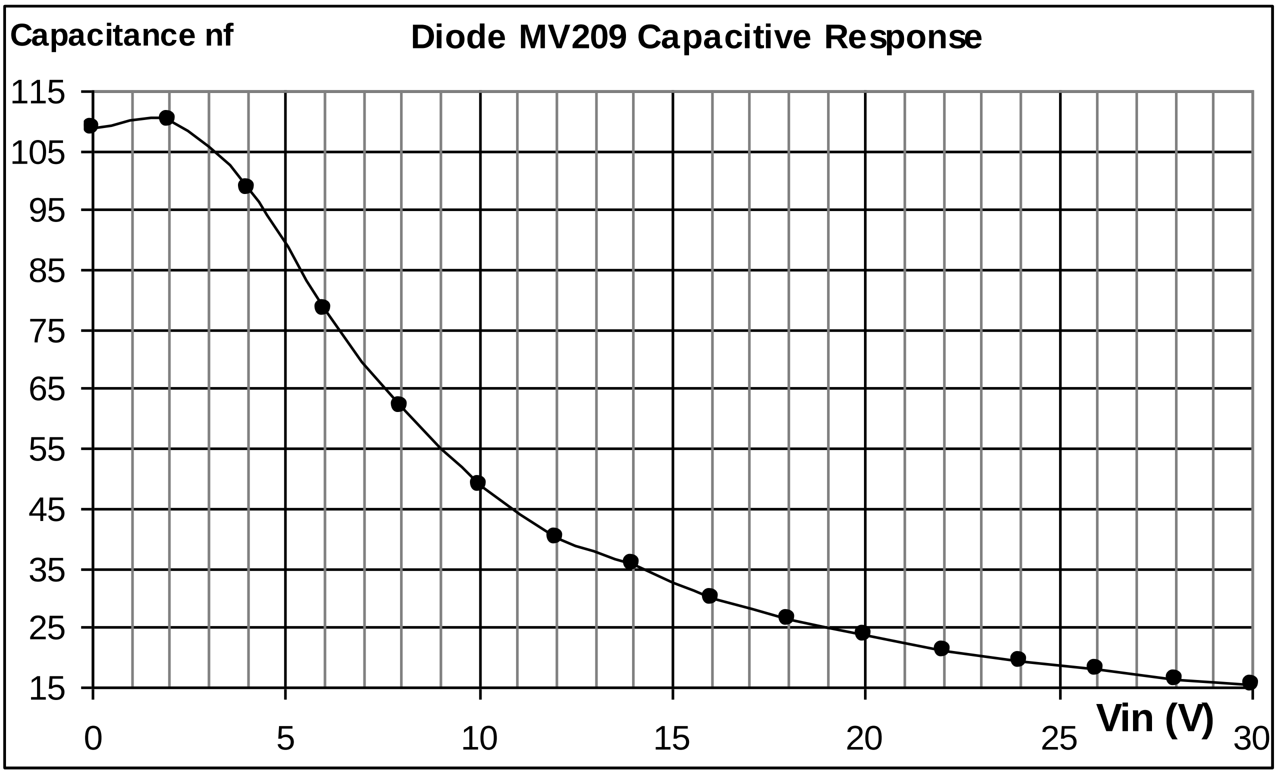




**Capacitor Vs Reverse voltage Characteristics:-**

**<u>DESIGN</u>:-**
Diode =MV209
Capacitor = 104 (0.1µf disc)

## OBSERVATION TABLE:-

| Sr. no. | Dc Voltage Source | Capacitance |
|---|---|---|
| | Vin (V) | nf |
| 1 | 0 | 109 |
| 2 | 2 | 110.2 |
| 3 | 4 | 98.6 |
| 4 | 6 | 78.5 |
| 5 | 8 | 62.2 |
| 6 | 10 | 48.8 |
| 7 | 12 | 40.1 |
| 8 | 14 | 35.6 |
| 9 | 16 | 30.1 |
| 10 | 18 | 26.46 |
| 11 | 20 | 23.61 |
| 12 | 22 | 21.29 |
| 13 | 24 | 19.44 |
| 14 | 26 | 17.87 |
| 15 | 28 | 16.51 |
| 16 | 30 | 15.37 |




## CONCLUSION:-

In forward bias after $V_F$=0.8V as voltage increases large current starts flowing through the diode and small amount of current flows when diode is in reverse bias. In Capacitor Vs Reverse voltage Characteristics as reverse voltage increases capacitance of varctor diode decreases.

# EXPERIMENT NO. 13

# EXPERIMENT NO.13

**AIM:-**
TO STUDY THE CHARACTERISTICS OF VDR

**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF VDR

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter

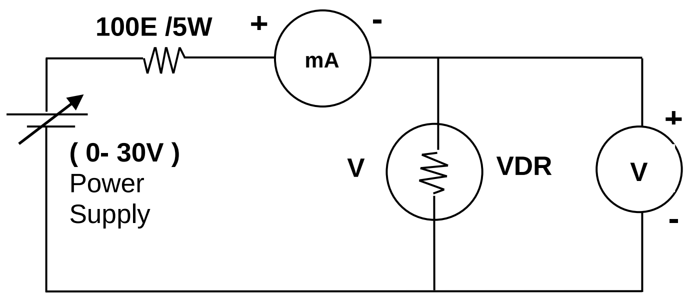
**THEORY:-**
A two-electrode semiconductor device having a voltage-dependent nonlinear resistance; its resistance drops as the applied voltage is increased, also known as voltage-dependent resistor.

Any two-terminal solid-state device in which the electric current I increases considerably faster than the voltage V. This nonlinear effect may occur over all, or only part, of the current-voltage characteristic. It is generally specified as $I \propto V^n$, where n is a number ranging from 3 to 35 depending on the type of varistor. Varistors are often used to protect circuits against excessive transient voltages by incorporating them into the circuit in such a way that, when triggered, they will shunt the current created by the high voltage away from the sensitive components.

The metal-oxide varistor, is made of a ceramic like material comprising zinc oxide grains and a complex amorphous inter granular material. It has a high resistance (about $10^9$ ohms) at low voltage due to the high resistance of the inter granular phase, which becomes nonlinearly conducting in its control range (100–1000 V) with *n* > 25.

**PROCEDURE:-**
1. Make the connections as shown in following circuit diagram on BEL-COT board.

**VDR Characteristics**




2. Apply dc voltage to VDR as per the observation table.
3. Note down the corresponding voltage across the VDR, current flowing through it and the resistance.
4. Plot the Resistive response for the corresponding voltage across VDR.

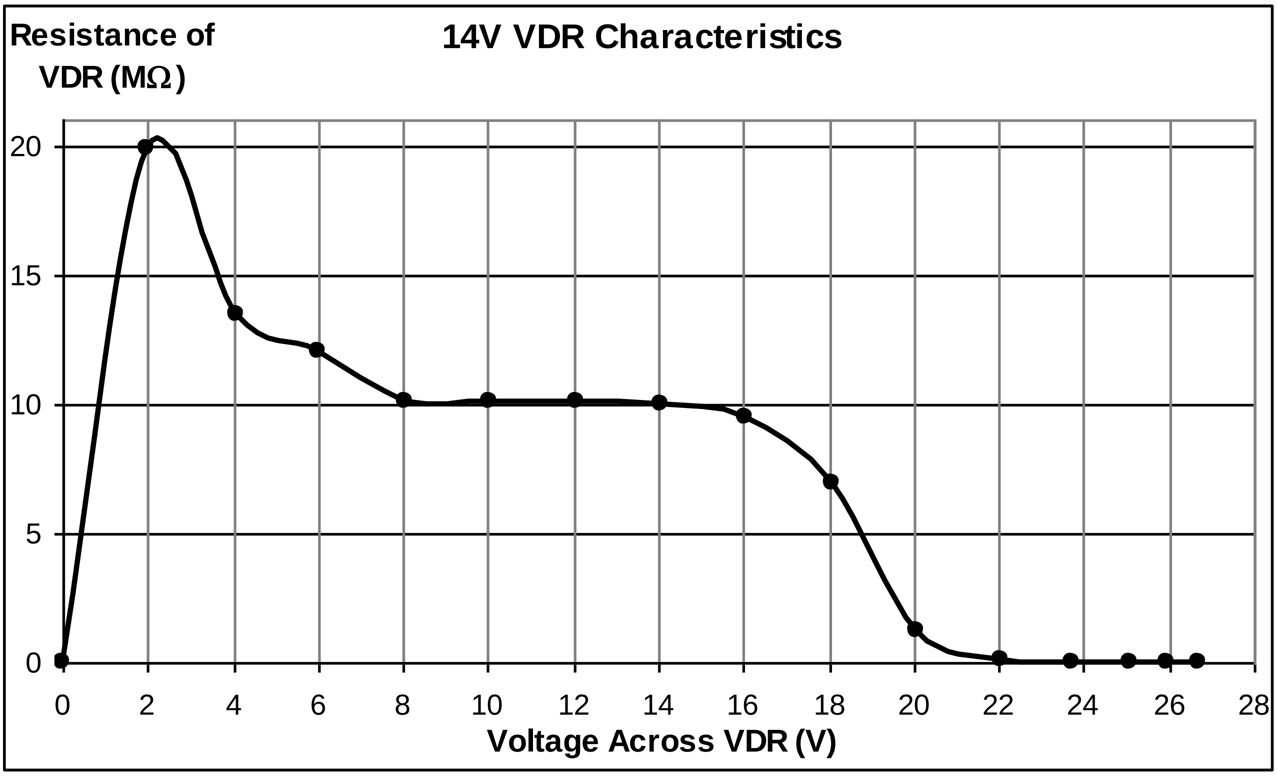
**DESIGN:-**
Rin = 100 E / 5W

Diode = 14V VDR

**OBSERVATION TABLE:-**

| Sr. no. | Applied Voltage | Voltage Across VDR (V) | Current through VDR | Resistance of VDR |
|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 |
| 2 | 2 | 1.99 | 0.1 μA | 19.9 MΩ |
| 3 | 4 | 4.079 | 0.3 μA | 13.5 MΩ |
| 4 | 6 | 6.00 | 0.5 μA | 12 MΩ |
| 5 | 8 | 8.08 | 0.8 μA | 10.1 MΩ |
| 6 | 10 | 10.06 | 1.0 μA | 10.06 MΩ |
| 7 | 12 | 12.06 | 1.2 μA | 10.05 MΩ |
| 8 | 14 | 14.06 | 1.4 μA | 10.04 MΩ |
| 9 | 16 | 16.04 | 1.7 μA | 9.43 MΩ |
| 10 | 18 | 18.07 | 2.6 μA | 6.95 MΩ |
| 11 | 20 | 20.09 | 16.3 μA | 1.23 MΩ |
| 12 | 22 | 22.03 | 234.1 μA | 94.10 KΩ |
| 13 | 24 | 23.70 | 2.68 mA | 8.84 KΩ |
| 14 | 26 | 25.10 | 9.5 mA | 2.64 KΩ |
| 15 | 28 | 25.96 | 20.37 mA | 1.27 KΩ |
| 16 | 30 | 26.70 | 33.68 mA | 792.7 Ω |

**Voltage Vs Resistance graphs:-**




**CONCLUSION:-**

As an applied voltage across the VDR is increases resistance of VDR decreases gradually after rated voltage. Here rated voltage is 14V.

# EXPERIMENT NO. 14

# EXPERIMENT NO.14

**AIM:-**
TO STUDY THE CHARACTERISTICS OF BJT IN CE MODE

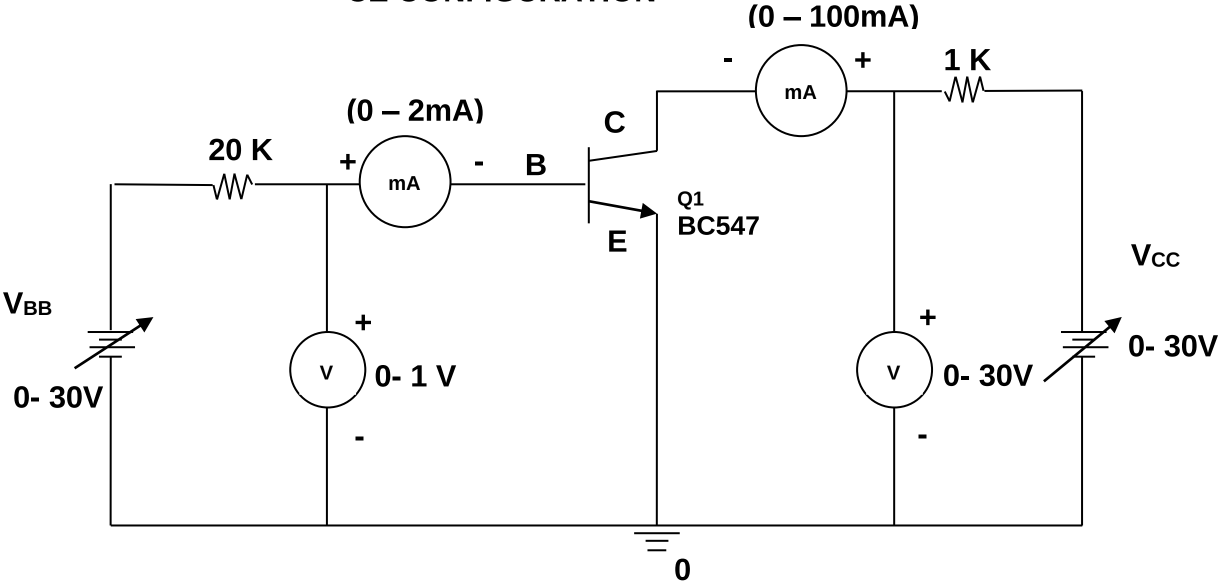
**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF BJT IN CE MODE

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter as voltmeter and as a current meter.

**THEORY:-**
The **COMMON-EMITTER CONFIGURATION (CE)** is the most frequently used configuration in practical amplifier circuits, since it provides good voltage, current, and power gain. The input to the CE is applied to the base-emitter circuit and the output is taken from the collector-emitter circuit, making the emitter the element "common" to both input and output. The CE is set apart from the other configurations, because it is the only configuration that provides a phase reversal between input and output signals.

**PROCEDURE:-**
**Input Characteristics (CE):-**
1. Connect the circuit for BJT in CE (common emitter) mode as per circuit diagram.
2. Switch on power supply.
3. Keep output voltage $V_{CE}$ = 0V by varying $V_{CC}$.
4. Varying $V_{BB}$ gradually, note down the corresponding base-emitter voltage $V_{BE}$ and base current $I_B$ as per the observation table.
5. Repeat the above procedure (step 3) for $V_{CE}$ = 5V and then 10V.
6. Plot the BJT input characteristics for its CE mode.

**CE CONFIGURATION**



**Output Characteristics (CE):-**

1. Now keep $I_B$ = 50 μA by varying $V_{BB}$.
2. Varying $V_{cc}$ gradually, note down the corresponding collector-emitter voltage $V_{CE}$ and collector current $I_C$ as per the observation table.
3. Repeat the above (step 2) for $I_B$ = 75 μA and then 100 μA.
4. Plot the BJT output characteristics for its CE mode.

**DESIGN:-**

BJT = BC547

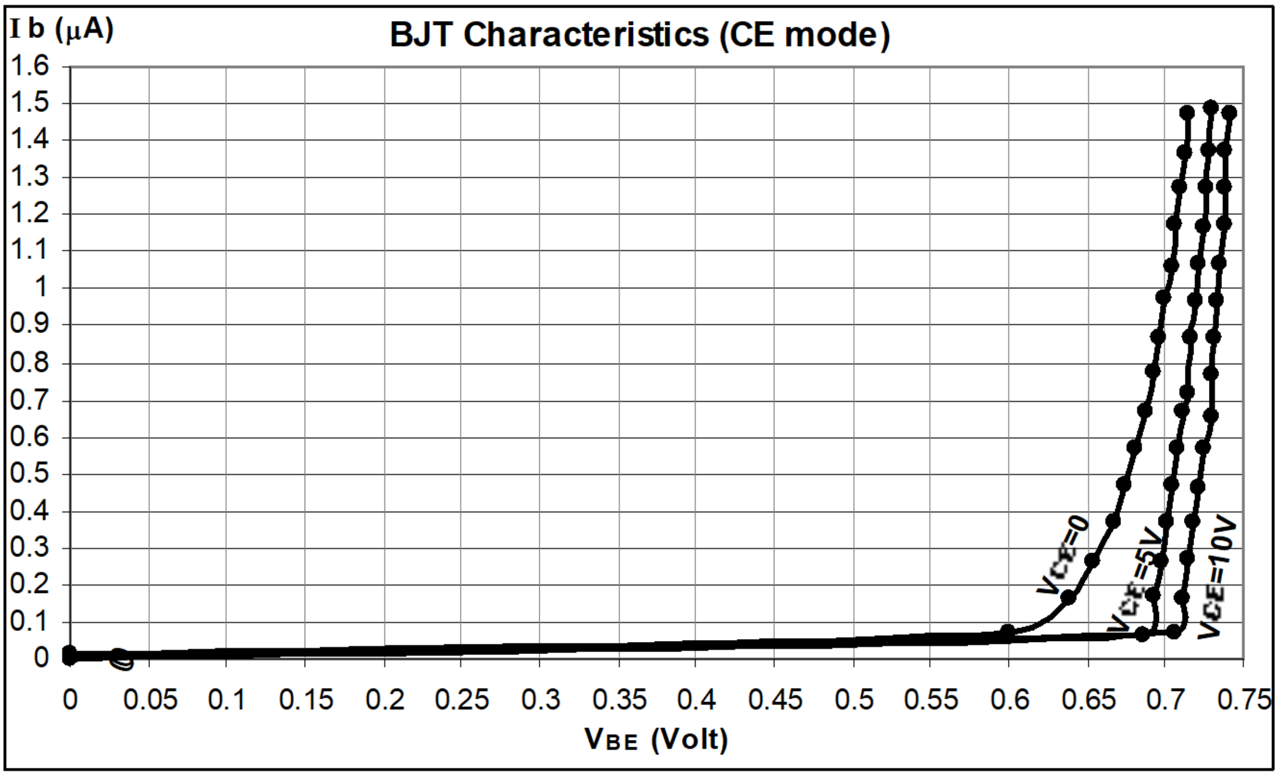
Resistor = 1KΩ and 20KΩ.

**OBSERVATION TABLE:-**

Transistor Input Characteristics (CE):

| **Sr. no.** | **Vin ($V_{BB}$)** | **VCE = 0V** | | **VCE = 5V** | | **VCE = 10V** | |
|---|---|---|---|---|---|---|---|
| | | $V_{BE}$ (Volt) | $I_B$ (μA) | $V_{BE}$ (Volt) | $I_B$ (μA) | $V_{BE}$ (Volt) | $I_B$ (μA) |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0.013 |
| 2 | 2 | 0.6 | 0.070 | 0.686 | 0.067 | 0.707 | 0.072 |
| 3 | 4 | 0.638 | 0.167 | 0.692 | 0.170 | 0.711 | 0.166 |
| 4 | 6 | 0.654 | 0.262 | 0.697 | 0.263 | 0.715 | 0.268 |
| 5 | 8 | 0..667 | 0.370 | 0.701 | 0.368 | 0.718 | 0.372 |
| 6 | 10 | 0.674 | 0.472 | 0.705 | 0.467 | 0.722 | 0.462 |
| 7 | 12 | 0.681 | 0.570 | 0.708 | 0.566 | 0.724 | 0.572 |
| 8 | 14 | 0.687 | 0.671 | 0.712 | 0.665 | 0.729 | 0.655 |
| 9 | 16 | 0.692 | 0.772 | 0.715 | 0.719 | 0.729 | 0.769 |
| 10 | 18 | 0.696 | 0.870 | 0.717 | 0.870 | 0.731 | 0.867 |
| 11 | 20 | 0.700 | 0.972 | 0.720 | 0.967 | 0.733 | 0.969 |
| 12 | 22 | 0.704 | 1.061 | 0.722 | 1.069 | 0.735 | 1.068 |
| 13 | 24 | 0.707 | 1.170 | 0.724 | 1.168 | 0.738 | 1.173 |
| 14 | 26 | 0.710 | 1.270 | 0.726 | 1.273 | 0.738 | 1.273 |
| 15 | 28 | 0.713 | 1.368 | 0.728 | 1.374 | 0.739 | 1.369 |
| 16 | 30 | 0.715 | 1.475 | 0.730 | 1.485 | 0.741 | 1.471 |

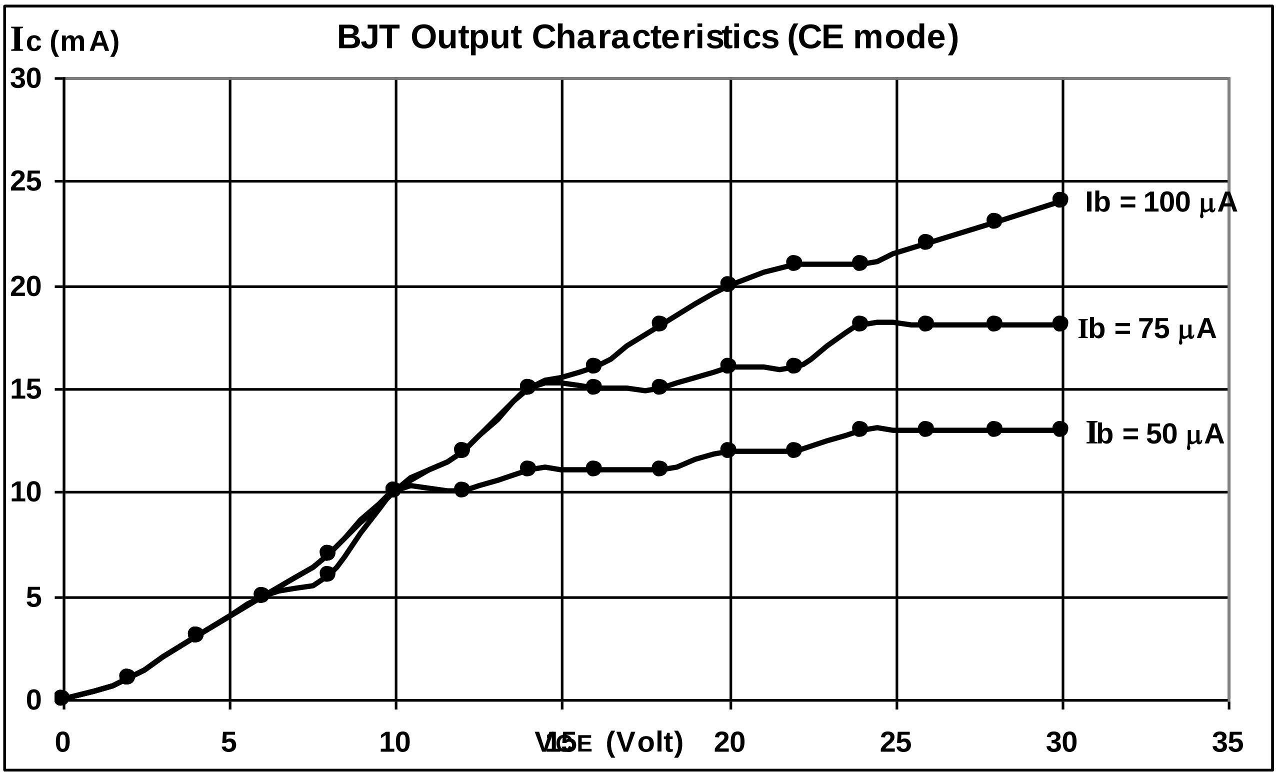
### Input Characteristics (CE) graph:-




### Transistor Output Characteristics (CE):

| Sr. no. | Vin ($V_{CC}$) | $I_B$ = 50 μA | | $I_B$ = 75 μA | | $I_B$ = 100 μA | |
|---|---|---|---|---|---|---|---|
| | | $V_{CE}$ (Volt) | Ic (mA) | Vce (Volt) | Ic (mA) | $V_{CE}$ (Volt) | Ic (mA) |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | 2 | 2 | 1 | 2 | 1 | 2 | 1 |
| 3 | 4 | 4 | 3 | 4 | 3 | 4 | 3 |
| 4 | 6 | 6 | 5 | 6 | 5 | 6 | 5 |
| 5 | 8 | 8 | 7 | 8 | 6 | 8 | 7 |
| 6 | 10 | 10 | 10 | 10 | 10 | 10 | 10 |
| 7 | 12 | 12 | 10 | 12 | 12 | 12 | 12 |
| 8 | 14 | 14 | 11 | 14 | 15 | 14 | 15 |
| 9 | 16 | 16 | 11 | 16 | 15 | 16 | 16 |
| 10 | 18 | 18 | 11 | 18 | 15 | 18 | 18 |
| 11 | 20 | 20 | 12 | 20 | 16 | 20 | 20 |
| 12 | 22 | 22 | 12 | 22 | 16 | 22 | 21 |
| 13 | 24 | 24 | 13 | 24 | 18 | 24 | 21 |
| 14 | 26 | 26 | 13 | 26 | 18 | 26 | 22 |
| 15 | 28 | 28 | 13 | 28 | 18 | 28 | 23 |
| 16 | 30 | 30 | 13 | 30 | 18 | 30 | 24 |

**Output Characteristics (CE) graph:-**




## CALCULATION FROM GRAPH:-

**1. Input resistance:**

To obtain input resistance find $\Delta V_{BE}$ and $\Delta I_B$ at constant $V_{CE}$ on one of the input characteristics.

Then Ri = $\Delta V_{BE} / \Delta I_B$ ($V_{CE}$ constant)

**2. Output resistance:**

To obtain output resistance, find $\Delta I_C$ and $\Delta V_{CE}$ at constant $I_B$.

Ro = $\Delta V_{CE} / \Delta I_C$ ($I_B$ constant)

**Calculations from graph:**

a) Input impedance (hic) = = $\Delta V_{BE} / \Delta I_B$, $V_{CE}$ constant.

b) Forward current gain (hfc) = = $\Delta I_C / \Delta I_B$, $V_{CE}$ constant.

c) Output admittance (hoe) = = $\Delta Ic / \Delta V_{EC}$, $I_B$ constant.

d) Reverse voltage gain (hrc) = $\Delta V_{BE} / \Delta V_{EC}$, $I_B$ constant.

**<u>CONCLUSION</u>:-**

Thus the input and output characteristics of CE configuration is plotted.

1. Input Resistance (Ri) = ……………Ω
2. Output Resistance (Ro) = ……………Ω

# EXPERIMENT NO. 15

# EXPERIMENT NO.15

**AIM:-**
TO STUDY THE CHARACTERISTICS OF BJT IN CB MODE

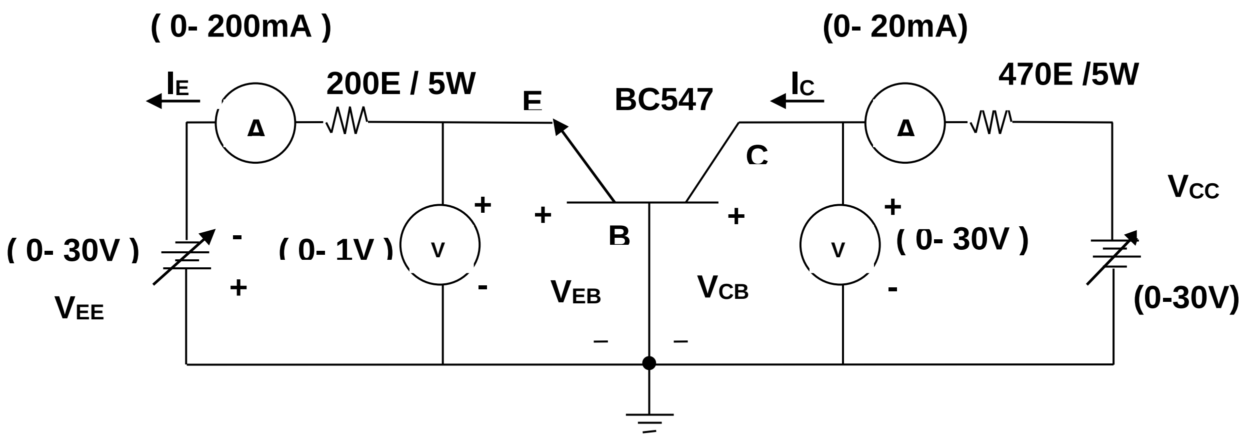
**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF BJT IN CB MODE

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter as voltmeter and as a current meter.

**THEORY:-**
The **COMMON-BASE CONFIGURATION (CB)** is mainly used for impedance matching, since it has a low input resistance and a high output resistance. It also has a current gain of less than 1. In the CB, the input is applied to the emitter, the output is taken from the collector, and the base is the element common to both input and output.

**PROCEDURE:-**
**Input Characteristics (CB):-**
1. Connect the circuit for BJT in CB (common Base) mode as per circuit diagram.
2. Switch on power supply.
3. Keep $V_{CB}$ = 0V by varying $V_{CC}$.
4. Vary the $V_{EE}$ gradually, note down the corresponding collector-base voltage $V_{EB}$ and emitter current $I_E$ as per the observation table.
5. Repeat the above (step3) for $V_{CB}$ = 4V and then 10V.
6. Plot the BJT input characteristics for its CB mode.

**CB CONFIGURATION**



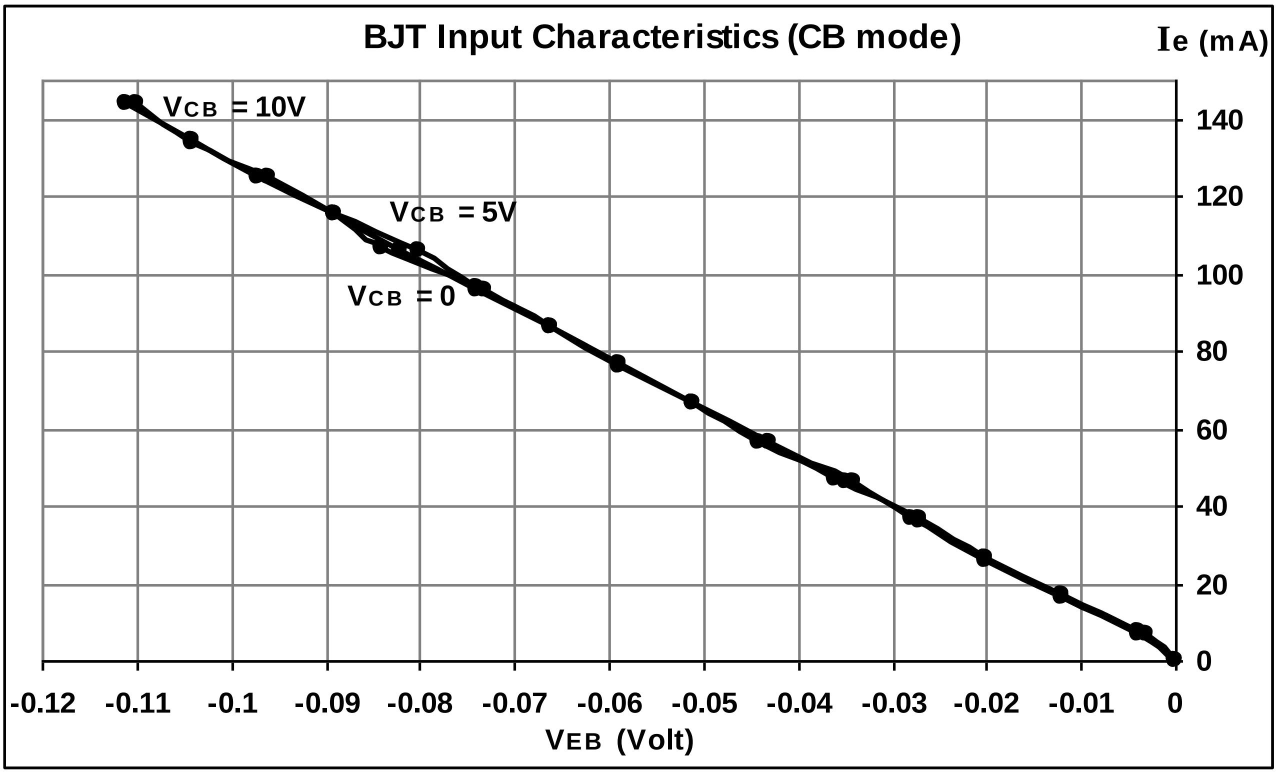

**Output Characteristics (CB):-**
1. Now keep $I_E$ = 2 mA by varying $V_{EE}$.
2. Vary $V_{CC}$ gradually, note down the corresponding collector-base voltage $V_{CB}$ and collector current $I_C$ as per the observation table.
3. Repeat the above (step 2) for $I_E$ = 5mA and then 10mA.
4. Plot the BJT output characteristics for its CB mode.

**DESIGN:-**
BJT = BC547
Resistor = 200Ω / 5W and 470Ω/5W.

**OBSERVATION TABLE:-**
**Transistor Input Characteristics (CB):**

| Sr. no. | Vin ($V_{EE}$) | $V_{CB}$ = 0V | | $V_{CB}$ = 4V | | $V_{CB}$ = 10V | |
|---|---|---|---|---|---|---|---|
| | | $V_{EB}$ (Volt) | $I_E$ (mA) | $V_{EB}$ (Volt) | $I_E$ (mA) | $V_{EB}$ (Volt) | $I_E$ (mA) |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | 2 | -0.004 | 6.49 | -0.003 | 6.48 | -0.004 | 7.74 |
| 3 | 4 | -0.012 | 16.29 | -0.012 | 16.56 | -0.012 | 16.91 |
| 4 | 6 | -0.020 | 26.08 | -0.020 | 26.34 | -0.020 | 26.55 |
| 5 | 8 | -0.027 | 36.23 | -0.028 | 36.49 | -0.027 | 36.35 |
| 6 | 10 | -0.035 | 46.2 | -0.034 | 45.9 | -0.036 | 46.5 |
| 7 | 12 | -0.044 | 56.4 | -0.043 | 56.1 | -0.043 | 56.5 |
| 8 | 14 | -0.051 | 66.3 | -0.051 | 66.2 | -0.051 | 66.2 |
| 9 | 16 | -0.059 | 76.2 | -0.059 | 75.8 | -0.059 | 76.5 |
| 10 | 18 | -0.066 | 86.1 | -0.066 | 86.2 | -0.066 | 86.0 |
| 11 | 20 | -0.074 | 96.2 | -0.073 | 95.9 | -0.074 | 95.7 |
| 12 | 22 | -0.080 | 105.8 | -0.084 | 106.3 | -0.082 | 105.7 |
| 13 | 24 | -0.089 | 115.2 | -0.089 | 115.1 | -0.089 | 115.7 |
| 14 | 26 | -0.097 | 125.0 | -0.096 | 124.9 | -0.097 | 125.2 |
| 15 | 28 | -0.104 | 134.2 | -0.104 | 133.9 | -0.104 | 134.6 |
| 16 | 30 | -0.111 | 143.7 | -0.110 | 143.7 | -0.111 | 143.9 |

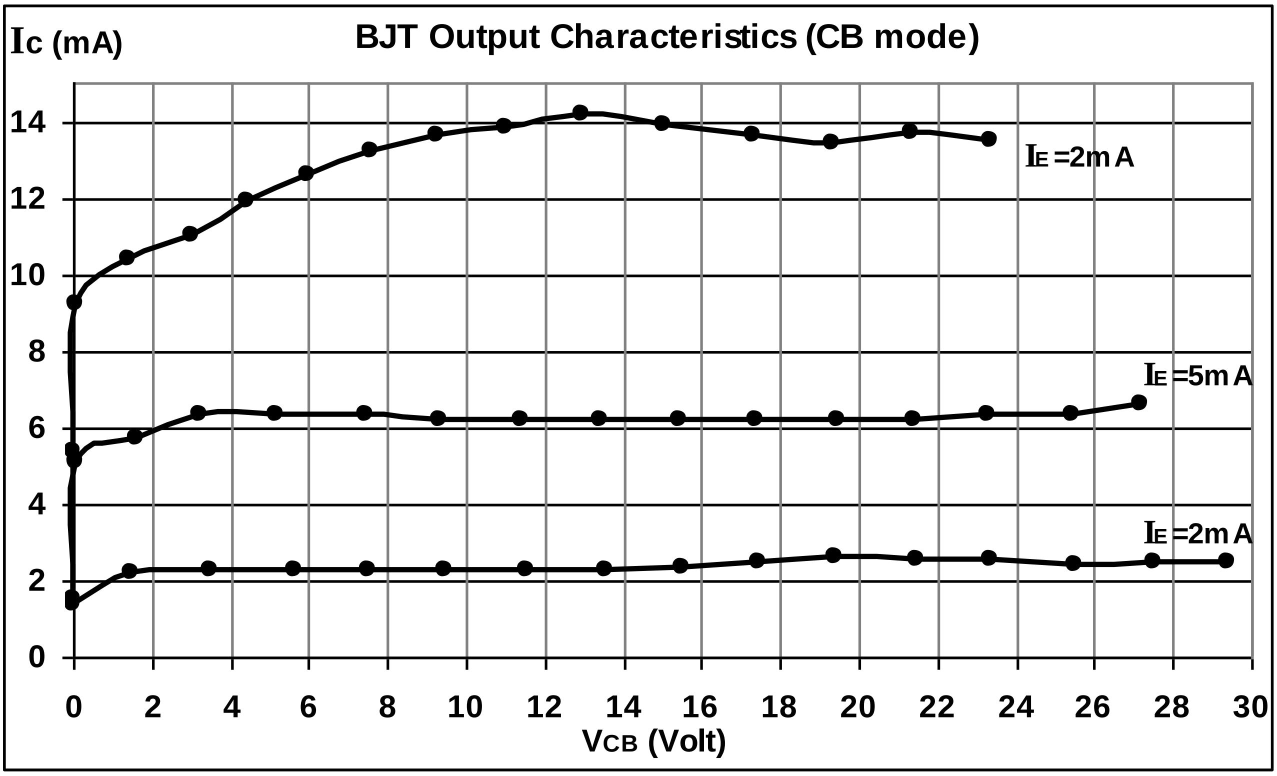
**Input Characteristics (CB) graph:-**



## Transistor Output Characteristics (CB):

| Sr. no. | Vin ($V_{CC}$) | $I_E$ = 2 mA | | $I_E$ = 5 mA | | $I_E$ = 10 mA | |
|---|---|---|---|---|---|---|---|
| | | $V_{CB}$ (Volt) | $I_C$ (mA) | $V_{CB}$ (Volt) | $I_C$ (mA) | $V_{CB}$ (Volt) | $I_C$ (mA) |
| 1 | 0 | 0.013 | 1.359 | 0.009 | 1.459 | 0.012 | 1.545 |
| 2 | 2 | 1.5 | 2.2 | 0.063 | 5.1 | 0.017 | 5.4 |
| 3 | 4 | 3.466 | 2.252 | 1.634 | 5.7 | 0.036 | 9.2 |
| 4 | 6 | 5.622 | 2.283 | 3.227 | 6.3 | 1.406 | 10.4 |
| 5 | 8 | 7.55 | 2.3 | 5.175 | 6.3 | 3.002 | 11.0 |
| 6 | 10 | 9.47 | 2.289 | 7.44 | 6.3 | 4.420 | 11.9 |
| 7 | 12 | 11.53 | 2.280 | 9.34 | 6.2 | 5.987 | 12.6 |
| 8 | 14 | 13.54 | 2.286 | 11.41 | 6.2 | 7.61 | 13.2 |
| 9 | 16 | 15.47 | 2.329 | 13.41 | 6.2 | 9.28 | 13.6 |
| 10 | 18 | 17.48 | 2.489 | 15.45 | 6.2 | 11.00 | 13.8 |
| 11 | 20 | 19.38 | 2.613 | 17.39 | 6.2 | 12.92 | 14.2 |
| 12 | 22 | 21.45 | 2.554 | 19.46 | 6.2 | 15.01 | 13.9 |
| 13 | 24 | 23.33 | 2.560 | 21.43 | 6.2 | 17.29 | 13.6 |
| 14 | 26 | 25.48 | 2.429 | 23.32 | 6.3 | 19.35 | 13.4 |
| 15 | 28 | 27.50 | 2.449 | 25.42 | 6.3 | 21.35 | 13.7 |
| 16 | 30 | 29.42 | 2.464 | 27.21 | 6.6 | 23.35 | 13.5 |

## Output Characteristics (CB) graph:-



**CALCULATION FROM GRAPH:-**

a) Input dynamic resistance (r1) = = $\Delta V_{EB} / \Delta I_E$, $V_{CB}$ constant.
b) Output dynamic resistance (ro) = = $\Delta V_{CB} / \Delta I_C$, $I_E$ constant.
c) DC Current gain ($\alpha$ dc) = = $I_C / I_E$.
d) AC Current gain ($\alpha$) = $\Delta I_C / \Delta I_E$, $V_{CB}$ constant.

**CONCLUSION:-**

Thus the input and output characteristics of CB configuration is plotted.
1. Input dynamic Resistance (ri) = ……………Ω
2. Output dynamic Resistance (ro) = ……………Ω

# EXPERIMENT NO. 16

# EXPERIMENT NO.16

**AIM:-**
TO STUDY THE CHARACTERISTICS OF BJT IN CC MODE

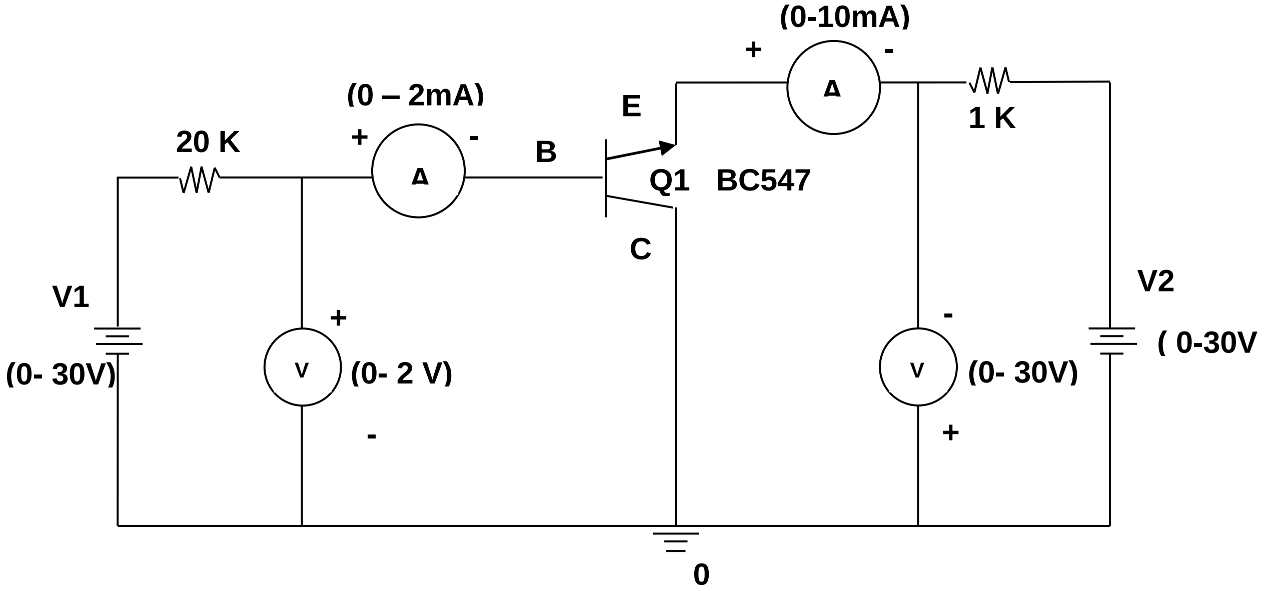
**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF BJT IN CC MODE

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter as voltmeter and as a current meter.

**THEORY:-**
The **COMMON-COLLECTOR CONFIGURATION (CC)** is used as a current driver for impedance matching and is particularly useful in switching circuits. The CC is also referred to as an emitter-follower and is equivalent to the electron-tube cathode follower. Both have high input impedance and low output impedance. In the CC, the input is applied to the base, the output is taken from the emitter, and the collector is the element common to both input and output.

**PROCEDURE:-**
**Input Characteristics (CC):-**
1. Connect the circuit for BJT in CC (Common Collector) mode as per circuit diagram.
2. Switch on power supply.
3. Keep $V_{CE}$ = 0V by varying V2.
4. Vary the V1 gradually, note down the corresponding base-collector voltage $V_{BC}$ and base current $I_B$ as per the observation table.
5. Repeat the above (step 3) for $V_{CE}$ = 1.045V and 1.53V.
6. Plot the BJT input characteristics for its CC mode.

**CC CONFIGURATION**




**Output Characteristics (CC):-**
1. Keep $I_B$ = 50μA by varying V1.

2. Now vary the V2 gradually, note down the corresponding collector-emitter voltage $V_{CE}$ and emitter current $I_E$ as per the observation table.
3. Repeat the above (step2) for $I_B$ = 75μA and then 100μA.
4. Plot the BJT output characteristics for its CC mode.

**DESIGN:-**

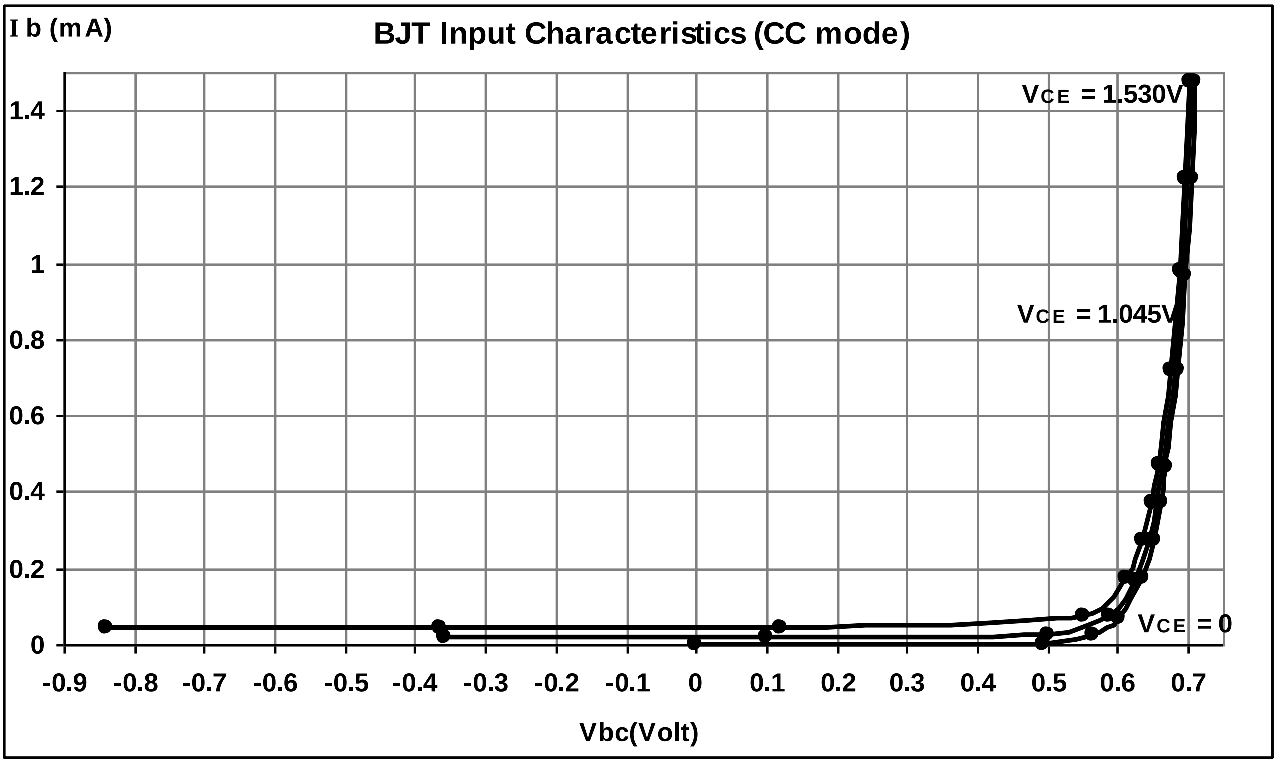
BJT = BC547
Resistor = 1KΩ and 20KΩ.

**OBSERVATION TABLE:-**

Transistor Input Characteristics (CC):

| **Sr. no.** | **Vin (V1)** | **$V_{CE}$ = 0V** | | **$V_{CE}$ = 1.045V @ Vcc =5V** | | **$V_{CE}$ = 1.530V @ Vcc =10V** | |
|---|---|---|---|---|---|---|---|
| | | $V_{BC}$(Volt) | $I_B$ (mA) | $V_{BC}$ (Volt) | $I_B$ (mA) | Vbc(Volt) | $I_B$ (mA) |
| 1 | 0 | 0 | 0 | -0.357 | 0.020 | -0.841 | 0.042 |
| 2 | 0.5 | 0.493 | 0.002 | 0.099 | 0.021 | -0.364 | 0.044 |
| 3 | 1 | 0.566 | 0.022 | 0.500 | 0.026 | 0.120 | 0.045 |
| 4 | 2 | 0.603 | 0.070 | 0.588 | 0.074 | 0.553 | 0.075 |
| 5 | 4 | 0.634 | 0.172 | 0.625 | 0.17 | 0.612 | 0.172 |
| 6 | 6 | 0.651 | 0.272 | 0.644 | 0.270 | 0.634 | 0.270 |
| 7 | 8 | 0.662 | 0.373 | 0.656 | 0.372 | 0.649 | 0.373 |
| 8 | 10 | 0.670 | 0.467 | 0.665 | 0.468 | 0.659 | 0.472 |
| 9 | 15 | 0.686 | 0.720 | 0.681 | 0.718 | 0.676 | 0.720 |
| 10 | 20 | 0.696 | 0.970 | 0.693 | 0.974 | 0.688 | 0.977 |
| 11 | 25 | 0.705 | 1.222 | 0.701 | 1.223 | 0.696 | 1.219 |
| 12 | 30 | 0.711 | 1.475 | 0.707 | 1.477 | 0.703 | 1.474 |

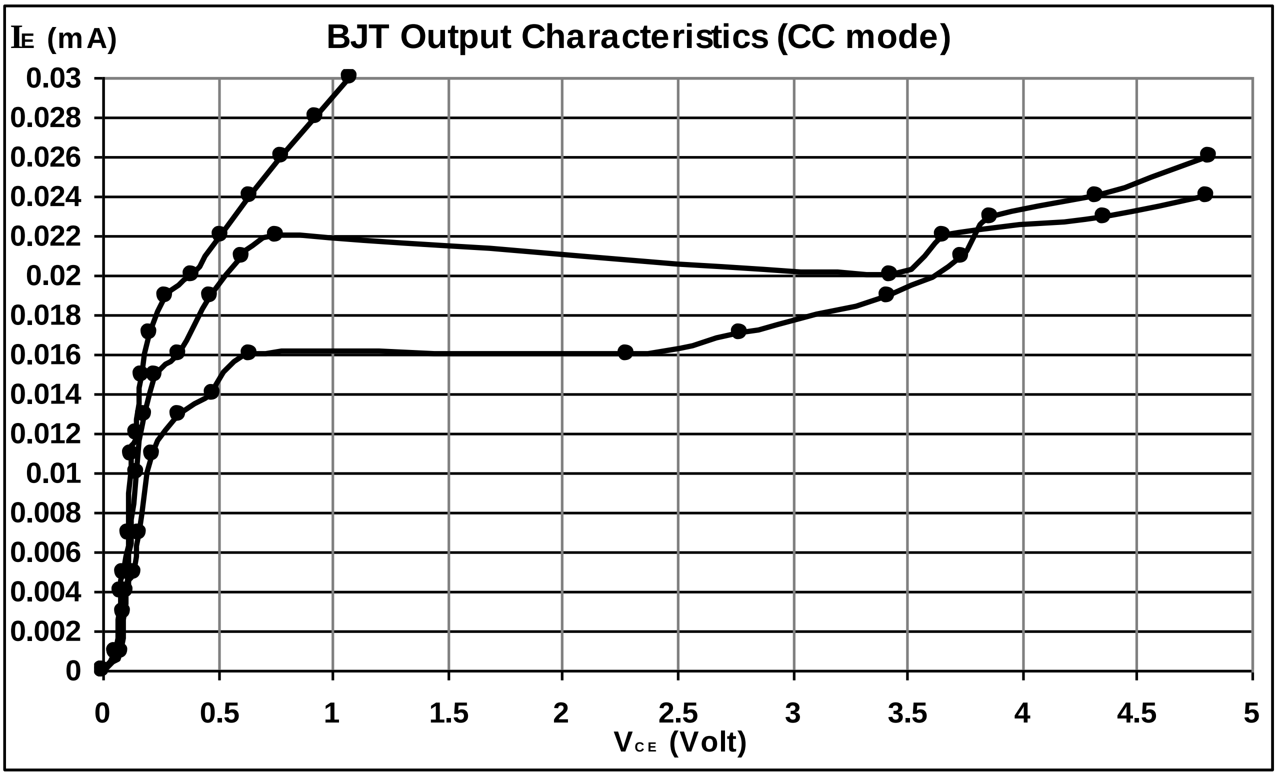
## Input Characteristics (CC) graph:-




## Transistor Output Characteristics (CC):

| Sr. no. | Vin (V2) | $I_B$ = 50 µA@ $V_{BC}$ = 0.591V | | $I_B$ = 75 µA @ $V_{BC}$ = 0.602V | | $I_B$ = 100 µA @ $V_{BC}$ = 0.608V | |
|---|---|---|---|---|---|---|---|
| | | $V_{CE}$ (Volt) | $I_E$ (mA) | $V_{CE}$ (Volt) | $I_E$ (mA) | $V_{CE}$ (Volt) | $I_E$ (mA) |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | 2 | 0.076 | 0.001 | 0.063 | 0.001 | 0.055 | 0.001 |
| 3 | 4 | 0.107 | 0.004 | 0.090 | 0.003 | 0.078 | 0.004 |
| 4 | 6 | 0.132 | 0.005 | 0.109 | 0.005 | 0.096 | 0.005 |
| 5 | 8 | 0.163 | 0.007 | 0.129 | 0.007 | 0.113 | 0.007 |
| 6 | 10 | 0.217 | 0.011 | 0.152 | 0.010 | 0.130 | 0.011 |
| 7 | 12 | 0.330 | 0.013 | 0.180 | 0.013 | 0.149 | 0.012 |
| 8 | 14 | 0.479 | 0.014 | 0.228 | 0.015 | 0.172 | 0.015 |
| 9 | 16 | 0.645 | 0.016 | 0.337 | 0.016 | 0.206 | 0.017 |
| 10 | 18 | 2.286 | 0.016 | 0.468 | 0.019 | 0.278 | 0.019 |
| 11 | 20 | 2.774 | 0.017 | 0.605 | 0.021 | 0.395 | 0.020 |
| 12 | 22 | 3.42 | 0.019 | 0.761 | 0.022 | 0.512 | 0.022 |
| 13 | 24 | 3.74 | 0.021 | 3.43 | 0.020 | 0.647 | 0.024 |
| 14 | 26 | 3.86 | 0.023 | 3.66 | 0.022 | 0.782 | 0.026 |
| 15 | 28 | 4.32 | 0.024 | 4.36 | 0.023 | 0.930 | 0.028 |
| 16 | 30 | 4.82 | 0.026 | 4.80 | 0.024 | 1.081 | 0.030 |

### Output Characteristics (CC) graph:-




### CALCULATION FROM GRAPH:-

a) Input impedance (hic) = $\Delta V_{BC} / \Delta I_B$
b) Forward current gain (hfc) = $\Delta I_E / \Delta I_B$
c) Output admittance (hoc) = $\Delta I_E / \Delta V_{EC}$
d) Reverse voltage gain (hrc) = $\Delta V_{BC} / \Delta V_{EC}$

### <u>CONCLUSION</u>:-

Thus the input and output characteristics of CC configuration are plotted and h Parameters are found.

a) Input impedance (hic) =
b) Forward current gain (hfc) =
c) Output admittance (hoc) =
d) Reverse voltage gain (hrc) =

# EXPERIMENT NO. 17

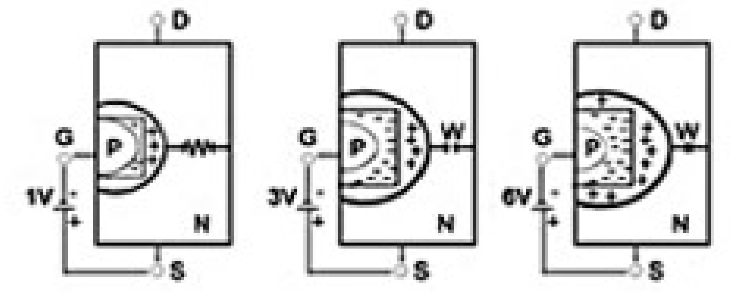
# EXPERIMENT NO.17

**AIM:-**
TO STUDY THE CHARACTERISTICS OF JFET

**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF JFET

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter.

**THEORY:-**
In a junction field effect transistor, the controlled current passes from source to drain or from drain to source. In normal usage, a voltage is applied across the channel, with the drain being made positive with respect to the source. Thus, electrons will passes through the n-type channel from source to drain. Using the voltage applied and the resistance of the channel the magnitude of current flowing through the channel can be determined. Therefore the effective width of the channel is restricted by the depletion region, and its effective resistance is higher in this part of the channel than in any other part of the channel.

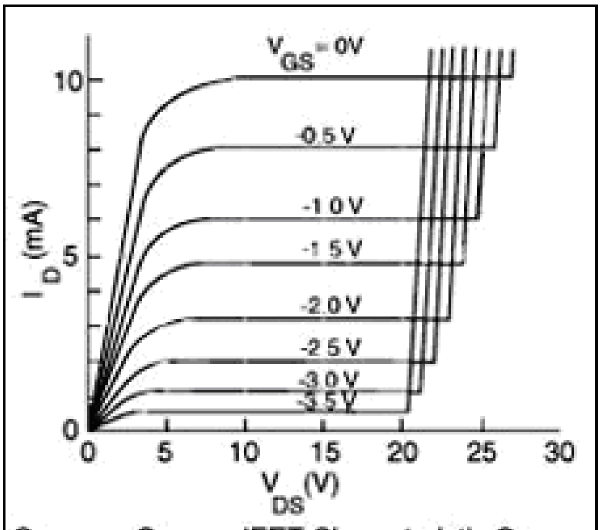


Effect of Reverse bias on the width of the channel

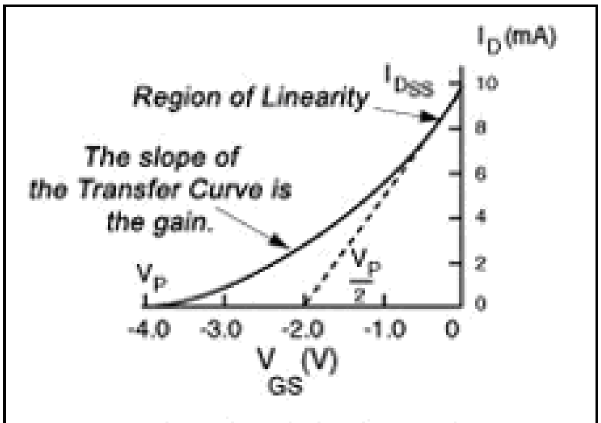
If now the gate is forward biased with respect to the channel the depletion region starts reducing thereby decreasing the flow of current through the channel. However, if the gate becomes reverse biased with respect to the channel the applied electric field will increase the depletion region resulting in the narrowing of channel width, and the channel resistance is increased accordingly. Hence it is seen that a small voltage applied to the gate can have a profound effect on the current flowing through the channel.

If the applied reverse bias becomes very high the channel width will become very narrow stopping the current flow completely.

Characteristic curves for the JFET are shown below. The application of a voltage VDS from drain to source will cause electrons to flow through the channel. It can be seen that for a given value of gate voltage, the current is nearly constant over a wide range of source-to Drain voltages. But, when the Gate is made more negative, it depletes the majority carriers from a larger depletion zone around the gate reducing the current flow for a given value of source-to Drain voltage.



Common Source JFET Characteristic Curves



Transfer Curve for a JFET

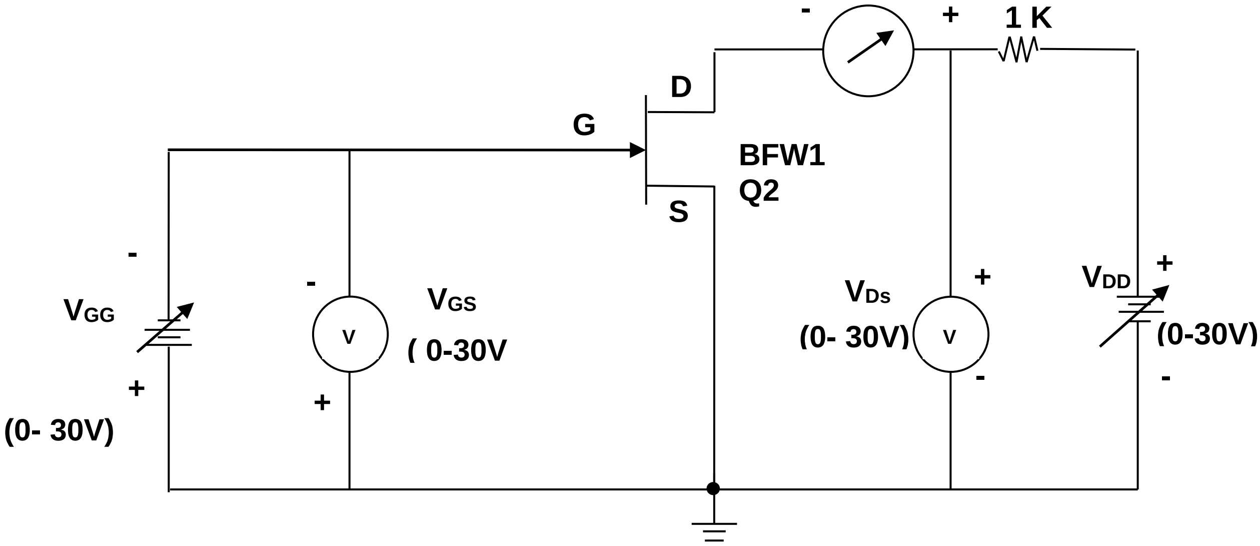
The transfer characteristic for the JFET could be used for visualizing the gain from the device and identifying the region of linearity. Hence, the gain is proportional to the slope of the transfer curve. The current value IDSS represents the value when the Gate is shorted to ground, which is the maximum current for the device. The Gate voltage at which the current becomes zero is called the "pinch voltage", VP. Note that the dashed line represents the gain the linear region of operation that touches the zero current line at about half the pinch voltage.

**<u>PROCEDURE</u>:-**

**Drain Characteristics:**

1. Connect the circuit for JFET Characteristics mode as per circuit diagram.
2. Switch ON power supply.
3. Keep $V_{GS}$ = 0 by varying $V_{GG}$.
4. Vary the $V_{DD}$ step by step and note down the corresponding Drain-source voltage $V_{DS}$ and Drain current $I_D$ as per shown in the observation table.
5. Repeat the above (step3) for $V_{GS}$ = 1V and 2V.
6. Plot the JFET drain characteristics.

**JFET Characteristics**



**Transfer Characteristics:**

1. Now keep $V_{DS}$ = 1V by varying $V_{DD}$.
2. Vary the $V_{GG}$ step by step and note down the corresponding Gate-source voltage $V_{GS}$ and Drain current $I_D$ as per shown in the observation table.
3. Repeat the above step2 for $V_{DS}$ = 3V and 5V.
4. Plot the JFET transfer characteristics.

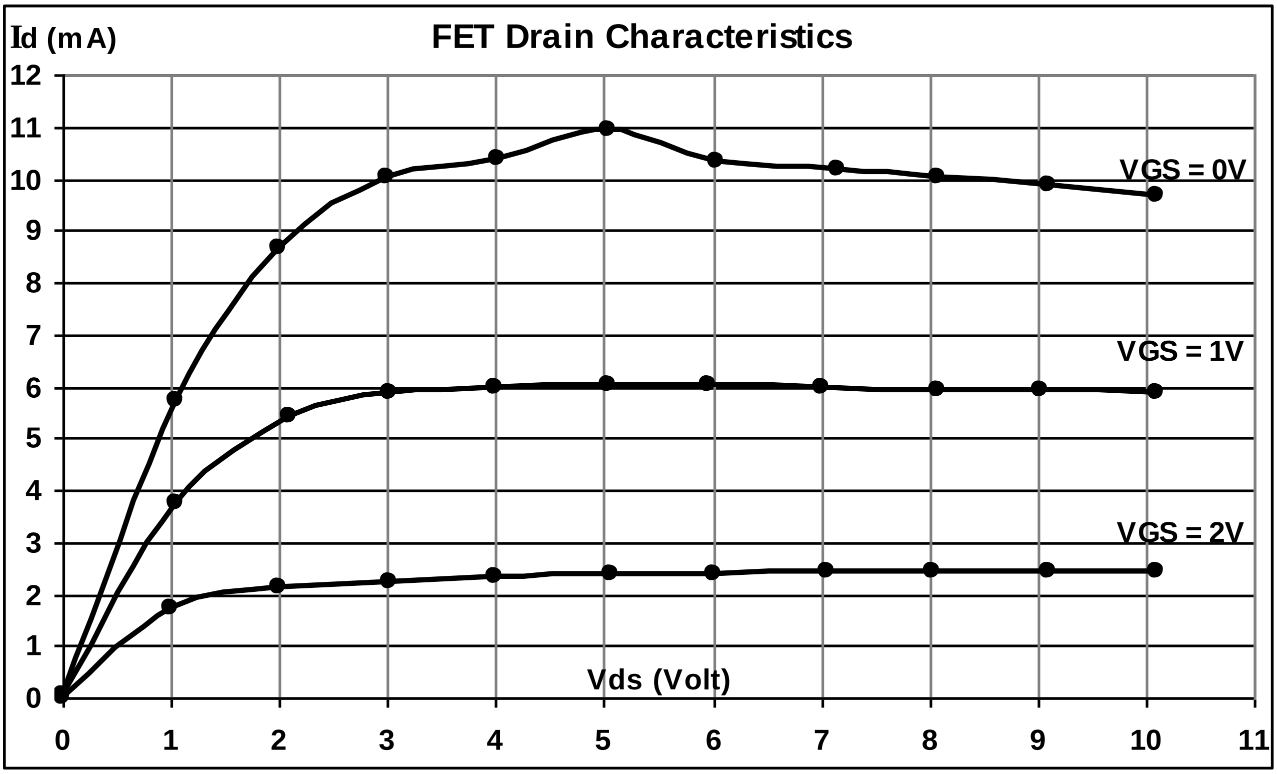
**DESIGN:-**

JFET = BF245C
Resistor = 1KΩ

**OBSERVATION TABLE:-**

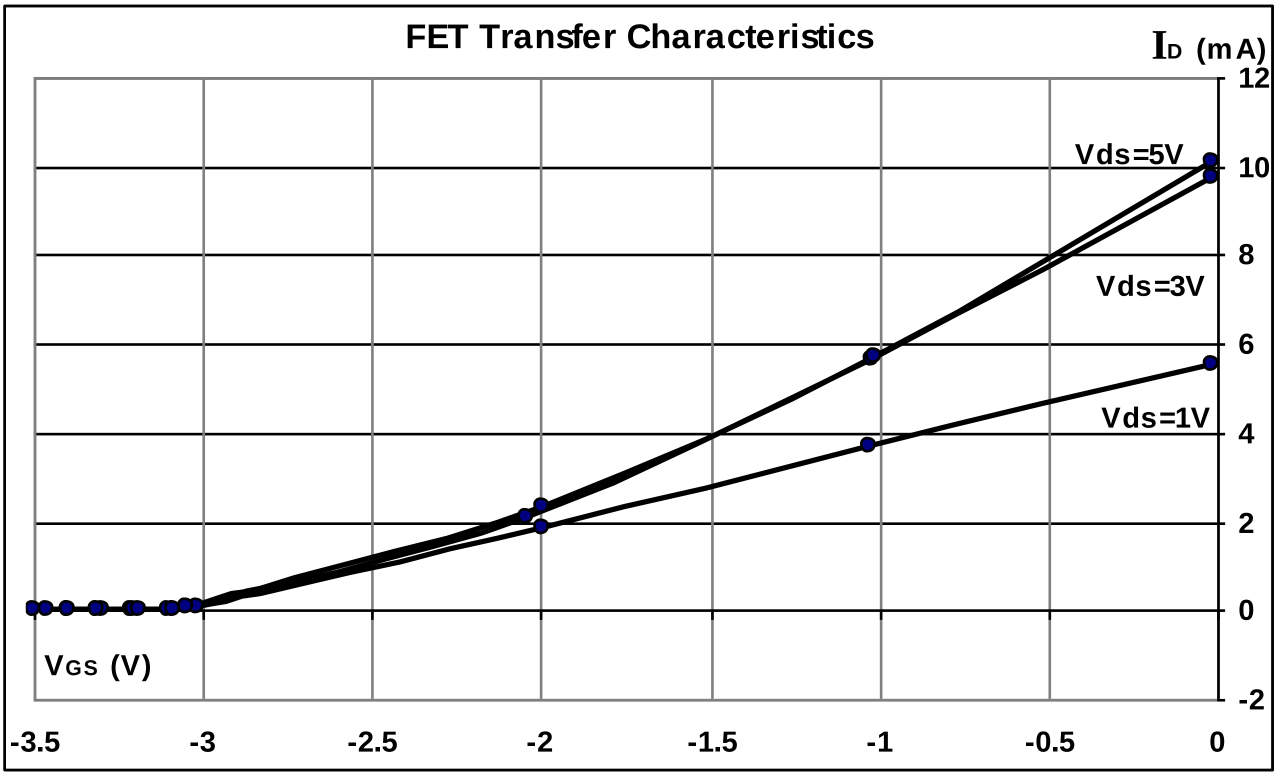
**Drain Characteristics:**

| Sr. no. | $V_{IN}$ ($V_{DD}$) | $V_{GS}$ = 0V | | $V_{GS}$ = 1V | | $V_{GS}$ = 2V | |
|---|---|---|---|---|---|---|---|
| | | $V_{DS}$ (Volt) | $I_D$ (mA) | $V_{DS}$ (Volt) | $I_D$ (mA) | $V_{DS}$ (Volt) | $I_D$ (mA) |
| 1 | 0 | 0.008 | 0.05 | 0.008 | 0.04 | 0.008 | 0.02 |
| 2 | 1 | 1.035 | 5.72 | 1.039 | 3.77 | 1.006 | 1.74 |
| 3 | 2 | 1.992 | 8.68 | 2.087 | 5.40 | 2.006 | 2.13 |
| 4 | 3 | 2.996 | 10.01 | 3.018 | 5.88 | 3.020 | 2.25 |
| 5 | 4 | 4.009 | 10.36 | 3.991 | 5.97 | 4.001 | 2.32 |
| 6 | 5 | 5.040 | 10.92 | 5.035 | 6.03 | 5.058 | 2.36 |
| 7 | 6 | 6.036 | 10.31 | 5.972 | 6.02 | 6.006 | 2.39 |
| 8 | 7 | 7.16 | 10.17 | 7.01 | 5.99 | 7.05 | 2.42 |
| 9 | 8 | 8.09 | 10.01 | 8.08 | 5.94 | 8.04 | 2.43 |
| 10 | 9 | 9.11 | 9.85 | 9.04 | 5.91 | 9.11 | 2.45 |
| 11 | 10 | 10.09 | 9.69 | 10.09 | 5.87 | 10.09 | 2.45 |




**Transfer Characteristics:**

| Sr. no. | Vin ($V_{GG}$) | $V_{DS}$ = 1V | | $V_{DS}$ = 3V | | $V_{DS}$ = 5V | |
|---|---|---|---|---|---|---|---|
| | | $V_{GS}$ (Volt) | $I_D$ | $V_{GS}$ (Volt) | $I_D$ | $V_{GS}$ (Volt) | $I_D$ |
| 1 | 0 | -0.013 | 5.563 mA | -0.012 | 9.749 mA | -0.012 | 10.10 mA |
| 2 | 1 | -1.028 | 3.697 mA | -1.020 | 5.668 mA | -1.017 | 5.74 mA |
| 3 | 2 | -1.996 | 1.848 mA | -2.041 | 2.109 mA | -1.996 | 2.31 mA |
| 4 | 3 | -3.02 | 0.055 mA | -3.05 | 0.053 mA | -3.044 | 0.07 mA |
| 5 | 3.1 | -3.1 | 0.018 mA | -3.09 | .0376 mA | -3.09 | 42.7uA |
| 6 | 3.2 | -3.21 | 1.74 μA | -3.20 | 0.0028 mA | -3.19 | 6.98 μA |
| 7 | 3.3 | -3.31 | 0.10 μA | -3.30 | 0.27 μA | -3.31 | 0.35 μA |
| 8 | 3.3 | -3.4 | 0 | -3.40 | .02 μA | -3.40 | 0.01 μA |
| 9 | 3.5 | | | -3.46 | 0 | -3.50 | 0 |




**Calculations from Graph:**

**Drain Resistance (rd):**

It is given by the ratio of small change in drain to source voltage ($\Delta V_{DS}$) to the corresponding change in Drain current ($\Delta I_D$) for a constant gate to source voltage ($V_{GS}$), when the JFET is operating in pinch-off or saturation region.

**Trans-Conductance (gm):**

Ratio of small change in drain current ($\Delta I_D$) to the corresponding change in gate to source voltage ($\Delta V_{GS}$) for a constant $V_{DS}$. gm = $\Delta I_D$ / $\Delta V_{GS}$ at constant $V_{DS}$. (From transfer characteristics) The value of gm is expressed in mho's or siemens (s).

**Amplification Factor (μ):**

It is given by the ratio of small change in drain to source Voltage ($\Delta V_{DS}$) to the corresponding change in gate to source Voltage ($\Delta V_{GS}$) for a constant drain current.

$\mu = \Delta V_{DS} / \Delta V_{GS}$.

$\mu = (\Delta V_{DS} / \Delta I_D) \times (\Delta I_D / \Delta V_{GS})$

$\mu = rd \times gm$.

**CONCLUSION:-**

1. Drain Resistance (rd) = ………….
2. Tran-conductance (gm) = ………….
3. Amplification factor (μ) = ……………

# EXPERIMENT NO. 18

# EXPERIMENT NO.18

## AIM:-
TO STUDY THE CHARACTERISTICS OF MOSFET

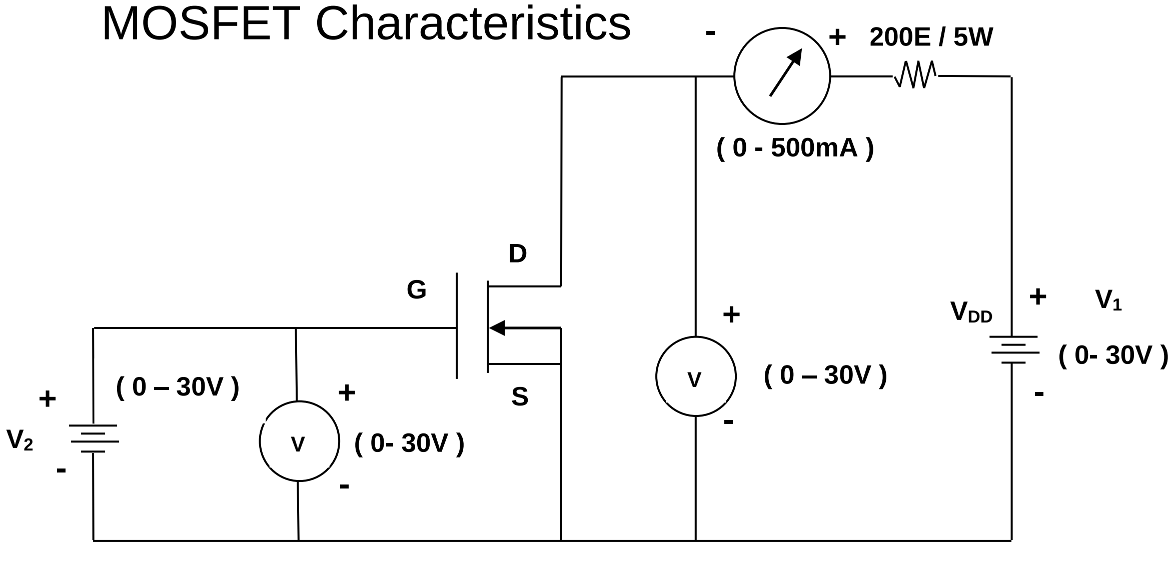
**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF MOSFET

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter

**THEORY:-**
The metal–oxide–semiconductor field-effect transistor (MOSFET) is a device used for amplifying or switching electronic signals. In MOSFETs, a voltage on the oxide-insulated gate electrode can induce a conducting channel between the two other contacts called source and drain. The channel can be of n-type or p-type (see article on semiconductor devices), and is accordingly called an NMOSFET or a PMOSFET (also commonly nMOS, pMOS).

**PROCEDURE:-**
**Drain Characteristics:**
1. Connect the circuit for MOSFET Characteristics mode as per circuit diagram.
2. Switch ON power supply.
3. Keep $V_{GS}$ = 3V by varying V2.
4. Vary the $V_1$ step by step and note down the corresponding Drain-source voltage $V_{DS}$ and Drain current $I_D$ as per shown in the observation table.
5. Repeat the above (step3) for $V_{GS}$ = 5V and 8V.
6. Plot the MOSFET drain characteristics.

MOSFET Characteristics



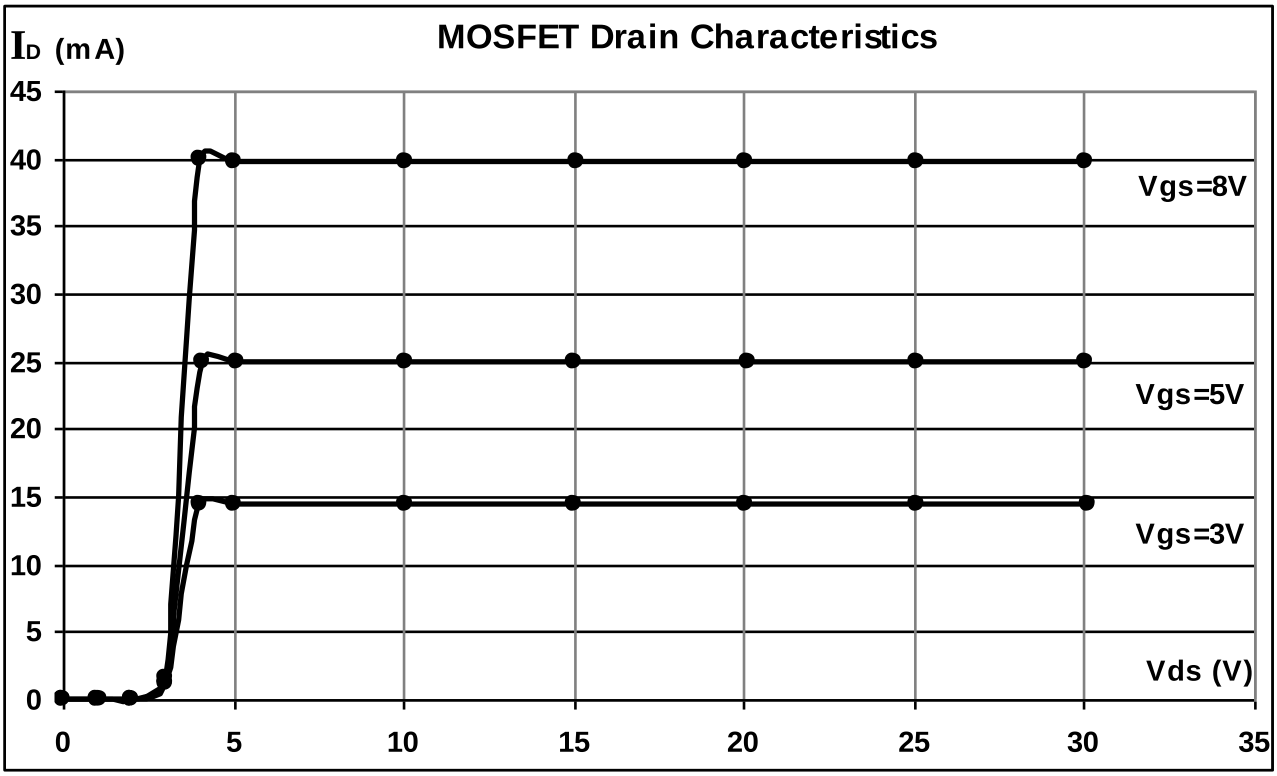

**Transfer Characteristics:**
1. Now keep $V_{DS}$ = 4V by varying $V_1$.

2. Vary the $V_2$ step by step and note down the corresponding Gate-source voltage $V_{GS}$ and Drain current $I_D$ as per shown in the observation table.
3. Repeat the above step2 for $V_{DS}$ = 8V and 10V.
4. Plot the MOSFET transfer characteristics.

**DESIGN:-**
MOSFET = IRF Z44N.
Resistor = 200E/5W.

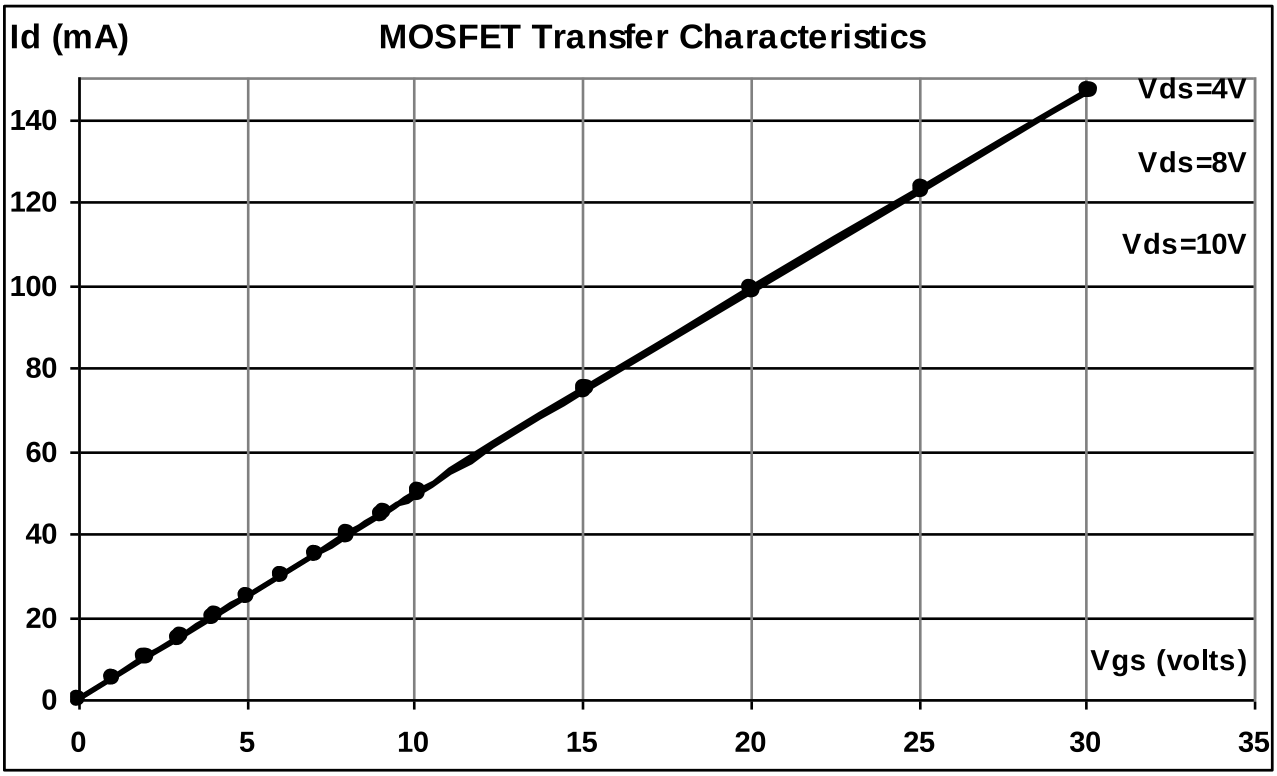
**OBSERVATION TABLE:-**
**MOSFET Drain Characteristics:**

| Sr. no. | Vin ($V_{GG}$) | Vgs = 3V | | Vgs = 5V | | Vgs = 8V | |
|---|---|---|---|---|---|---|---|
| | | Vds (V) | Id | Vds (V) | Id | Vds (V) | Id |
| 1 | 0 | 0.012 | 0.0 µA | 0.012 | 0.0 µA | 0.012 | 0.0 µA |
| 2 | 1 | 1.031 | 0.0 µA | 1.095 | 0.0 µA | 1.014 | 0.0 µA |
| 3 | 2 | 2.005 | 11 µA | 2.012 | 12. µA | 2.004 | 0.11 µA |
| 4 | 3 | 3.03 | 1.25 mA | 3.01 | 1.134 mA | 3.038 | 1.5 mA |
| 5 | 4 | 4.06 | 14.39 mA | 4.08 | 24.94 mA | 4.01 | 39.87 mA |
| 6 | 5 | 5.06 | 14.39 mA | 5.11 | 24.94 mA | 5.00 | 39.78 mA |
| 7 | 10 | 10.05 | 14.39 mA | 10.06 | 24.94 mA | 10.07 | 39.77 mA |
| 8 | 15 | 15.03 | 14.39 mA | 15.05 | 24.94 mA | 15.07 | 39.76 mA |
| 9 | 20 | 20.04 | 14.39 mA | 20.10 | 24.94 mA | 20.02 | 39.76 mA |
| 10 | 25 | 25.10 | 14.39 mA | 25.11 | 24.94 mA | 25.06 | 39.76 mA |
| 11 | 30 | 30.11 | 14.39 mA | 30.05 | 24.94 mA | 30.08 | 39.76 mA |



**MOSFET Transfer Characteristics:-**

| Sr. no. | Vin ($V_{DD}$) | VDS = 4V | | VDS = 8V | | VDS = 10V | |
|---|---|---|---|---|---|---|---|
| | | Vgs (Volt) | Id (mA) | Vgs (Volt) | Id (mA) | Vgs (Volt) | Id (mA) |
| 1 | 0 | 0.008 | 0.04 | 0.009 | 0.04 | 0.008 | 0.04 |
| 2 | 1 | 1.022 | 5.06 | 1.009 | 5 | 1.002 | 5.01 |
| 3 | 2 | 2.00 | 9.91 | 2.028 | 10.32 | 2.016 | 9.96 |
| 4 | 3 | 2.985 | 14.81 | 3.050 | 15.13 | 3 | 14.82 |
| 5 | 4 | 4.043 | 20.08 | 4.022 | 19.95 | 4.021 | 19.88 |
| 6 | 5 | 5.008 | 24.87 | 5.000 | 24.80 | 5.007 | 24.77 |
| 7 | 6 | 6.016 | 29.88 | 6.040 | 29.97 | 6.021 | 29.80 |
| 8 | 7 | 7.06 | 34.94 | 7.08 | 35.04 | 7.04 | 34.71 |
| 9 | 8 | 8.02 | 39.74 | 8.02 | 39.69 | 8.02 | 39.55 |
| 10 | 9 | 9.05 | 44.78 | 9.03 | 44.66 | 9.07 | 44.82 |
| 11 | 10 | 10.09 | 49.98 | 10.12 | 50.10 | 10.09 | 49.75 |
| 12 | 15 | 15.16 | 74.94 | 15.04 | 74.80 | 15.09 | 74.41 |
| 13 | 20 | 20.05 | 98.85 | 20.07 | 98.71 | 20.06 | 98.69 |
| 14 | 25 | 25.11 | 123.18 | 25.10 | 123.40 | 25.10 | 122.59 |
| 15 | 30 | 30.09 | 146.78 | 30.17 | 146.66 | 30.07 | 146.53 |




**<u>CONCLUSION</u>:-**

In drain characteristics as applied voltage increases current increases up to certain limit then it remain constant for specified gate voltage. Also as gate-source voltage increases with increases in drain current.

# EXPERIMENT NO. 19

# EXPERIMENT NO.19

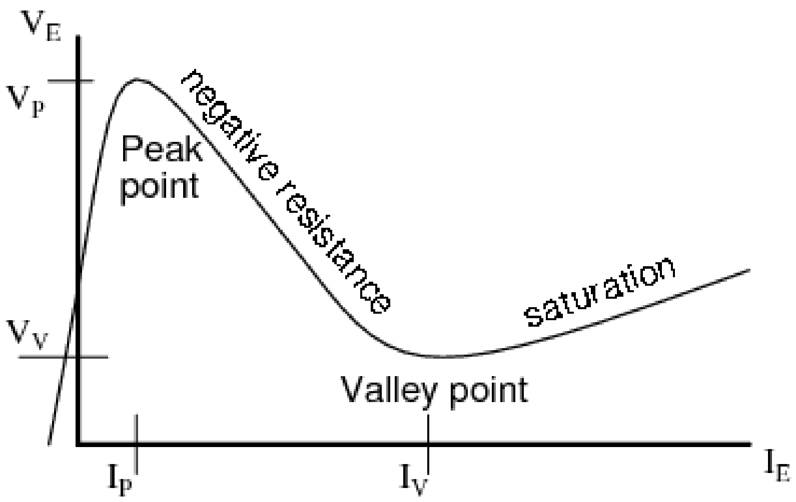
**AIM:-**
TO STUDY THE CHARACTERISTICS OF UJT

**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF UJT

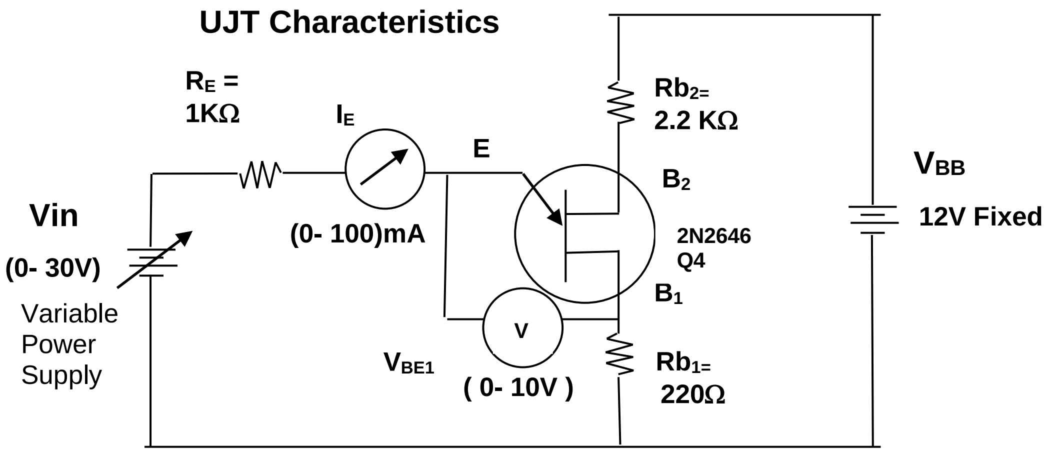
**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter

**THEORY:-**
A **unijunction transistor** (**UJT**) is an electronic semiconductor device that has only one junction. The UJT has three terminals: an emitter (E) and two bases (B1 and B2). The base is formed by lightly doped n-type bar of silicon. Two ohmic contacts B1 and B2 are attached at its ends. The emitter is of p-type and it is heavily doped. The resistance between B1 and B2, when the emitter is open-circuit is called interbase resistance.

The Unijunction emitter current vs voltage characteristic curve shows that as $V_E$ increases, current $I_E$ increases up $I_P$ at the peak point. Beyond the peak point, current increases as voltage decreases in the negative resistance region. The voltage reaches a minimum at the valley point. The resistance of $R_{B1}$, the saturation resistance is lowest at the valley point.



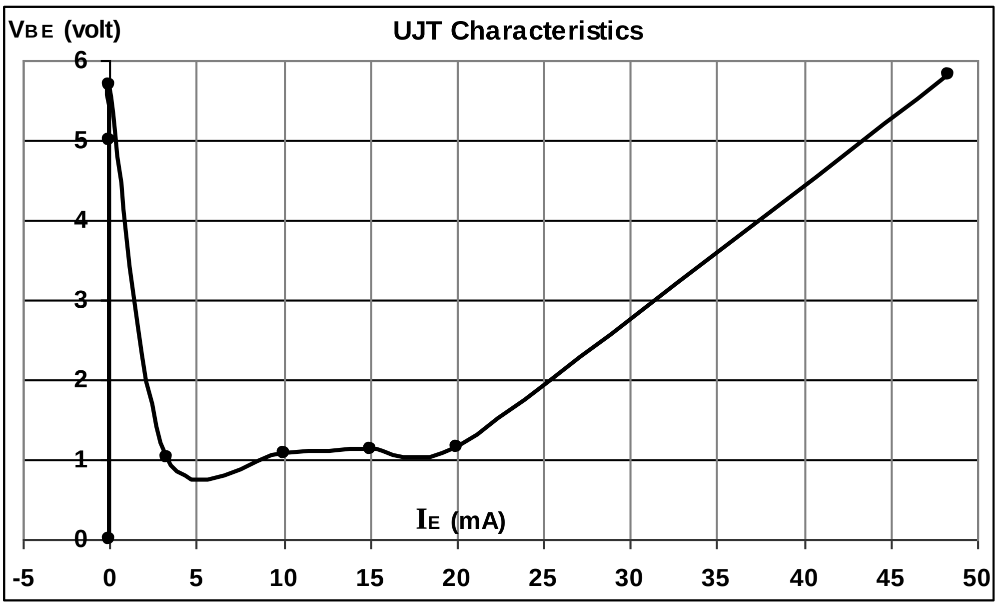

**PROCEDURE:-**
1. Connect the circuit for UJT Characteristics as per circuit diagram.
2. Switch ON power supply.
3. Apply fix $V_{BB}$= 12V.
4. By varying Vin, note the corresponding emitter-base1 voltage $V_{BE}$ and emitter current $I_E$, as per shown in the observation table.
5. Plot the UJT characteristics.




**DESIGN:-**
UJT = 2N2646.
Resistors
RE = 1KΩ.
Rb1= 220Ω.
Rb2= 2.2KΩ.

**OBSERVATION TABLE:-**

| Sr.no. | Vin (volt) | $V_{BE}$ (volt) | $I_E$ (mA) | Remark |
|---|---|---|---|---|
| 1 | 0 | 0 | 0 | |
| 2 | 5 | 5 | 0 | |
| 3 | 5.7 | 5.7 | 0 | |
| 4 | 5.82 | 5.82 | 48.3 | |
| 5 | 10 | 1.014 | 3.3 | |
| 6 | 15 | 1.084 | 10 | |
| 7 | 20 | 1.127 | 15 | |
| 8 | 25 | 1.163 | 20 | |



**CONCLUSION:-**

In the negative resistance region of V-I characteristics of UJT the emitter current increases as emitter to base1 voltage decreases.

# EXPERIMENT NO. 20

# EXPERIMENT NO.20

**AIM:-**
TO STUDY THE CHARACTERISTICS OF PUT

**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF PUT

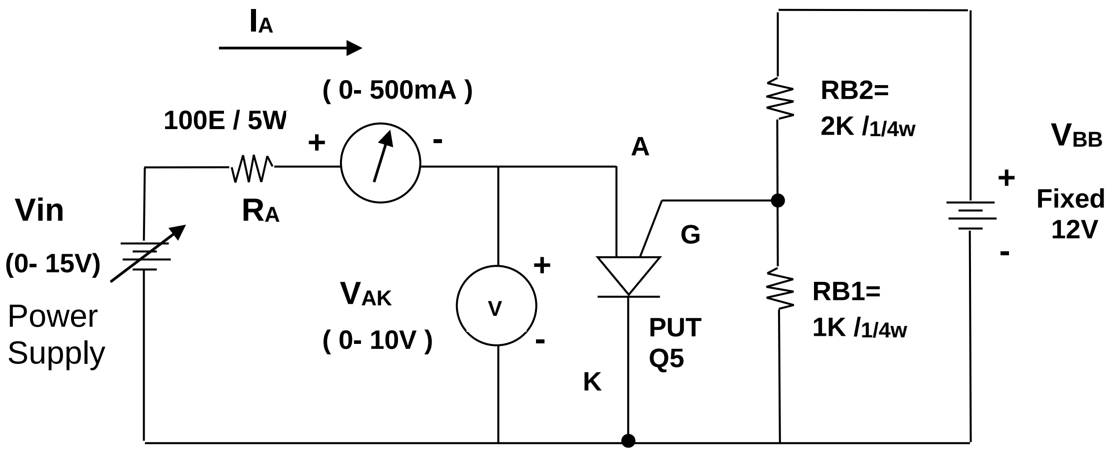
**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter

**THEORY:-**
A **unijunction transistor** (**UJT**) is an electronic semiconductor device that has only one junction. The UJT has three terminals: an emitter (E) and two bases (B1 and B2). The base is formed by lightly doped n-type bar of silicon. Two ohmic contacts B1 and B2 are attached at its ends. The emitter is of p-type and it is heavily doped. The resistance between B1 and B2, when the emitter is open-circuit is called interbase resistance.

The **programmable unijunction transistor**, or PUT, is a close cousin to the thyristor. Like the thyristor it consists of four P-N layers and has an anode and a cathode connected to the first and the last layer, and a gate connected to one of the inner layers. They are not directly interchangeable with conventional UJTs but perform a similar function. In a proper circuit configuration with two "programming" resistors for setting the parameter η, they behave like a conventional UJT. The 2N6027 is an example of such a device.

**PROCEDURE:-**
1. Connect the circuit for PUT Characteristics as per circuit diagram.
2. Switch ON power supply.
3. Apply fix $V_{BB}$= 12V.
4. By varying Vin, note the corresponding anode-cathode voltage $V_{AK}$ and anode current $I_A$, as per shown in the observation table.
5. Plot the PUT characteristics.

## PUT Characteristics

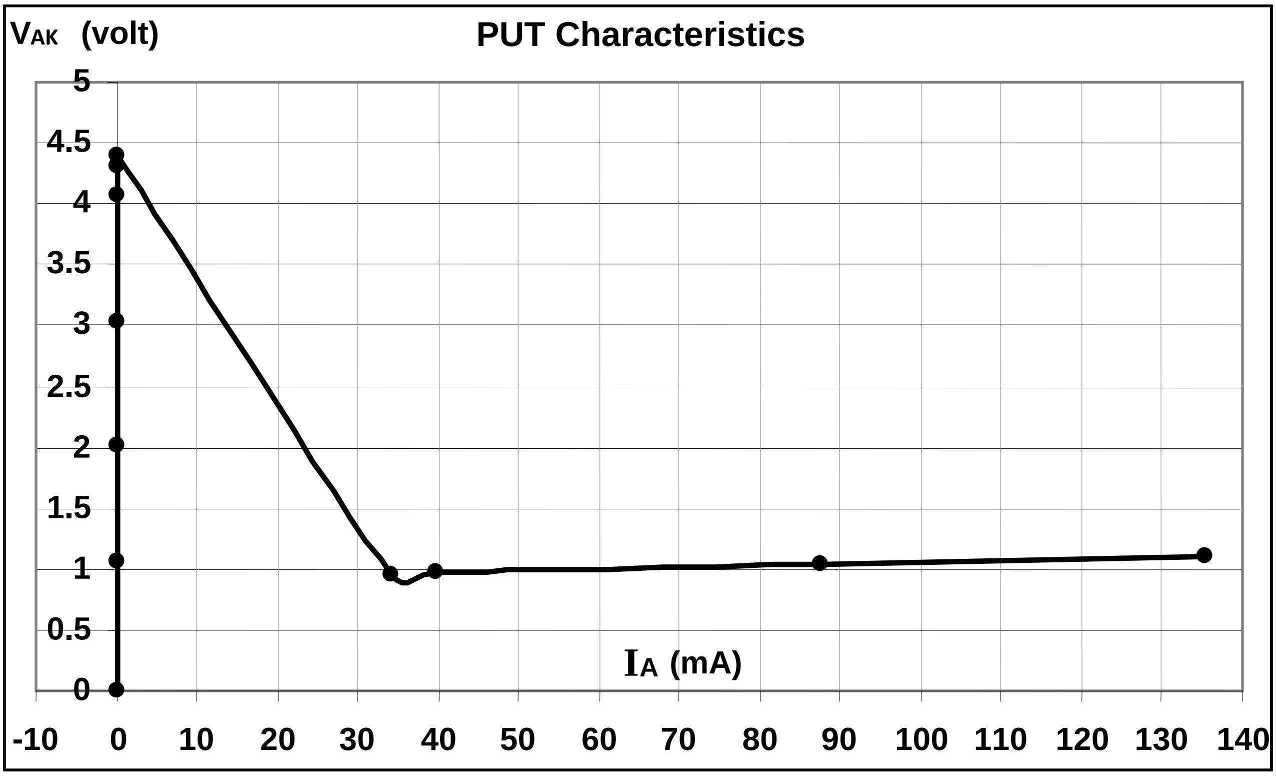



PUT= 2N6027

**DESIGN:-**
PUT = 2N6027.
Resistors $R_A$ = 1KΩ, Rb1= 1KΩ, Rb2= 2KΩ.

**OBSERVATION TABLE:-**

| Sr.no. | Vin (volt) | $V_{AK}$ (volt) | $I_A$ | Remark |
|---|---|---|---|---|
| 1 | 0 | 0 | 0 | |
| 2 | 1 | 1.052 | 0.09 μA | |
| 3 | 2 | 2.018 | 0.18 μA | |
| 4 | 3 | 3.03 | 0.29 μA | |
| 5 | 4 | 4.07 | 0.40 μA | |
| 6 | 4.3 | 4.30 | 0.43 μA | |
| 7 | 4.4 | 4.39 | 0.48 μA | |
| 8 | 4.46 | 0.957 | 34.16 mA | |
| 9 | 5 | 0.969 | 39.80 mA | |
| 10 | 10 | 1.047 | 87.44 mA | |
| 11 | 15 | 1.106 | 135.42 mA | |




**CONCLUSION:-**
In the negative resistance region of V-I characteristics of PUT the emitter current increases as emitter to base1 voltage decreases.

# EXPERIMENT NO. 21

# EXPERIMENT NO.21

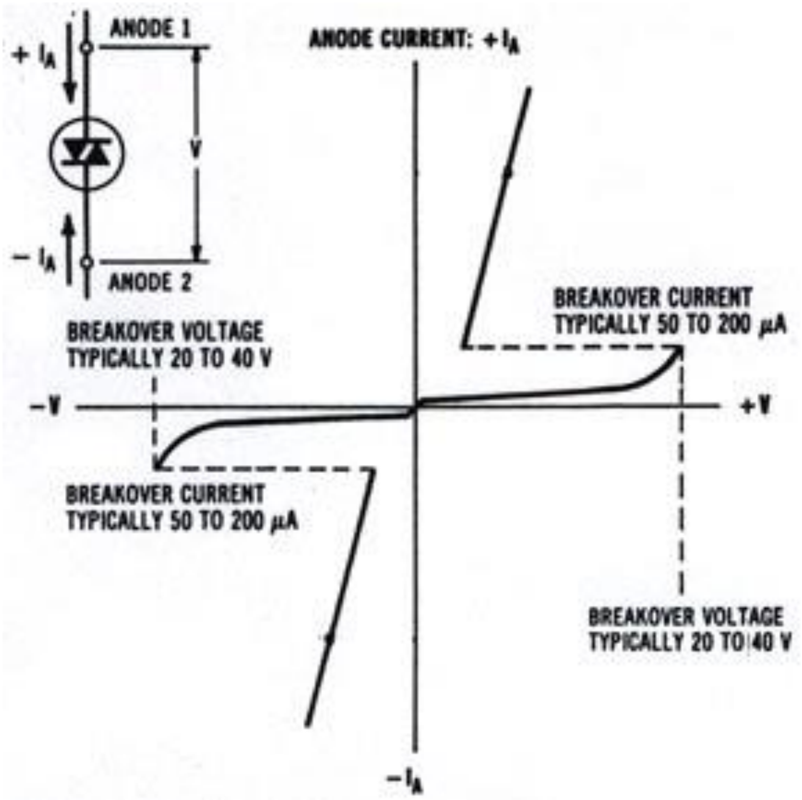
**AIM:-**
TO STUDY THE CHARACTERISTICS OF DIAC

**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF DIAC

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter

**THEORY:-**
The **DIAC**, or **diode for alternating current**, is a trigger diode that conducts current only after its breakdown voltage has been exceeded momentarily. When this occurs, the resistance of the diode abruptly decreases, leading to a sharp decrease in the voltage drop across the diode and, usually, a sharp increase in current flow through the diode. The diode remains "in conduction" until the current flow through it drops below a value characteristic for the device, called the holding current. Below this value, the diode switches back to its high-resistance (non-conducting) state. This behavior is bidirectional, meaning typically the same for both directions of current flow.

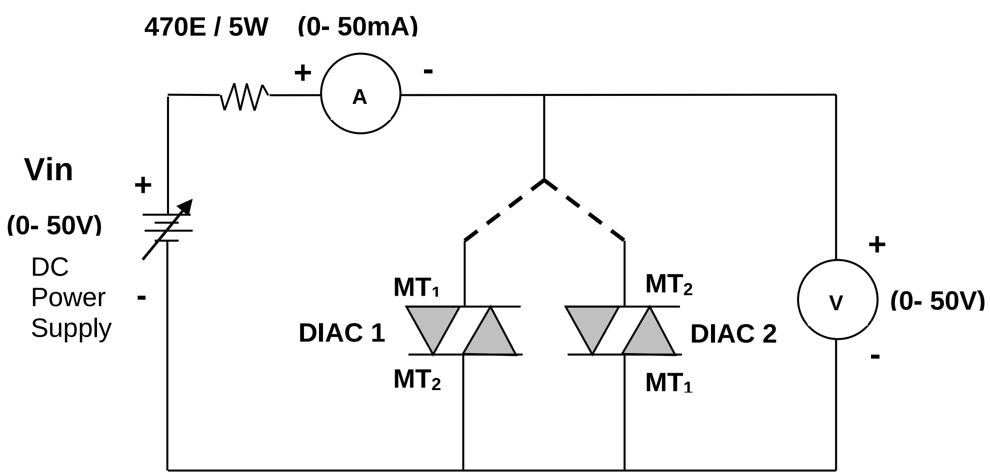
DIACs are also called symmetrical trigger diodes due to the symmetry of their characteristic curve. Because DIACs are bidirectional devices, their terminals are not labeled as anode and cathode but as A1 and A2 or MT1 ("Main Terminal") and MT2. Most DIACs have a breakdown voltage around 30 V. In this way, their behavior is somewhat similar to (but much more precisely controlled and taking place at lower voltages than) a neon lamp.
DIACs have no gate electrode, unlike some other thyristors they are commonly used to trigger, such as TRIACs. Some TRIACs contain a built-in DIAC in series with the TRIAC's "gate" terminal for this purpose.




**<u>PROCEDURE</u>:-**

1. Voltage requirement is more than 30V; therefore connect on-board power sec-1in series with power sec-2. So we get 0-60V variable dc power source. Connect gnd terminal of power sec-1 to +ve terminal of power sec-1. Take 0 to 60V from +ve terminal of power sec-1 and gnd terminal of power sec-2.
2. Connect the circuit for DIAC Characteristics as per circuit diagram.
3. Switch ON power supply.
4. Connect DIAC terminals in forward bias and by varying Vin, note the corresponding $V_{MT1MT2}$ and current through DIAC, as per shown in the observation table.
5. Plot the DIAC forward characteristics.
6. Reverse the terminals of DIAC, by varying Vin, note the corresponding $V_{MT2MT1}$ and current through DIAC, as per shown in the observation table.
7. Plot the DIAC reverse characteristics.

**DIAC Characteristics**



**NOTE: - Voltage requirement is more than 30V; therefore connect on-board power sec-1 in series with power sec-2. So we get 0-60V variable dc power source. Connect gnd terminal of power sec-1 to +ve terminal of power sec-1. Take 0 to 60V from +ve terminal of power sec-1 and gnd terminal of power sec-2.**

**<u>DESIGN</u>:-**

DIAC = DB3.

Resistor = 470E / 5W.

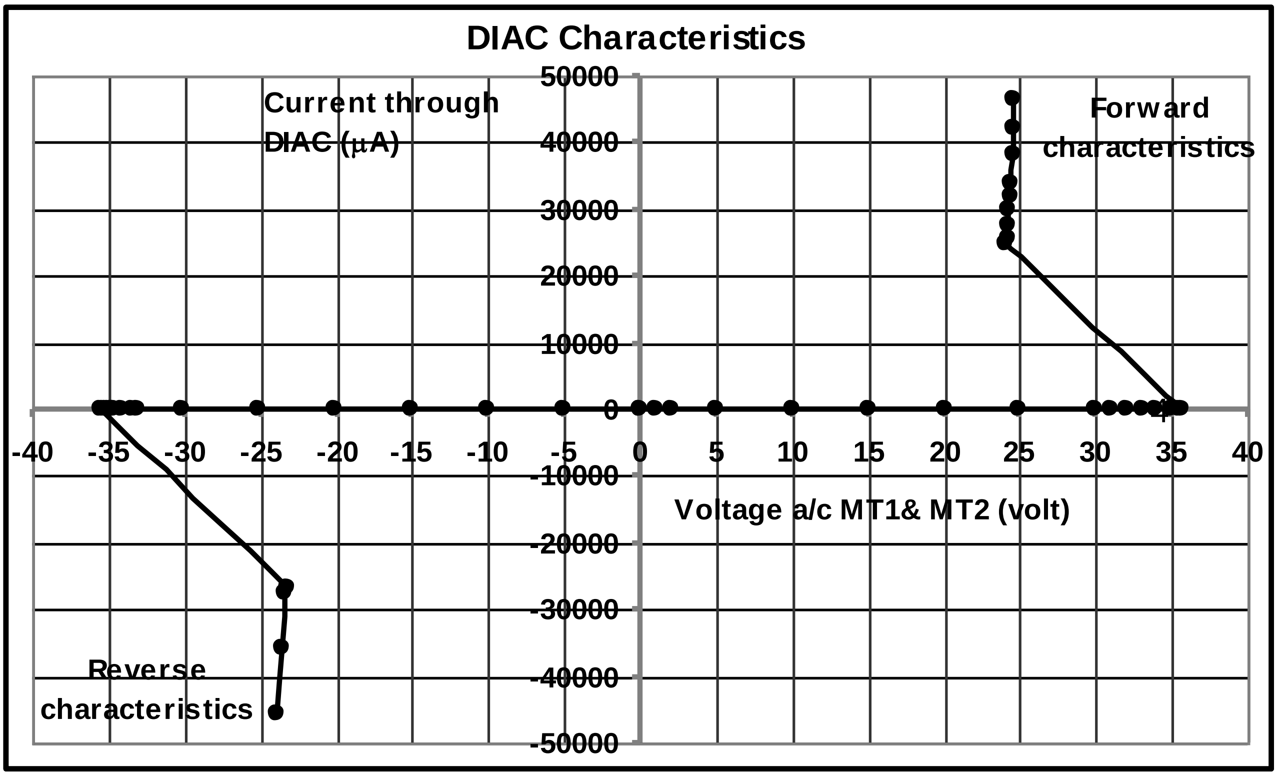
**<u>OBSERVATION TABLE</u>:-**

**Forward characteristics:**

| Sr. no. | Vin (volt) | Voltage MT1& MT2 (volt) | Current through DIAC |
|---|---|---|---|
| 1 | 0 | 0 | 0 |
| 2 | 1.0 | 1.0 | 0.3 μA |
| 3 | 2.0 | 2.0 | 0.5 μA |
| 4 | 5.0 | 5.0 | 0.9 μA |
| 5 | 10.0 | 10.0 | 1.6 μA |
| 6 | 15.0 | 15.0 | 2.1 μA |
| 7 | 20.0 | 20.0 | 2.5 μA |
| 8 | 25.0 | 25.0 | 3.1 μA |
| 9 | 30.0 | 30.0 | 3.7 μA |
| 10 | 31.0 | 31.0 | 3.8 μA |
| 11 | 32.0 | 32.0 | 4.0 μA |
| 12 | 33.0 | 33.0 | 4.1 μ |
| 13 | 34.0 | 34.0 | 4.3 μA |
| 14 | 35.0 | 35.0 | 4.6 μA |
| 15 | 35.5 | 35.5 | 4.7 μA |
| 16 | 35.6 | 35.6 | 4.9 μA |
| 17 | 35.7 | 24.10 | 25 mA |
| 18 | 36 | 24.26 | 25.78 mA |
| 19 | 37 | 24.28 | 27.58 mA |
| 20 | 38 | 24.30 | 30.06 mA |
| 21 | 39 | 24.38 | 31.77 mA |
| 22 | 40 | 24.43 | 33.95 mA |
| 23 | 42 | 24.53 | 38.05 mA |
| 24 | 44 | 24.59 | 42.2 mA |
| 25 | 46 | 24.56 | 46.5 mA |

**Reverse characteristics:**

| Sr. no. | Vin (volt) | Voltage MT1& MT2 (volt) | Current through DIAC |
|---|---|---|---|
| 1 | 0 | 0 | 0 |
| 2 | 5 | 5.036 | 0.4 μA |
| 3 | 10 | 10.02 | 0.9 μA |
| 4 | 15 | 15.14 | 1.5 μA |
| 5 | 20 | 20.16 | 2.0 μA |
| 6 | 25 | 25.08 | 2.5 μA |
| 7 | 30 | 30.11 | 3.2 μA |
| 8 | 33 | 33.01 | 3.9 μA |
| 9 | 33.5 | 33.50 | 4.1 μA |
| 10 | 34 | 34.07 | 4.5 μA |
| 11 | 34.5 | 34.61 | 4.9 μA |
| 12 | 34.6 | 34.69 | 5.0 μA |
| 13 | 34.7 | 34.81 | 5.2 μA |
| 14 | 34.8 | 34.91 | 5.3 μA |
| 15 | 35 | 35.11 | 5.9 μA |
| 16 | 35.2 | 35.29 | 7.4 μA |
| 17 | 35.3 | 35.42 | 12.2 μA |
| 18 | 35.5 | 23.15 | 26.80 mA |
| 19 | 36 | 23.31 | 27.63 mA |
| 20 | 40 | 23.59 | 35.68 mA |
| 21 | 45 | 23.98 | 45.5 mA |



**CONCLUSION:-**

In the V-I Characteristics of DIAC, when voltage increases current increases in conduction state. Diac is a bi-directional device.

# EXPERIMENT NO. 22

# EXPERIMENT NO.22

**AIM:-**
TO STUDY THE CHARACTERISTICS OF TRIAC

**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF TRIAC

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter

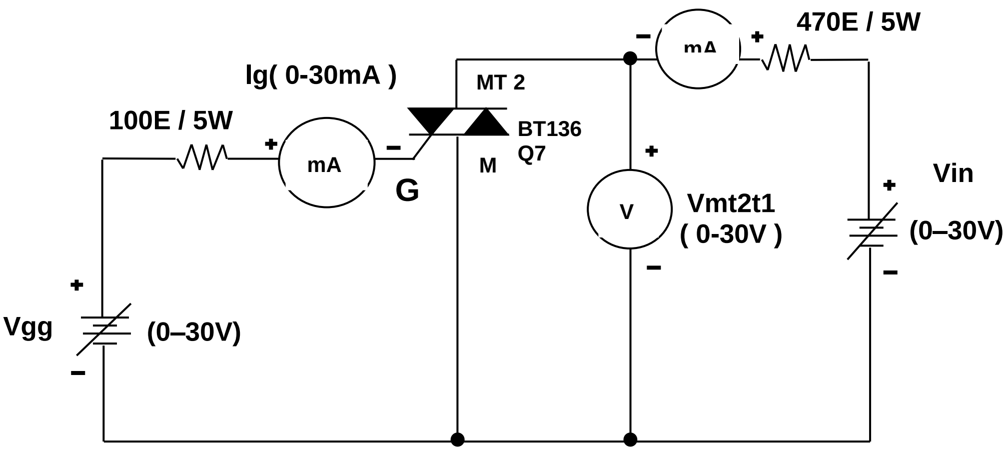
**THEORY:-**
The drawback of the diac is the same as it was for the four-layer diode: it cannot be triggered at just any point in the ac power cycle; it triggers at its preset break over voltage only. If we could add a gate to the diac, we could have variable control of the trigger point, and therefore a greater degree of control over just how much power will be applied to the line-powered device.

A **TRIAC**, or **Triode for Alternating Current** is an electronic component approximately equivalent to two silicon-controlled rectifiers (SCRs / thyristors) joined in inverse parallel (paralleled but with the polarity reversed) and with their gates connected together. The formal name for a TRIAC is **bidirectional triode thyristor**. This results in a bidirectional electronic switch which can conduct current in either direction when it is triggered (turned on) and thus doesn't have any polarity. It can be triggered by either a positive or a negative voltage being applied to its gate electrode (with respect to A1, otherwise known as MT1). Once triggered, the device continues to conduct until the current through it drops below a certain threshold value, the holding current, such as at the end of a half-cycle of alternating current (AC) mains power. This makes the TRIAC a very convenient switch for AC circuits, allowing the control of very large power flows with mill ampere-scale control currents. In addition, applying a trigger pulse at a controllable point in an AC cycle allows one to control the percentage of current that flows through the TRIAC to the load (phase control).

The useful feature of the triac is that it not only carries current in either direction, but the gate trigger pulse can be either polarity regardless of the polarity of the main applied voltage. The gate can inject either free electrons or holes into the body of the triac to trigger conduction either way. For this reason, you may see the triac referred to as a "four-quadrant" device.

**PROCEDURE:-**
**TRIAC CHARACTERISTICS = MODE 1:-**
1. Connect the circuit for TRIAC Characteristics mode 1 as per circuit diagram.
2. Switch on power supply.
3. Connect TRIAC terminals in forward bias and keep $I_G$ = 1.3mA using Vgg.
4. By varying Vin, note the corresponding $V_{MT1MT2}$ and forward current through TRIAC, $I_A$, as per shown in the observation table.
5. Plot the TRIAC forward characteristics for mode 1.

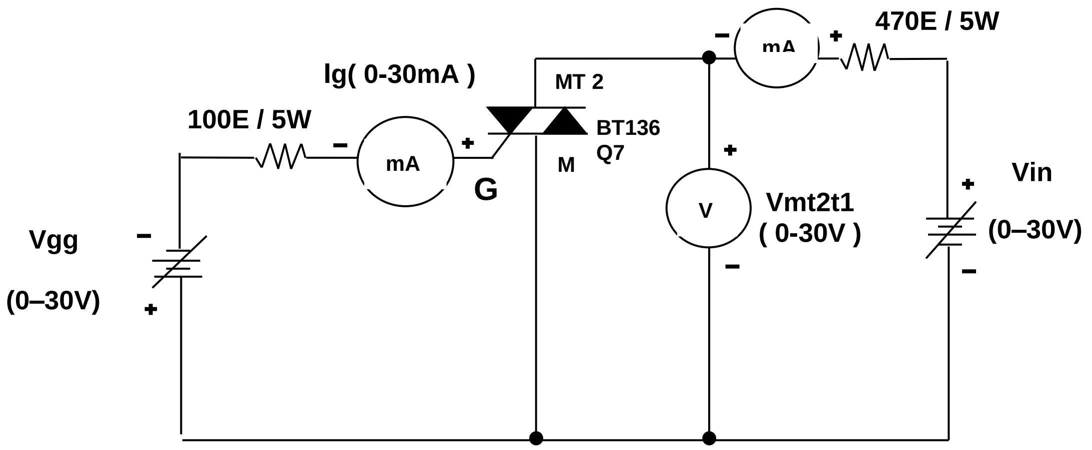
**TRIAC Characteristics = Mode 1** **$I_A$(0-100mA)**




**TRIAC CHARACTERISTICS = MODE 2:-**

1. Connect the circuit for TRIAC Characteristics mode 2 as per circuit diagram.
2. Switch on power supply.
3. Connect TRIAC terminals in forward bias and keep $I_G$ = 2.4mA using Vgg.
4. By varying Vin, note the corresponding $V_{MT1MT2}$ and forward current through TRIAC, $I_A$, as per shown in the observation table.
5. Plot the TRIAC forward characteristics for mode 2.

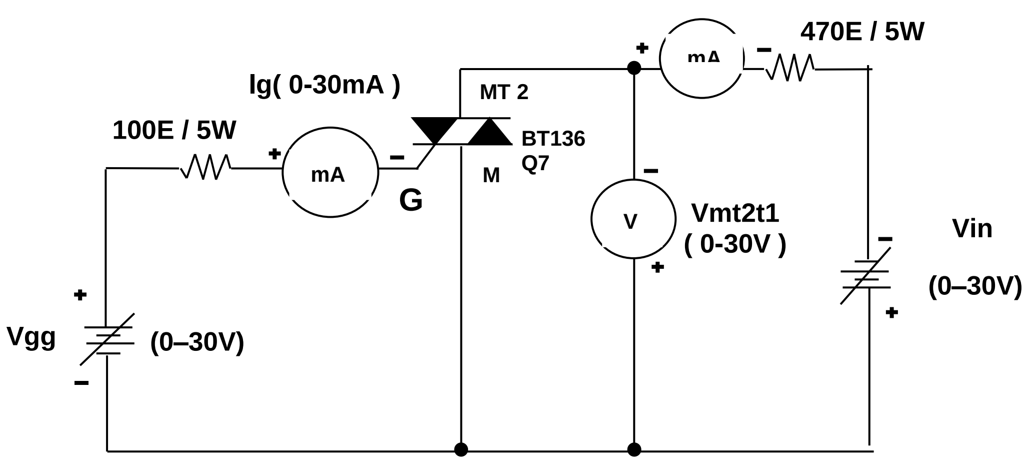
**TRIAC Characteristics = Mode 2** **$I_A$(0-100mA)**




**TRIAC CHARACTERISTICS = MODE 3:-**

1. Connect the circuit for TRIAC Characteristics mode 3 as per circuit diagram.
2. Switch on power supply.
3. Connect TRIAC terminals in forward bias and keep $I_G$ = 5.9mA using Vgg.
4. By varying Vin, note the corresponding $V_{MT1MT2}$ and reverse current through TRIAC, $I_A$, as per shown in the observation table.
5. Plot the TRIAC reverse characteristics for mode 3.

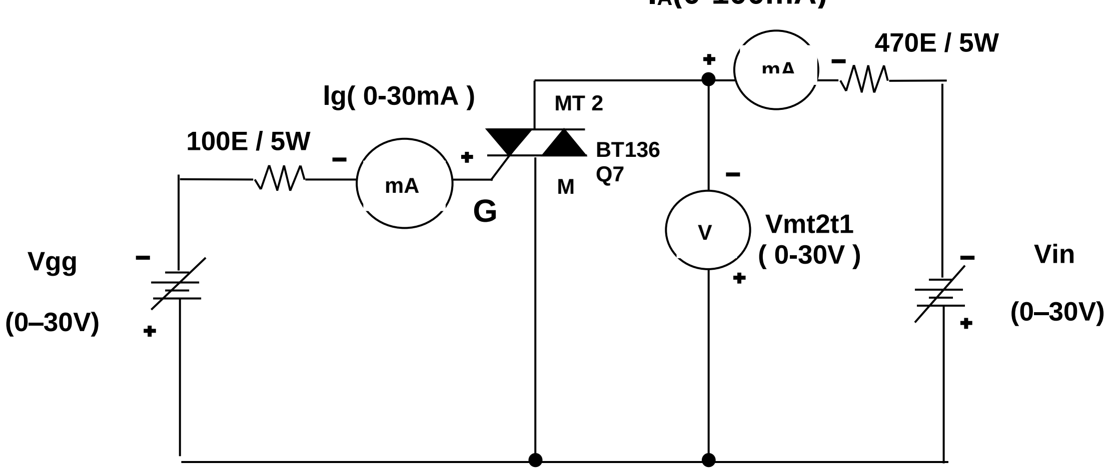
TRIAC Characteristics = Mode 3




**TRIAC CHARACTERISTICS = MODE 4:-**

1. Connect the circuit for TRIAC Characteristics mode 4 as per circuit diagram.
2. Switch on power supply.
3. Connect TRIAC terminals in forward bias and keep $I_G$ = 2.89mA using Vgg.
4. By varying Vin, note the corresponding $V_{MT1MT2}$ and reverse current through TRIAC, $I_A$, as per shown in the observation table.
5. Plot the TRIAC reverse characteristics for mode 4.

TRIAC Characteristics = Mode 4




**<u>DESIGN</u>:-**

TRIAC = BT136.

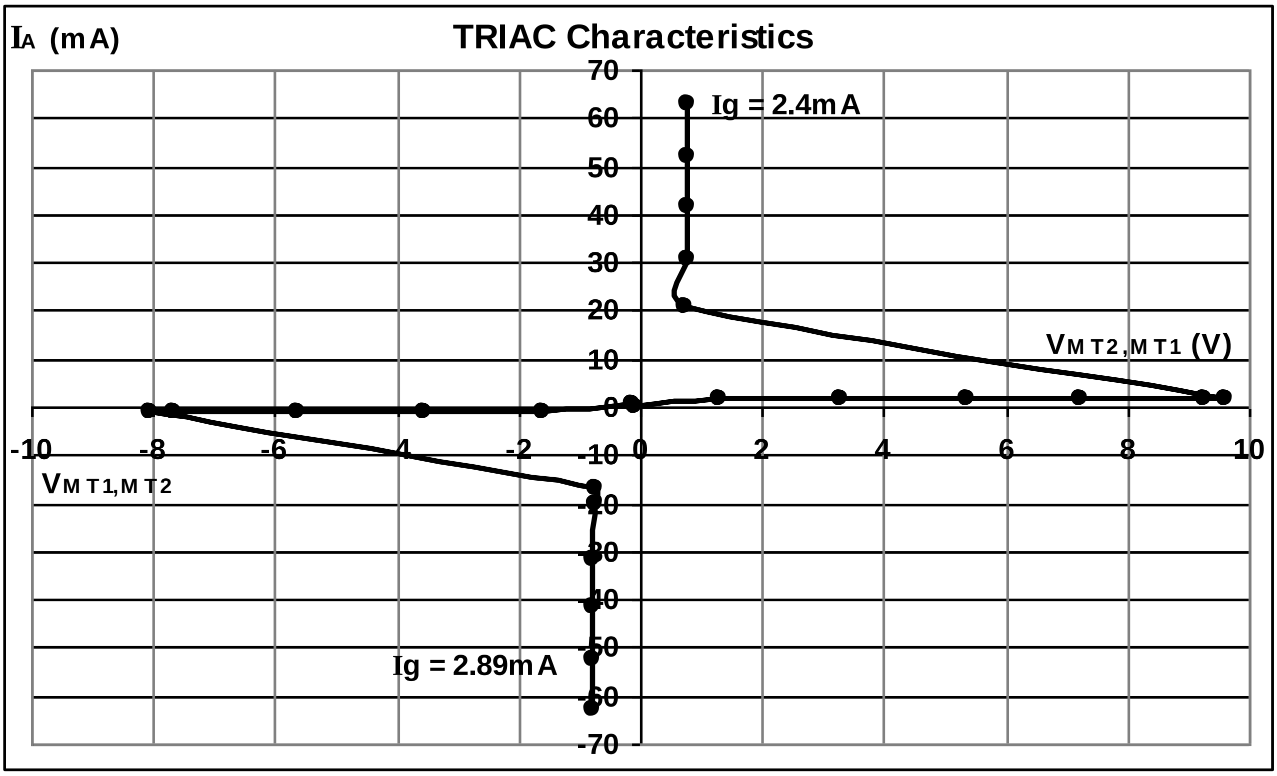
Resistors = 100E/5W and 470E/5W.

## OBSERVATION TABLE:-

| Sr. no | Vin (V) | MODE 1 (Ig = 1.3mA) | | Vin (V) | MODE 2 (Ig = 2.4mA) | | Vin (V) | MODE 3 (Ig = 5.9mA) | | Vin (V) | MODE 4 (Ig = 2.89mA) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | $V_{MT2,MT1}$ | $I_A$ | | $V_{MT2,MT1}$ | $I_A$ (mA) | | $V_{MT1,MT2}$ | $I_A$ (mA) | | $V_{MT1,MT2}$ | $I_A$(mA) |
| 1 | 0 | 0.023 | -24.8 uA | 0 | -0.108 | 0.25 | 0 | -0.376 | 0.82 | 0 | 0.142 | -0.28 |
| 2 | 2 | 1.979 | 76.5 uA | 2 | 1.297 | 1.52 | 2 | 1.436 | 1.21 | 2 | 1.589 | 0.94 |
| 3 | 4 | 3.974 | 75.7 uA | 4 | 3.282 | 1.58 | 4 | 3.415 | 1.23 | 4 | 3.546 | 0.95 |
| 4 | 6 | 5.974 | 75.6 uA | 6 | 5.35 | 1.62 | 6 | 5.451 | 1.24 | 6 | 5.632 | 0.96 |
| 5 | 8 | 8.05 | 76.0 uA | 8 | 7.25 | 1.65 | 8 | 7.44 | 1.25 | 8 | 7.66 | 0.98 |
| 6 | 10 | 9.98 | 76.1 uA | 10 | 9.26 | 1.71 | 10 | 9.52 | 1.27 | 8.5 | 8.06 | 1.01 |
| 7 | 10.1 | 10.19 | 75.2 uA | 10.2 | 9.59 | 1.77 | 10.6 | 10.09 | 1.29 | 8.6 | 0.732 | 16.79 |
| 8 | 10.7 | 10.68 | 75.1 uA | 10.3 | 0.744 | 20.65 | 10.7 | 10.20 | 1.30 | 10 | 0.740 | 20.06 |
| 9 | 10.8 | 0.738 | 21.72 mA | 15 | 0.762 | 30.84 | 10.8 | 0.709 | 21.74 | 15 | 0.761 | 31.62 |
| 10 | 15 | 0.761 | 30.83 mA | 20 | 0.777 | 41.6 | 15 | 0.737 | 30.98 | 20 | 0.776 | 41.4 |
| 11 | 20 | 0.775 | 41.5 mA | 25 | 0.788 | 52.2 | 20 | 0.758 | 41.4 | 25 | 0.788 | 52.5 |
| 12 | 25 | 0.786 | 52.1 mA | 30 | 0.797 | 62.9 | 25 | 0.773 | 52.3 | 30 | 0.797 | 63.0 |
| 13 | 30 | 0.795 | 62.9 mA | | | | 30 | 0.785 | 63.1 | | | |

**TRIAC CHARACTERISTICS GRAPH:-**

Here only mode2 forward characteristics and mode 4 reverse characteristics are plotted.




## CONCLUSION:-

Mode 1 and 2 are the forward characteristic mode and mode 3 and 4 are revere characteristics mode. Mode 1, 2 are same and mode 3, 4 are same. Therefore Triac is a four quadrant device.

# EXPERIMENT NO. 23

# EXPERIMENT NO.23

**AIM:-**
TO STUDY THE CHARACTERISTICS OF SCR

**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF SCR

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter.

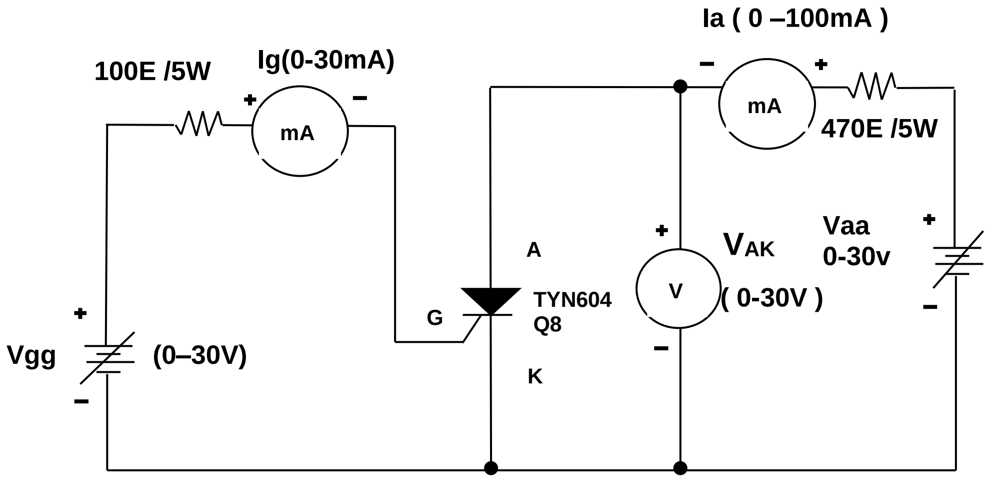
**THEORY:-**
As already mentioned, the **SCR** is a four-layer device with three terminals, namely, the anode, the cathode and the gate. When the anode is made positive with respect to the cathode, junctions $J_1$ and $J_3$ are forward biased and junction $J_2$ is reverse-biased and only the leakage current will flow through the device. The SCR is then said to be in the forward blocking state or in the forward mode or off state. But when the cathode is made positive with respect to the anode, junctions $J_1$ and $J_3$ are reverse-biased, a small reverse leakage current will flow through the SCR and the SGR is said to be in the reverse blocking state or in reverse mode.

When the anode is positive with respect to cathode i.e. when the SCR is in forward mode, the SCR does not conduct unless the forward voltage exceeds certain value, called the forward break over voltage, $V_{FB0}$. In non-conducting state, the current through the SCR is the leakage current which is very small and is negligible. If a positive gate current is supplied, the SCR can become conducting at a voltage much lesser than forward break-over voltage. The larger the gate current, lower the break-over voltage. With sufficiently large gate current, the SCR behaves identical to PN rectifier. Once the SCR is switched on, the forward voltage drop across it is suddenly reduced to very small value, say about 1 volt. In the conducting or on-state, the current through the SCR is limited by the external impedance.

When the anode is negative with respect to cathode that is when the SCR is in reverse mode or in blocking state no current flows through the SCR except very small leakage current of the order of few micro-amperes. But if the reverse voltage is increased beyond a certain value, called the reverse break-over voltage, $V_{RB0}$ avalanche break down takes place. Forward break-over voltage $V_{FB0}$ is usually higher than reverse break over voltage, $V_{RBO}$.

**PROCEDURE:-**
1. Connect the circuit for SCR forward Characteristics as per circuit diagram.
2. Switch ON power supply.
3. Connect SCR terminals in forward bias and keep $I_G$ = 2.937mA by varying Vgg.
4. By varying Vaa, note the corresponding $V_{AK}$ and forward current through SCR, $I_G$, as per shown in the observation table.
5. Similarly, repeat the above step 4 for $I_G$ = 3.5mA by varying Vgg.
6. Plot the SCR forward characteristics.

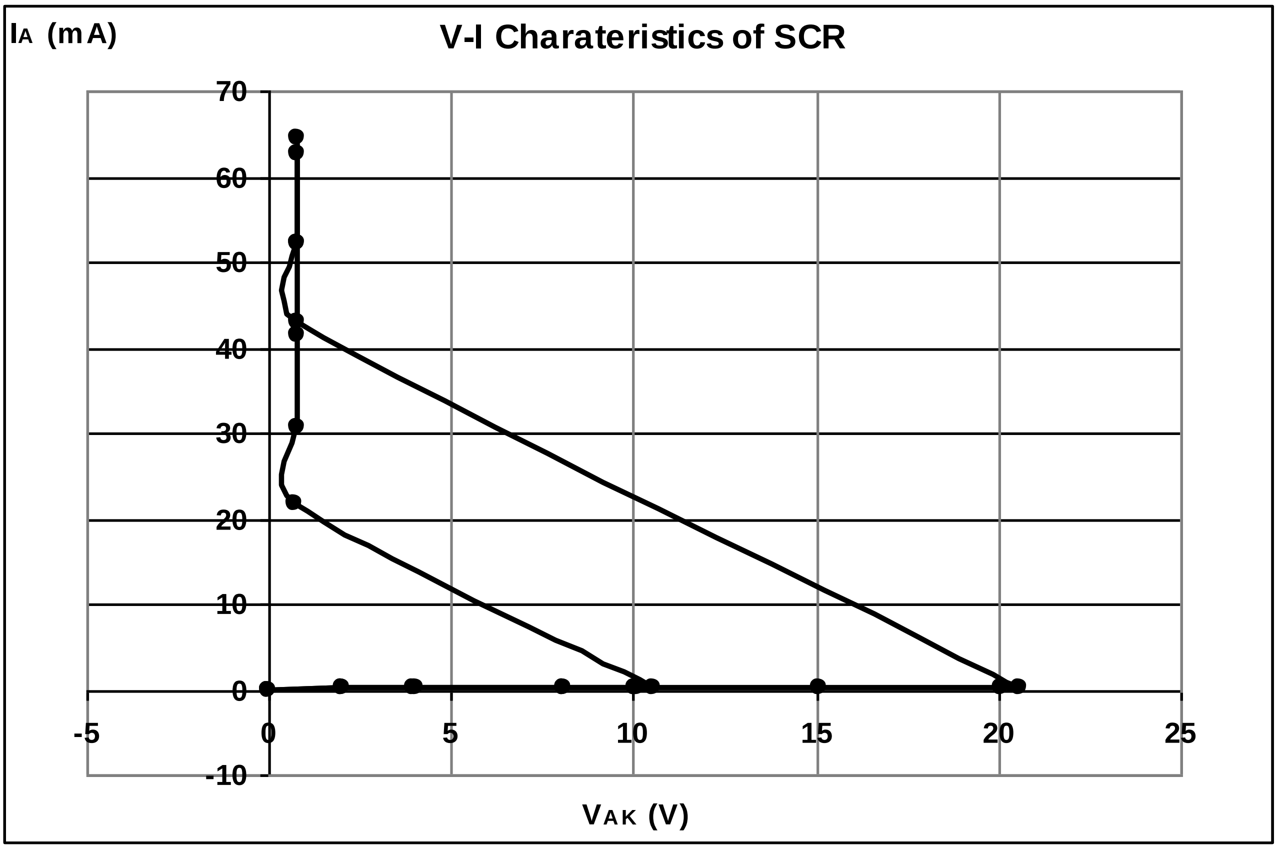
## SCR Characteristics




**DESIGN**:-
SCR = TYN604.
Resistors = 100E/5W and 470E/5W.

**OBSERVATION TABLE**:-

| Sr. no. | Vin (volt) Vgg | $I_G$ = 2.937mA | | Vin (volt) Vgg | $I_G$ = 3.5mA | |
|---|---|---|---|---|---|---|
| | | $V_{AK}$ | $I_A$ | | $V_{AK}$ | $I_A$ |
| 1 | 0 | 0.008 | -0.27 μA | 0 | 0.016 | -0.24 μA |
| 2 | 2 | 1.988 | 0.99 μA | 2 | 1.979 | 1.1 μA |
| 3 | 4 | 4.023 | 1.18 μA | 4 | 3.996 | 1.2 μA |
| 4 | 8 | 8.05 | 1.64 μA | 8 | 8.05 | 1.68 μA |
| 5 | 10 | 10.08 | 1.84 μA | 10 | 10.03 | 1.88 μA |
| 6 | 15 | 15.09 | 2.3 μA | 10.6 | 10.56 | 2.5 μA |
| 7 | 20 | 20.09 | 2.8 μA | 10.8 | 0.741 | 21.6 mA |
| 8 | 20.5 | 20.6 | 2.9 μA | 15 | 0.758 | 30.73 mA |
| 9 | 20.6 | 0.769 | 42.8 mA | 20 | 0.772 | 41.34 mA |
| 10 | 25 | 0.783 | 52.2 mA | 25 | 0.782 | 52.09 mA |
| 11 | 30 | 0.795 | 64.5 mA | 30 | 0.791 | 62.68 mA |




**CONCLUSION:-**

In VI Characteristics of SCR, the current increases slowly with increase in $V_{AK}$ till $V_{BO}$. After $V_{BO}$ as small increases in voltage produces large current in SCR.

# EXPERIMENT NO. 24

# EXPERIMENT NO.24

**AIM:-**
TO STUDY THE CHARACTERISTICS OF IGBT

**OBJECTIVE:-**
TO STUDY THE CHARACTERISTICS OF IGBT

**EQUIPMENTS:-**
1. BEL-COT Trainer kit.
2. Power Supply.
3. Patch cords.
4. Multi-meter.

**THEORY:-**
The structure of a typical n-channel IGBT. All discussion here will be concerned with the n-channel type but p-channel IGBT's can be considered in just the same way.

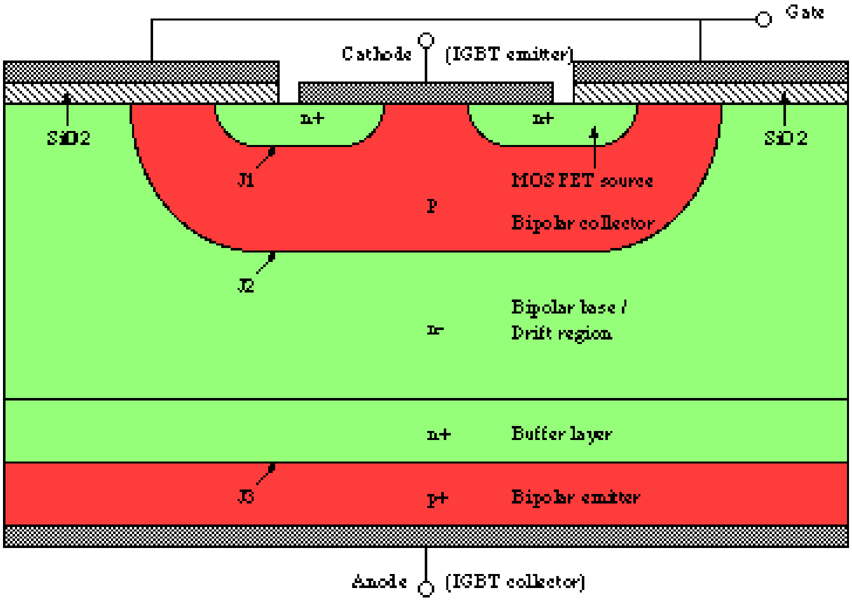


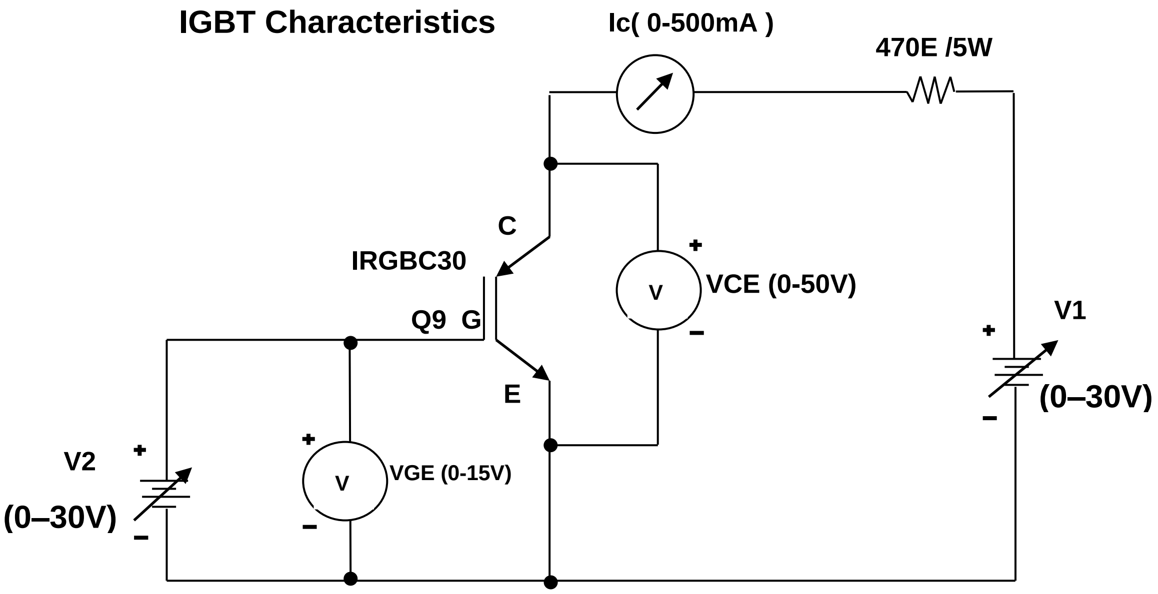

Fig.1: A typical IGBT structure

The structure is very similar to that of a vertically diffused MOSFET featuring a double diffusion of a p-type region and an n-type region. An inversion layer can be formed under the gate by applying the correct voltage to the gate contact as with a MOSFET. The main difference is the use of a $p^+$ substrate layer for the drain. The effect is to change this into a bipolar device as this p-type region injects holes into the n-type drift region.

**PROCEDURE:-**
**COLLECTOR CHARACTERISTICS:-**
1. Connect the circuit for IGBT Characteristics as per circuit diagram.
2. Switch ON power supply.
3. Connect IGBT terminals in forward bias and keep $V_{GE}$ = 5.5V by varying V2.
4. By varying V1, note the collector–emitter voltage $V_{CE}$ corresponding current through IGBT, $I_C$, as per shown in the observation table.
5. Similarly, repeat the above step 3 for $V_{GE}$ = 6V by varying V2.
6. Plot the IGBT characteristics.

**TRANSCONDUCTANCE CHARACTERISTICS:-**

1. Connect the circuit for IGBT Characteristics as per circuit diagram.
2. Switch ON power supply.
3. Connect IGBT terminals in forward bias and keep $V_{CE}$ = 0.8V by varying V1.
4. By varying V2, note the gate–emitter voltage $V_{GE}$ corresponding current through IGBT, $I_C$, as per shown in the observation table.
5. Similarly, repeat the above step 3 for $V_{CE}$ = 1V by varying V1.
6. Plot the IGBT characteristics.




**DESIGN:-**

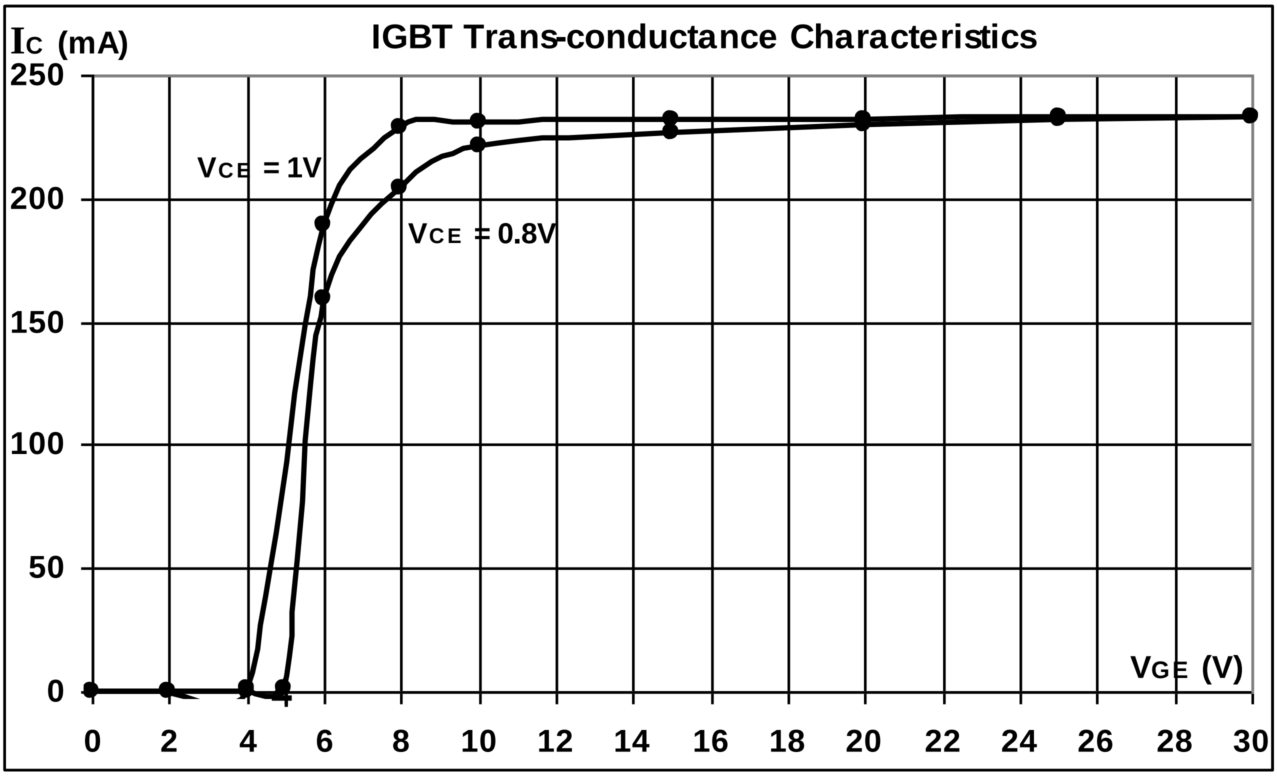
IGBT = IRGBC305.
Resistor = 470E/5W.

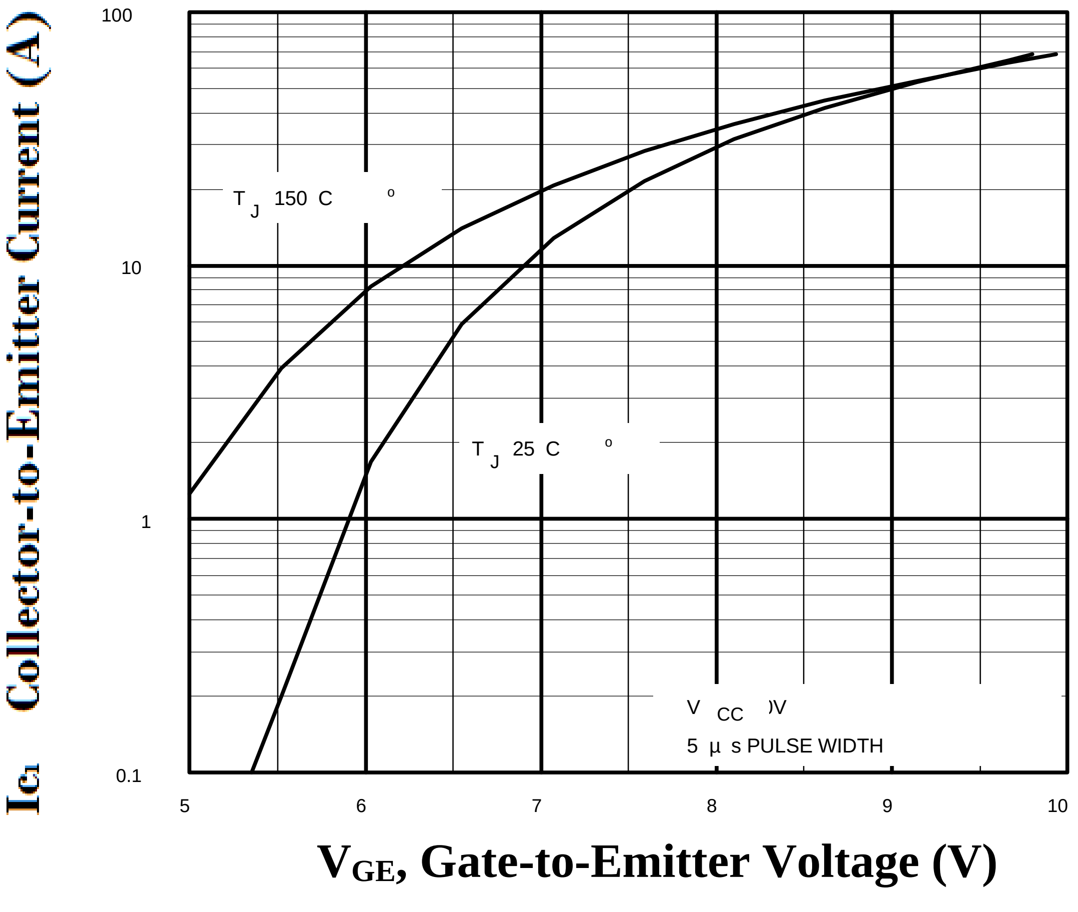
**OBSERVATION TABLE:-**

**Trans-conductance Characteristics:-**

| Sr. no. | $V_{CE}$ = 0.8V | | $V_{CE}$ = 1V | |
|---|---|---|---|---|
| | $V_{GE}$ (V)=V2 | | $V_{GE}$ (V) =V2 | |
| 1 | 0 | | 0 | |
| 2 | 2 | | 2 | |
| 3 | 4 | | 4 | |
| 4 | 5 | | 6 | |
| 5 | 6 | | 8 | |
| 6 | 8 | | 10 | |
| 7 | 10 | | 15 | |
| 8 | 15 | | 20 | |
| 9 | 20 | | 25 | |
| 10 | 25 | | 30 | |
| 11 | 30 | | | |

**Trans-conductance Characteristics:**



## **CONCLUSION**:-

In collector Characteristics of IGBT, as collector-emitter voltage increases, increases the collector current through IGBT and also in Trans-conductance characteristics collector current increases with increase in gate-emitter voltage.